Debt Capacity and the Anatomy of Sovereign Crises*

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Abstract

Sovereign defaults are rarely complete: governments withhold part of their debt service, the missed payments accumulate as arrears, and borrowing continues. We study the partial default model of Arellano, Mateos-Planas, and Ríos-Rull (2023) in continuous time and find that it changes the account of how sovereign debt crises begin, unfold, and end. Withholding debt service raises funds at a price fixed by the recovery on arrears, so a sovereign's debt capacity, the most it will service in full, is set in closed form by the cost of defaulting and the price of its bonds, and not by how much it values market access. Default episodes are triggered by shocks that overwhelm a precautionary buffer, and they begin with a jump whose size the default costs pin down. Once an episode begins, it is a financing regime: the sovereign keeps borrowing, and the regime ends when output recovers rather than when the debt is repaid. Pessimism alone can bring on a slow-moving default, without a run. The model is tractable because the sovereign never loses market access: given the bond price, the default decision is static. Calibrated to the same targets, it fits the data about as well as the discrete-time original.

Keywords: sovereign debt, partial default, arrears, debt capacity, self-fulfilling crises
JEL classification: E44, F34, F41, H63

Latest version: https://gmihalache.com/docs/KM_partial.pdf
Replication package: https://github.com/gabrielgggg/Partial-Default-Replication

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1 Introduction

Sovereign defaults are rarely complete. Governments in distress typically continue to service some of their obligations while suspending payments on others, the missed payments accumulate as arrears that add to the stock of debt, and countries often emerge from a default episode owing more than when it began. Arellano, Mateos-Planas, and Ríos-Rull (2023) document these facts and build a theory of *partial default* around them. A sovereign chooses, period by period, what share of its debt service to withhold, at an output cost that rises with the share withheld. It never loses access to the market, where the price of its bonds reflects the default intensity lenders expect. Their framework replaces the binary default decision of the standard model, introduced by Eaton and Gersovitz (1981) and quantified by Arellano (2008), with a continuous one, and exclusion from financial markets with continued access. We regard partial default as the right description of sovereign distress and the natural baseline for the quantitative theory of sovereign debt.[1]

This paper asks what the partial default model teaches about the economics of sovereign crises, and finds four lessons. The debt a sovereign will service in full is limited by the cost of defaulting, not by the value it attaches to market access. Defaults begin with a jump, whose size the cost structure fixes, and they are triggered by shocks that overwhelm a precautionary buffer, not by borrowing into default. Once a crisis episode has begun, the sovereign keeps borrowing, and the episode ends when output recovers, not when the debt is repaid. Pessimism alone can produce a slow crisis, without a run or loss of market access. We draw our lessons from three versions of the model. The first has a constant endowment, where a sharp characterization in closed form is possible. The second puts the endowment on a Markov chain, which adds precautionary behavior. The third has a diffusion calibrated to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023). We write the model in continuous time, where the default intensity is a flow decision and arrears accrue continuously. The sovereign never loses market access, so there is no stopping problem, and given the bond price the default decision is static. Tractability does not come at the cost of quantitative performance: the calibrated diffusion model fits the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023) about as well as their discrete-time model. Figure 1 shows one default episode in that model, with the first three lessons at work.

The first lesson concerns the sovereign's debt limit. In the binary model the limit is the outcome of the whole dynamic problem: the debt at which the value of staying in the market no longer covers repayment. Under partial default it reflects a static comparison. Withholding a unit of debt service raises consumption by one minus the market value of the arrears it creates, at the cost of lost output. It is a source of funds whose price is fixed by the recovery on arrears and not by the market, which is why arrears are attractive precisely when market prices are low. Given debt, the endowment, and the price, the share withheld maximizes a resource function that de-

1. In their data, emerging economies are in partial default about a third of the time and withhold about forty percent of the debt service when they are. The framework has since been applied to the debt crises that the COVID-19 pandemic set off in emerging markets (Arellano, Bai, and Mihalache 2024).

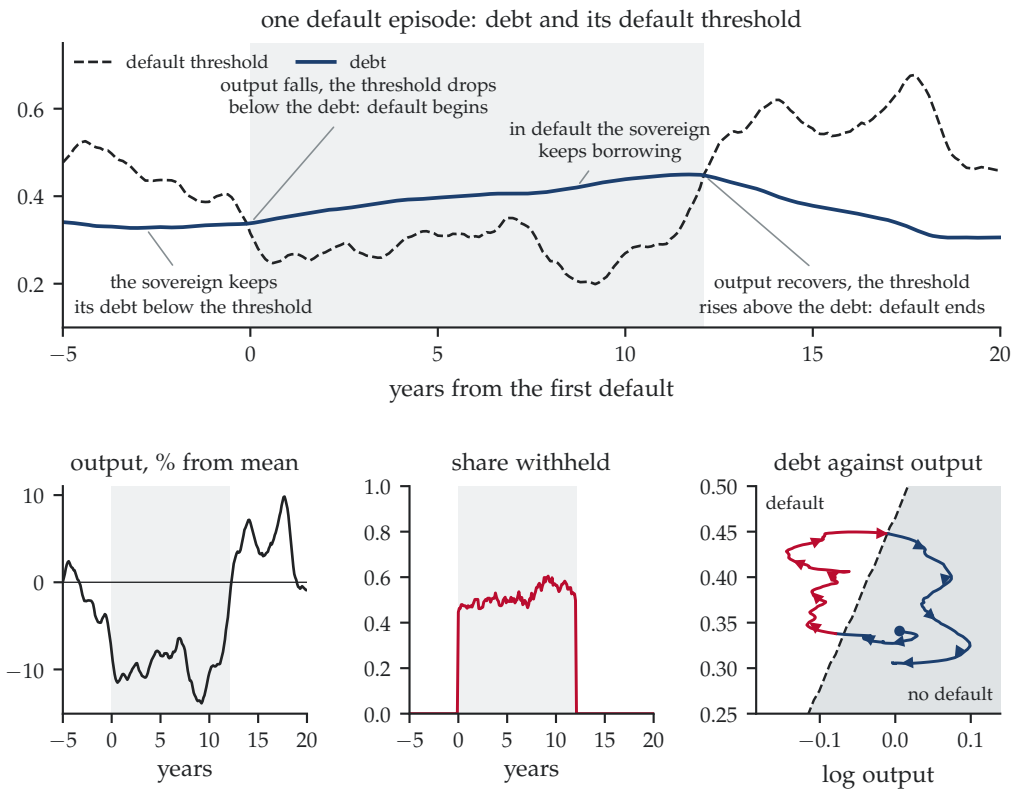


Figure 1: One default episode in the calibrated model, from a simulated path. Shaded: the years in default. The default threshold is the most debt the sovereign services in full at the current output. Output and the threshold are one-year moving averages. Last panel: the same path as debt against output, with the no-default region shaded.

pends neither on the value function nor on the direction in which debt is moving (Proposition 1). So the regions of no, partial, and full default are explicit in the state and the price (Proposition 2 and Figure 2). In the deterministic model the impatient sovereign borrows at the risk-free price up to the largest debt it is willing to service in full and stops there. This debt capacity is a closed-form function of the default costs, of the recovery on arrears, and of the maturity of the debt, and it does not move with the sovereign's impatience or the curvature of its utility (Theorem 1). With shocks the threshold is the same closed form at a price below par. Patience and risk aversion then matter through that price and through the buffer the sovereign keeps below the threshold. The value of market access still does not enter the decision to default.[2]

The second lesson is about how defaults begin. A sovereign never defaults a little. A fixed cost makes entry into default lumpy: the first default is a jump whose size the endowment and the cost parameters alone determine (Proposition 2). At the calibrated parameters it is half of the debt service. A shock triggers the jump. In a state that can switch to a worse one the

2. Amador and Phelan (2021, 2023) develop the reputational theory of sovereign default. The latter does so in a setting with partial default, the closest theoretical treatment of a partial default decision we know of. Tsoy (2026) adds reputation to the model of DeMarzo, He, and Tourre (2023). There, a government builds a reputation for debt discipline by reducing its debt after large negative shocks, not by repaying on time.

sovereign holds its debt below its threshold, with a buffer set by an Euler equation that balances the impatience gain from borrowing against the loss from arriving in a worse state with more debt (Proposition 5). In the state with the lowest consumption it borrows to its edge. Default occurs when a shock moves the threshold below the sovereign's current debt, further than its buffer guards against. How often a sovereign defaults depends on how the tail of its shocks compares with the buffer it holds, not on a willingness to borrow into default.

The third lesson speaks to the nature of default episodes. In the binary model a sovereign past its limit defaults, and the adjustment is a jump into exclusion or renegotiation. Under partial default the adjustment is gradual, and in the deterministic model we characterize it exactly. Above its capacity the sovereign is in partial default and repays, at a speed that vanishes like the cube root of the excess debt, while the price leaves par with a vertical tangent (Theorem 2). A sizable excess takes a few years to work off, but even a small excess is visibly priced at once. From below, the sovereign approaches its capacity along a square-root law and reaches it in finite time (Lemma 1).[3] With shocks the episode acquires a feature the deterministic model cannot show. Once a bad shock has moved the threshold below the current debt, the fixed cost is being paid regardless and no longer restrains borrowing. The sovereign's stationary debt is then set by the dilution of its own price, the force that Hatchondo, Martinez, and Sosa-Padilla (2016) show drives default risk with long-term debt, and it is higher than in good times. So the sovereign in partial default borrows. In the calibrated diffusion model, after a fall in output of about one standard deviation the default intensity jumps and stays elevated for two decades, and debt keeps rising for ten years. New issues are the larger addition to the debt in the first five years, and arrears thereafter. The episode ends when output recovers and the threshold rises back above the debt, not when the debt is repaid: default is lumpy at entry and sticky thereafter.

The fourth lesson is that pessimism alone can produce a crisis, and that it unfolds slowly. If lenders expect the sovereign to settle at a debt above its capacity and to withhold a constant share of its debt service there forever, they price the bond accordingly. At the lower price withholding is cheaper, so the sovereign starts defaulting at debt levels at which it would otherwise have repaid, and it can borrow past its old capacity. It drifts into a permanent partial default over years, through new issues and arrears, and the arrears validate the price. This is the slow-moving crisis of Lorenzoni and Werning (2019), with arrears in the place of rollover, and there is no run and no loss of market access.[4] A sovereign in a rollover crisis must roll over, whereas ours can stay out of the default region. So a pessimistic expectation is self-fulfilling only if borrowing into default at a discount beats staying out, all the way from zero debt. For a risk-neutral sovereign

3. Bornstein (2020) identifies a painful deleveraging effect specific to settings in which the stock of debt moves slowly. Hurtado, Nuño, and Thomas (2023) study the closest mechanism in continuous time, a sovereign that lowers the real value of its local-currency debt through inflation. Inflation acts on the whole stock, whereas partial default acts on the flow of debt service and raises the stock through arrears.

4. The idea that an expected repudiation raises the interest rate, and that the higher rate justifies the repudiation, goes back to Calvo (1988). The crisis is closer to the self-fulfilling dilution of Aguiar and Amador (2020) and to Aguiar et al. (2022), in which the government keeps issuing at depressed prices, than to the rollover crises of Cole and Kehoe (2000).

we give the condition: the gain from front-loading the borrowed funds must cover the output cost of the cheapest default. When the default cost is not too convex, the impatience wedge must exceed the spread implied by the first default (Proposition 4). Risk aversion and uncertainty both work against the crisis. In a numerical example a relative risk aversion of two roughly triples the impatience that a pessimistic expectation needs, and with shocks the range of impatience that supports it narrows as volatility rises and closes at moderate volatility (Appendix E.4, Figure E.2). At the calibrated parameters the impatience wedge is far below the threshold.

The quantitative section calibrates the diffusion model to the moments that Arellano, Mateos-Planas, and Ríos-Rull (2023) target. We take the recovery and the maturity of the debt from them and calibrate the other seven of their nine parameters. The model reproduces the frequency of partial default, the level and the dispersion of debt, the debt service and debt due ratios, and the volatility of output. The thresholds and the debt capacity of the theory are visible in the policies, and the ergodic distribution concentrates on the locus where debt is stationary, whose shape reflects the first three lessons at once. Spreads are low most of the time and high in the tail, and they track output and debt more tightly than in the data, since the model has no shocks on the lenders' side.

Two further contributions are technical. The first is computational. The default intensity is a continuous control, so the computed decision is the model's own and needs no smoothing, unlike in discrete-time models with long-term debt, which rely on i.i.d. shocks to make the price schedule converge (Chatterjee and Eyigungor 2012; Dvorkin et al. 2021; Mihalache 2020). The long-run distribution and the impulse responses come exactly from the generator of the discretized process.[5] The second concerns the model at high debt. Where default carries no fixed cost, a sovereign in full default faces a marginal cost of debt close to zero while lenders still pay a positive price, so it borrows up to whatever borrowing limit the model imposes. Quantitative models leave the limit implicit, as the top of the debt grid, so a choice about the computation becomes part of the model. We make the limit explicit and adopt an underwriting standard, a lower bound on the price of new issues as in Chatterjee and Eyigungor (2015), under which the long-run distribution of debt does not depend on the limit (Section 3.2).[6] The bonds of a sovereign in full default are claims on its recovery, and a standard above the price of such a claim rules out the accumulation (Proposition 3). The standard stops new placements, not market access as a whole. While the price is below it the bonds still trade and the sovereign can buy debt back, and placements resume when the price recovers. It binds on about 2 percent of the long-run distribution, and at some point in about one default episode in ten.

5. Bornstein (2020) finds that computation is faster in continuous time in the binary model. Rendahl (2022) discusses the extent to which similar results can be obtained in discrete time by adapting the sparse matrix methods at the core of the continuous-time solution. Tourre (2017) studies the binary model in continuous time with recursive utility and a rich pricing kernel. Aguiar et al. (2019) study default and debt maturity in continuous time, and DeMarzo, He, and Tourre (2023) a sovereign that cannot commit to a debt path. These papers solve optimal stopping problems with HJB variational inequalities, because default is a discrete transition into exclusion. The partial default model has no such transition, which is the source of its tractability.

6. Hatchondo, Martinez, and Roch (2022) study a related rule, a spread brake that caps the deficit when the sovereign spread exceeds a threshold.

2 The Partial Default Model

The environment consists of a small open economy governed by a risk-averse sovereign and of a unit measure of competitive, risk-neutral international lenders. Time is continuous. We describe the two problems in turn and define the equilibrium.

2.1 The Sovereign's Problem

The sovereign receives an endowment stream, borrows abroad with long-term bonds, and chooses at every instant what fraction of its scheduled debt service to pay. The part of the debt service it does not pay accrues as arrears and adds to the stock of debt.

Preferences and endowment. The sovereign maximizes

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t}\, u(c_t)\, dt,$$

where ρ is its discount rate and u is increasing and strictly concave. The log endowment z_t follows a Markov process with generator $\mathcal{L}_z$, which we leave general in this section: in Section 3 it is constant or a finite-state chain, in Section 4 an Ornstein–Uhlenbeck diffusion. We call the model with a general Markov process the *general model*, and its three special cases the *deterministic model* (constant endowment), the *chain model* (a finite-state Markov chain), and the *diffusion model* (an Ornstein–Uhlenbeck diffusion). Output is e^{z_t} when the sovereign is current on its debt. When it defaults, output is $\phi(d_t, z_t)\, e^{z_t}$, where the cost function ϕ is described below.

Debt. The stock of debt is B_t. Each unit of debt pays a coupon λ and matures at rate δ, so that the scheduled debt service is $(\delta + \lambda)B_t$ per unit of time.[7] The sovereign chooses a *default intensity* $d_t \in [0, 1]$, the share of the scheduled debt service it withholds. Of every unit withheld, a fraction κ is added to the stock of debt as arrears. The fraction $1 - \kappa$ is the haircut suffered by lenders on defaulted payments. The sovereign also issues ℓ_t units of new debt at the market price q_t (when $\ell_t < 0$ it buys debt back). The budget constraint and the law of motion of debt are

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1 - d_t)(\delta + \lambda)B_t + q_t\, \ell_t, \tag{1}$$

$$\dot{B}_t = \ell_t - \delta B_t + \kappa(\delta + \lambda)\, d_t\, B_t. \tag{2}$$

The three terms in (1) are income net of the default cost, the part of the debt service that is paid, and the proceeds from issuance. The three terms in (2) are issuance, amortization, and the

7. This is the continuous-time counterpart of the long-term bond of Hatchondo and Martinez (2009) and Chatterjee and Eyigungor (2012). The maturing fraction δ of the stock is repaid at par, and a single state variable summarizes the debt.

accrual of arrears. It is convenient to write

$$a(d) := \delta - \kappa(\delta + \lambda)\, d, \qquad a := a(1) = \delta - \kappa(\delta + \lambda),$$

so that $\dot{B}_t = \ell_t - a(d_t)B_t$: absent issuance, debt decays at rate $a(d)$, which is lowest in full default. We assume $a > 0$, so that even a sovereign in full default sees its debt shrink, slowly, when it does not issue. Debt is bounded, $B_t \in [0, \bar{B}]$: the sovereign does not lend, and, as in every incomplete-markets model, a borrowing limit $\bar{B}$ rules out Ponzi schemes. The restriction to borrowing follows Arellano, Mateos-Planas, and Ríos-Rull (2023), and Appendix F.2 shows that relaxing it leaves the behavior of the model qualitatively unchanged. We make the limit explicit because, without the underwriting standard of Section 2.4, its level matters.

The cost of default. Following Arellano, Mateos-Planas, and Ríos-Rull (2023) the cost of default has a variable and a fixed component,

$$\phi(d, z) = (1 - \gamma_0 d^{\gamma_1})\, F(z) \;\text{ for } d > 0, \qquad \phi(0, z) = 1, \qquad F(z) = 1 - \gamma_2\,(e^z - y^*)\,\mathbb{1}\{e^z \geq y^*\}, \tag{3}$$

with $\gamma_0 \in (0, 1)$, $\gamma_1 > 1$, and $\gamma_2 \geq 0$. The variable cost $\gamma_0 d^{\gamma_1}$ is convex in the default intensity with zero marginal cost at $d = 0$. The fixed cost $(1 - F(z))e^z$ is paid whenever the sovereign defaults at all, however little, and only when output e^z exceeds the threshold y^*. The fixed cost is larger in better times. Below the threshold default carries only the variable cost, and we call these states the *no-fixed-cost region*. The jump of ϕ at $d = 0$ above the threshold creates a region of no default: a small default is not worth its fixed cost.

Substituting issuance ℓ_t out of the budget constraint (1) using the law of motion (2), we write the drift of debt as a function of the state, the price, and the two controls,

$$S(B, z, c, d, q) = \frac{c - \phi(d, z)e^z + (1 - d)(\delta + \lambda)B}{q} - a(d)\, B,$$

and we treat consumption c and the default intensity d as the controls. Equivalently, for a given drift S,

$$c = \mathcal{R}(d; B, z, q) + q\, S, \qquad \mathcal{R}(d; B, z, q) := \phi(d, z)e^z - (1 - d)(\delta + \lambda)B + q\, a(d)\, B. \tag{4}$$

We call $\mathcal{R}$ the *resource function*: it is consumption at zero drift, and it is all that the default decision affects for a given drift. The characterization of Section 3 rests on this decomposition.

Arrears as borrowing. The decomposition has a second reading, which makes exact the description of partial default by Arellano, Mateos-Planas, and Ríos-Rull (2023) as a flexible way to raise funds. Let $n_t := \ell_t + \kappa(\delta + \lambda)d_t B_t$ be the total addition to the stock of debt, from issuance

and from arrears, so that $\dot{B}_t = n_t - \delta B_t$. Substituting into (1),

$$c_t = \phi(d_t, z_t)e^{z_t} - (\delta + \lambda)B_t + q_t\, n_t + (1 - \kappa q_t)(\delta + \lambda)B_t\, d_t. \tag{5}$$

Holding the path of debt fixed, withholding a unit of debt service raises consumption by $1 - \kappa q_t > 0$: the unit saved, net of the market value of the κ units of arrears that replace market issuance one for one. Equivalently, withholding raises funds at the implicit price $1/\kappa$ per unit of arrears, which exceeds the market price and does not depend on it, paid for with the output cost $(1 - \phi)e^z$ of default. Section 3.1 draws the consequences.

The HJB equation. Given a bond price function $q(B, z)$, the sovereign's value $V(B, z)$ solves

$$\rho V(B, z) = \max_{c,\ d \in [0,1]} \{u(c) + S(B, z, c, d, q(B, z))\, V_B(B, z) + \mathcal{L}_z V(B, z)\}, \tag{6}$$

subject to the state constraints $0 \leq B \leq \bar{B}$ and the underwriting standard of Section 2.4. Subscripts denote partial derivatives. Unlike in models with a binary default decision, there is no variational inequality and no second regime: the sovereign never loses market access, and a single value function with possibly discontinuous policies describes it at all states.

At an interior drift, the first-order condition for consumption is

$$u'(c)\, q(B, z) = -V_B(B, z), \tag{7}$$

the price of a bond times the marginal utility of the resources it brings equals the marginal cost of debt in utility units. By (4) the default intensity enters the objective only through $\mathcal{R}$, which is strictly concave on $(0, 1]$ with

$$\mathcal{R}_d(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1 - \kappa q)(\delta + \lambda)B.$$

The marginal gain from raising the default intensity is $(1 - \kappa q)(\delta + \lambda)B$: the payment saved, net of the market value κq of the arrears it creates. The interior maximizer is

$$d_{\text{int}}(B, z, q) = \min\left\{1,\ \left[\frac{(1 - \kappa q)(\delta + \lambda)B}{\gamma_0\gamma_1 F(z)e^z}\right]^{1/(\gamma_1 - 1)}\right\}, \tag{8}$$

and the optimal default intensity is either 0 or d_{int}, whichever gives the larger resource function. Section 3.1 makes this precise and shows that it holds for every maximizer of (6) wherever the underwriting standard is slack.

2.2 International Lenders

A unit measure of deep-pocketed, competitive, risk-neutral lenders with outside option r buy and hold the sovereign's bonds. A unit of debt held at t yields the flow $(1 - d_t)(\delta + \lambda)$ in

cash and $\kappa(\delta+\lambda)d_t$ in new units (the arrears), and δ of it matures. The break-even price is the expected discounted value of the cash flow, where the discount rate is adjusted for the growth of the position through arrears,

$$q(B,z) = \mathbb{E}_{B,z}\int_0^\infty \exp\left(-\int_0^t \xi(d_s)\,ds\right)(1-d_t)(\lambda+\delta)\,dt, \qquad \xi(d) := r+\delta-\kappa(\delta+\lambda)\,d. \tag{9}$$

The effective discount rate $\xi(d)$ combines the lenders' opportunity cost r, the amortization δ, and the accrual of arrears, which lowers it. By the Feynman–Kac formula, q solves the linear equation

$$\xi(d(B,z))\,q(B,z) = (1-d(B,z))(\lambda+\delta) + \tilde{S}(B,z)\,q_B(B,z) + \mathcal{L}_z q(B,z), \tag{10}$$

where $\tilde{S}(B,z)$ is the equilibrium drift of debt and $d(B,z)$ the equilibrium default policy. The evolution of the stock of debt is governed by $\tilde{S}$, which includes issuance. The cash flow of a single unit is governed by $\xi(d)$, which does not.

The price of a bond that never defaults, the *risk-free price*, is $q^{\mathrm{rf}} = (\delta+\lambda)/(r+\delta)$. We set the coupon equal to the risk-free rate, $\lambda = r$, so that $q^{\mathrm{rf}} = 1$ and the face value of a bond equals its risk-free value. We keep the symbol q^{rf} wherever a price is the risk-free one. The yield to maturity implied by a price q is $(\delta+\lambda)/q - \delta$, so the spread over the risk-free yield is

$$\mathrm{spread}(B,z) = \frac{\delta+\lambda}{q(B,z)} - (\delta+r),$$

and the risk-free Macaulay duration of the bond is $1/(\delta+r)$.

2.3 Equilibrium and the Ergodic Set

Definition 1. A *Markov equilibrium* consists of a value function $V(B,z)$, policies $c(B,z)$ and $d(B,z)$, and a bond price $q(B,z)$ such that

1. given q, the value V solves the HJB equation (6) and (c,d) attain the maximum on its right-hand side, subject to the state constraints $0 \le B \le \bar{B}$ and the underwriting standard of Section 2.4;
2. given the policies, q solves the pricing equation (10), equivalently (9).

Given an equilibrium, let $g(B,z)$ be the density of the state. Given the equilibrium drift $\tilde{S}$ and the generator $\mathcal{L}_z$ with adjoint $\mathcal{L}_z^*$, the density solves the Kolmogorov forward equation

$$\partial_t g = -\partial_B\left[\tilde{S}(B,z)\,g\right] + \mathcal{L}_z^* g, \tag{11}$$

and the stationary distribution solves $0 = -\partial_B[\tilde{S}g] + \mathcal{L}_z^* g$. The forward equation is the adjoint of the HJB equation, which is how we compute the long-run distribution (Appendix E). The stationary distribution can have atoms: at $B = 0$ when the sovereign repays in full at a high

endowment, and at stationary points of debt in the deterministic and chain models. We call its support the *ergodic set*. Its shape, in particular how far it extends into the default region, is a central object of the paper.

2.4 The Underwriting Standard

Without a further restriction the borrowing limit $B \leq \bar{B}$ is not innocuous: in the states where default carries no fixed cost, a sovereign in full default borrows up to the limit, whatever its level (Section 3.2). We therefore impose an *underwriting standard*: the banks that underwrite the sovereign's issues in the financial centers do not bring a new issue to market at a price below a floor $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}. \tag{12}$$

The restriction is on placements, not on holdings: the outstanding bonds continue to trade at the price q that competitive lenders set, buybacks remain allowed, and partial default and the accrual of arrears are unchanged. A lower bound of this kind on the price of new issues appears in Chatterjee and Eyigungor (2015). With $\lambda = r$ the floor corresponds to a spread of $(\delta + \lambda)(1/\underline{q} - 1)$. The restriction is on the price and not on the quantity of debt because the accumulation begins at the full-default line, at debt levels that differ across states. A floor on the price is a state-dependent debt limit expressed in the one variable both parties observe, the underwriters' counterpart of the spread brakes that Hatchondo, Martinez, and Roch (2022) study as fiscal rules. We set its level by a rule: the lowest level at which the sovereign does not issue in full default on the ergodic set (Section 4.2).

No trade at a zero price. A second, milder convention rules out the autarky equilibrium in which lenders expect full default forever, the price is zero, and the sovereign keeps itself above the full-default line by issuing at a zero price. We impose that no bonds change hands at a zero price. Section 3.1 shows that the price is then strictly positive at every state.

3 Characterization

This section collects what can be established analytically. Section 3.1 contains the results for any Markov endowment process, Section 3.2 turns to high debt and the underwriting standard, Section 3.3 to the deterministic model, and Section 3.4 to the chain model. Proofs are in Appendix A. The numerical illustrations borrow the parameters of the calibration of Section 4 (Table 1), except for Figure 4, which has parameter values of its own. Throughout we maintain the following.

Assumption 1.

(i) $a = \delta - \kappa(\delta + \lambda) > 0$, so that also $\xi(1) = r + a > 0$.

(ii) $\lambda = r$, so that $q^{\text{rf}} = 1$; $\rho > r$.

(iii) u is C^2, strictly increasing, and strictly concave; $\gamma_1 > 1$, $0 < \gamma_0 < 1$, and $F(z) > 0$.

(iv) The drift is bounded above, $S \leq \bar{S}$, consumption is bounded below, $c \geq c_{\min} > 0$, which bounds buybacks, and $B \in [0, \bar{B}]$.

3.1 Results for any Markov endowment process

For any Markov endowment process, the default decision is static and its thresholds are explicit. This makes debt capacity a closed form.

By (5), a unit of debt service withheld raises one unit of funds against κ units of arrears, at the implicit price $1/\kappa$, independent of the market price, in exchange for the output cost of default. The sovereign has two ways of raising funds, the market at the price q and arrears at the price $1/\kappa$. The second is always the cheaper per unit of debt but always costs output. For a given drift of debt the default intensity affects consumption only through the resource function $\mathcal{R}(d; B, z, q) = \phi(d, z)e^z - (1-d)(\delta+\lambda)B + q\,a(d)B$, with $\mathcal{R}_d = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1-\kappa q)(\delta+\lambda)B$ for $d > 0$. So the sovereign withholds until the marginal output cost of default per unit of debt service withheld, $\gamma_0\gamma_1 F(z)e^z d^{\gamma_1-1}/((\delta+\lambda)B)$, equals $1-\kappa q$, the unit of cash it saves net of the market value of the arrears it creates. A lower price makes the arrears cheaper, so the sovereign withholds more. A larger debt raises the cash saved per unit of intensity, so intensity rises with debt. The fixed cost decides only whether the sovereign is on this margin at all.

Proposition 1 (the default decision is static)**.** *For every state (B, z), price $q \geq 0$, and value function V, every maximizer (d^*, c^*) of the right-hand side of (6), subject to constraints on the drift that do not depend on d (in particular wherever the underwriting standard is slack), satisfies*

$$d^* \in \arg\max_{d \in [0,1]} \ \mathcal{R}(d; B, z, q).$$

The default policy depends on $(B, z, q(B, z))$ only, and not on V, on the sign of the drift, or on the constraints on the drift, as long as those constraints do not depend on d.

The qualification matters for the underwriting standard, whose constraint $\ell = S + a(d)B \leq 0$ does depend on d. Section 3.2 gives the rule in that case (Proposition A.4). Everywhere else the default decision is a static maximization of the function $\mathcal{R}$, concave on $(0, 1]$, which yields explicit thresholds.

Proposition 2 (default thresholds)**.** *In the general model, fix (B, z, q) with $B > 0$ and $\kappa q < 1$. The optimal default intensity is $d^* = 0$ if the fixed cost exceeds the gain from defaulting,*

$$(1 - F(z))\, e^z \ \geq \ \Gamma(B, z, q) := (1-\kappa q)(\delta+\lambda)B\, d_{\text{int}} - \gamma_0 F(z) e^z d_{\text{int}}^{\gamma_1}, \tag{13}$$

and $d^* = d_{\text{int}}(B, z, q)$ of (8) otherwise. Γ is continuous, strictly increasing in B, strictly decreasing in q, and zero at $B = 0$. Consequently:

(a) *there is a threshold* $B_{\text{pd}}(z,q)$*, nondecreasing in* q*, with* $d^* = 0$ *for* $B \leq B_{\text{pd}}$ *and* $d^* = d_{\text{int}} > 0$ *for* $B > B_{\text{pd}}$*;*

(b) *in the no-fixed-cost region* ($F(z) = 1$)*,* $B_{\text{pd}}(z,q) = 0$*: any positive debt is partially defaulted on;*

(c) *if* $F(z) < 1$ *and* $d_{\text{first}}(z) < 1$ *(below), the default intensity jumps at* B_{pd} *from* 0 *to the* first default intensity

$$d_{\text{first}}(z) = \left[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\right]^{1/\gamma_1}, \tag{14}$$

which depends only on the endowment and the cost parameters, not on debt, the price, or preferences;

(d) *full default,* $d^* = 1$*, occurs for* $B \geq B_{\text{fd}}(z,q)$ *and* $B > B_{\text{pd}}(z,q)$*, where*

$$B_{\text{fd}}(z,q) := \frac{\gamma_0\gamma_1 F(z)e^z}{(1-\kappa q)(\delta+\lambda)};$$

when $d_{\text{first}}(z) \geq 1$ *the intensity jumps from* 0 *directly to* 1 *at* B_{pd}*;*

(e) *the threshold has a closed form: when* $d_{\text{first}} < 1$*,*

$$B_{\text{pd}}(z,q)^{\frac{\gamma_1}{\gamma_1-1}} = \frac{\gamma_1}{\gamma_1 - 1}\,\frac{(1-F(z))\,e^z\,G^{\frac{1}{\gamma_1-1}}}{K^{\frac{\gamma_1}{\gamma_1-1}}}, \qquad G := \gamma_0\gamma_1 F(z)e^z, \quad K := (1-\kappa q)(\delta+\lambda), \tag{15}$$

and when $d_{\text{first}} \geq 1$*,* $B_{\text{pd}}(z,q) = (1-(1-\gamma_0)F(z))e^z/K$*.*

Two monotonicity properties follow, and both reflect the default margin at work. First, $d^*(B,z,q)$ is nondecreasing in B and nonincreasing in q, in γ_0, in κ, and in $F(z)e^z$ holding the fixed cost $(1-F(z))e^z$ fixed. Second, the thresholds B_{pd} and B_{fd} are increasing in γ_0, in κ, and in q, decreasing in $\delta+\lambda$, and B_{pd} is increasing in the fixed cost $1-F(z)$ (using $\gamma_0 < 1$). The thresholds make the regions of default explicit in the primitives and the price, which the discrete-time model obtains only numerically.

The static decision is a property of the environment of Arellano, Mateos-Planas, and Ríos-Rull (2023). Default brings no exclusion and no penalty on future values, and arrears are ordinary bonds, perfect substitutes for new issues. A punishment that worked through the continuation value would put V back into the default decision, as in the binary model. Continuous time makes the property explicit. In their discrete-time model the output cost of a default falls on the next period's income, and the price of new debt depends on the default. The sovereign there weighs cash today against value tomorrow, jointly with its borrowing. Here the cost is paid as the payment is withheld, so the trade-off lies within the instant, and the price that matters is the one at the current state.

The thresholds are stated at a given price. In equilibrium the no-default region is $\{B \leq B_{\text{pd}}(z, q(B,z))\}$, a fixed-point condition, which is explicit at the risk-free price. Two implications follow. First, the size of the first default depends only on the endowment and the cost parameters.

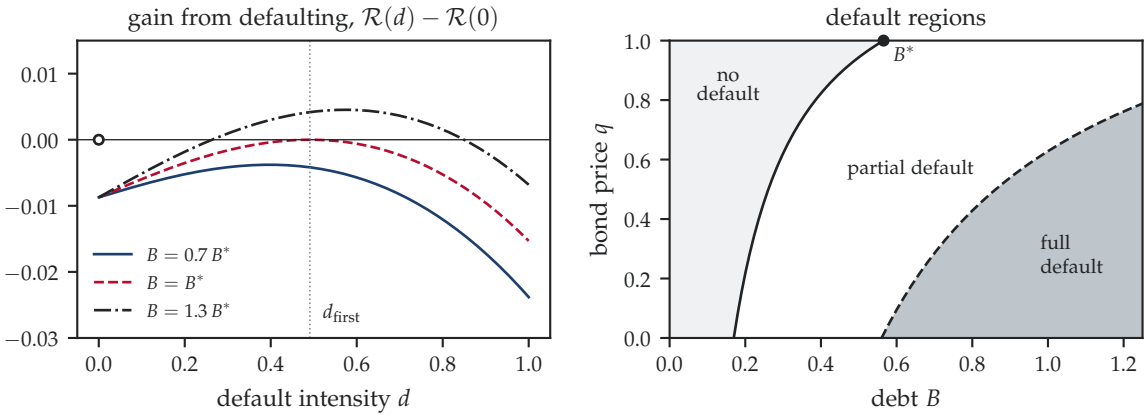


Figure 2: The default decision at the mean endowment, from the closed forms of Proposition 2 at the calibrated parameters. Left: the gain from defaulting at intensity d, $\mathcal{R}(d) - \mathcal{R}(0)$, at the risk-free price and three debt levels. At $d = 0$ the gain is zero (open circle). Any positive intensity pays the fixed cost, so the gain jumps down. At $B = B^*$ the best default, d_{first}, ties with full repayment. Right: the thresholds $B_{\text{pd}}(0, q)$ and $B_{\text{fd}}(0, q)$ divide the (B, q) plane into the regions of no, partial, and full default. The no-default region shrinks as the price falls, and its edge at the risk-free price is the debt capacity $B^* = B_{\text{pd}}(0, q^{\text{rf}})$.

At the calibrated parameters it is half of the debt service, $d_{\text{first}} \approx 0.49$ at the average endowment. The jump is large because the variable cost is steep, not because the fixed cost is large. The fixed cost is what makes full repayment possible: with $F(z) = 1$ the variable cost has zero marginal cost at $d = 0$, so the sovereign defaults a little at every positive debt level. Second, at the risk-free price the no-default region at the average endowment ends at $B_{\text{pd}}(0, q^{\text{rf}}) = 0.565$, which is where the steady state of the deterministic model sits (Theorem 1). Figure 2 draws both implications at the calibrated parameters.

Away from the risk-free price the threshold is the same closed form, scaled by a factor in which only the recovery and the price appear. For every z and every price with $\kappa q < 1$,

$$B_{\text{pd}}(z, q) = \frac{1 - \kappa q^{\text{rf}}}{1 - \kappa q}\, B_{\text{pd}}(z, q^{\text{rf}}), \tag{16}$$

so $B_{\text{pd}}(z, q)$ lies between $(1 - \kappa q^{\text{rf}})\, B_{\text{pd}}(z, q^{\text{rf}})$ and $B_{\text{pd}}(z, q^{\text{rf}})$ for every $q \in [0, q^{\text{rf}}]$. With shocks the price at the threshold is below q^{rf}, because default is expected after future shocks, and it depends on the sovereign's patience and risk aversion through its future borrowing. Preferences and the endowment process move the threshold only through the price, and never outside a band that the default costs, the recovery, and the maturity fix. The sovereign's continuation value does not enter. This is the first lesson: the debt limit is a static comparison, not the outcome of the dynamic problem.

Three further properties of the price hold in the general model (Appendix A.1). The bond price lies between 0 and q^{rf}. It equals q^{rf} only along paths without default, and 0 only along paths in full default forever (Proposition A.1). It can jump in debt only at points where debt stops

moving that are not reached in finite time from both sides (Proposition A.2). The convention against trade at a zero price makes it strictly positive at every state (Proposition A.3).

3.2 High Debt and the Underwriting Standard

In the no-fixed-cost region a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt close to zero, while lenders still pay a positive price because output will recover. By (7) it issues at any positive price, and with a borrowing limit it borrows up to the limit, whatever its level. It sells a claim on its recovery. If from the state (B, z) the sovereign is in full default until a stopping time τ, then, with $a := a(1)$ and $\xi(1) = r + a$,

$$q(B,z) \;=\; \mathbb{E}_{B,z}\left[e^{-(r+a)\tau}\, q(B_\tau, z_\tau)\right] \;\le\; q^{\mathrm{rf}}\, \mathbb{E}_{B,z}\left[e^{-(r+a)\tau}\right]$$

(Proposition A.7). In the chain model of Section 3.4 the endowment leaves state j at the rate $\eta_j := \sum_{k \neq j} \eta_{jk}$, and the bound is

$$q_j^* := q^{\mathrm{rf}}\, \frac{\eta_j}{r + a + \eta_j},$$

the price of a bond that pays nothing while the endowment is in state j, shrinks at the rate a while it waits, and is worth its risk-free value afterward.

Proposition 3 (the recovery-claim threshold)**.** *Consider the chain model with the underwriting standard, a state j, and an equilibrium in which V_j is nonincreasing in debt.*

(a) If $\underline{q} > q_j^*$, there is no debt level from which the sovereign stays in full default until the endowment leaves state j. In particular, debt is never stationary in full default in state j. If in addition q_j is nonincreasing in debt, the drift of debt is negative at every debt level at which the sovereign is in full default in state j, and debt falls until the sovereign leaves full default.

(b) Suppose that from (B, j) the debt is priced at q^{rf} at the date at which the endowment leaves state j, almost surely. Then $q_j(B) \ge q_j^*$, with equality if and only if the sovereign is in full default until that date. Hence if $\underline{q} < q_j^*$ the standard is slack at (B, j).

A sovereign that waits for the recovery in full default must issue to keep its debt from shrinking, and no lender pays more than q_j^* for a claim on the recovery. So a standard above q_j^* rules out the accumulation in every equilibrium. The threshold depends on the lenders' discount rate, on the debt contract through $a = \delta - \kappa(\delta + \lambda)$, and on the rate at which the crisis ends. It does not depend on the sovereign's preferences or on the default costs. In the deterministic model the endowment never changes, $q_j^* = 0$, and every positive standard rules out the accumulation (Section 3.3). So q_j^* is an upper bound on the lowest standard that does so. By part (b), a standard below q_j^* does not bind where the debt is priced at par after the crisis. With recurring crises the debt is priced below par after the switch, and the bound is not tight (Corollary C.1).

The underwriting standard removes the accumulation. With issuance switched off below $\underline{q}$, debt shrinks in full default, the full-default region is transient, and the long-run distribution of

debt is confined to the debt levels at which the price can still reach the floor (Proposition A.5). The thresholds of Proposition 2, as functions of the state and the price, are unaffected. Where the standard binds, its constraint $\ell \leq 0$ involves the default intensity through $\ell = S + a(d)B$, so the separation of Proposition 1 fails. The sovereign then applies the rule (8) at its own shadow price of debt, $\hat{q} = -V_B/u'(c)$, which is below the market price because it would like to issue and cannot (Proposition A.4). So it values its arrears less and defaults more.

3.3 The Deterministic Model

The deterministic model gives a sovereign's debt capacity, and the motion of its debt on either side of it, in closed form. For a risk-neutral sovereign it also answers whether lenders' pessimism alone can carry debt past that capacity. With z fixed, normalized to 0, the equilibrium is a pair of ordinary differential equations. Writing S^*, c^*, d^* for the policies, (6) and (10) read

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\xi(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*}, \tag{17}$$

with $d^* = d^*(B, q(B))$ from Proposition 2, $c^* = (u')^{-1}(-V'/q)$, and $S^* = (c^* - \mathcal{R}^*)/q$. The system is singular where the drift vanishes, and the stationary points of debt organize the solution. We call a zero of the drift a *steady state*. Theorem 1 pins down one of them within the class of no-default-basin equilibria, the no-default steady state B^*. A zero inside the partial-default region would be a self-fulfilling partial-default steady state, discussed at the end of this subsection.

Definition 2. An equilibrium of the deterministic model is of *no-default-basin type* with steady state $B^* \in (0, \bar{B})$ if $d \equiv 0$ on $[0, B^*]$, $S > 0$ on $[0, B^*)$, $S(B^*) = 0$, and $S < 0$ on $(B^*, \bar{B}]$. We say that debt reaches B^* from above in finite time if from $B^* + x$ the state reaches B^* in a time $\tau(x) \to 0$ as $x \downarrow 0$.

Theorem 1 (the deterministic steady state in closed form)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium in which debt reaches B^* from above in finite time. Then $q \equiv q^{\mathrm{rf}}$ on $[0, B^*]$, $V(B^*) = u(1 - rB^*)/\rho$, and*

$$B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}}),$$

the largest debt the sovereign is willing to service in full at the risk-free price, given explicitly by Proposition 2(e) with $q = q^{\mathrm{rf}}$.

The steady state is where the impatient sovereign, borrowing at the risk-free price, runs into its own default threshold. We call it the sovereign's *debt capacity*. It is a closed form in the primitives, because the default decision that defines it is a static comparison of the gain from the first default with the fixed cost. Capacity is increasing in the default costs γ_0 and $1 - F$, in income, and in the recovery κ, since a higher recovery makes withholding more expensive. It is decreasing in $\delta + \lambda$, and independent of ρ (as long as $\rho > r$), of the curvature of u, of the drift cap, and of the borrowing limit. Steady-state consumption is $1 - rB^*$ and the first partial default

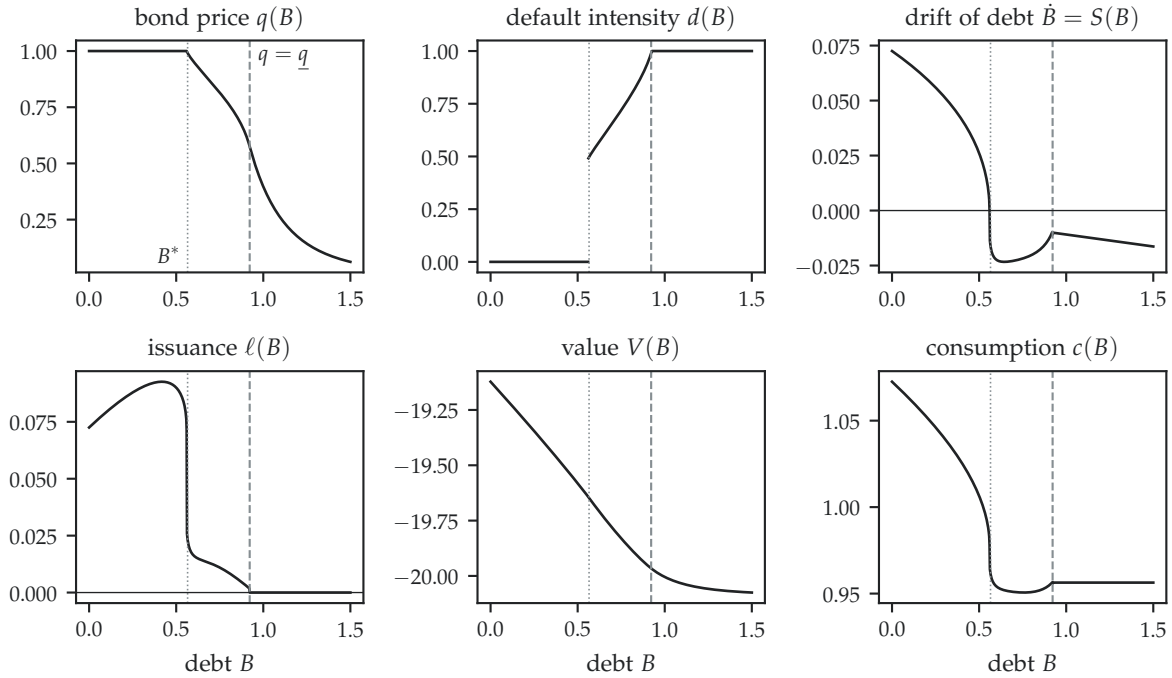


Figure 3: The deterministic model: bond price, default intensity, drift of debt, issuance, value, and consumption as functions of debt. Parameters of Table 1 with the endowment at its mean. Dotted vertical line: the steady state $B^* = B_{\text{pd}}(0, q^{\text{rf}})$. Dashed vertical line: the debt at which the price reaches the underwriting standard $\underline{q}$.

beyond B^* has intensity d_{first}. On the basin $[0, B^*)$ the sovereign faces the constant risk-free price and borrows up to the steady state, which it reaches in finite time. The approach is governed by a square-root law.

Lemma 1 (the square-root law)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium with V twice continuously differentiable on $(0, B^*)$ and S continuous at B^*, and write $\psi := u'/|u''|$ for the absolute risk tolerance and $\bar{c} = 1 - rB^*$. Then as $x \downarrow 0$*

$$S(B^* - x) = \left(2(\rho - r)\,\psi(\bar{c})\right)^{1/2} x^{1/2}\,(1 + o(1)),$$

and the state reaches B^* from the left in finite time.

Proof. On the basin $q \equiv q^{\text{rf}}$ and $d \equiv 0$, so $c = 1 - rB + S$ and the HJB equation reads $\rho V = u(c) + SV'$ with $V' = -u'(c)$. Differentiating in B and substituting the first-order condition gives $SS' = rS - (\rho - r)\,\psi(c)$. As $B \uparrow B^*$ the drift vanishes and $c \to \bar{c}$, so $(S^2)' \to -2(\rho - r)\psi(\bar{c})$, which integrates to the stated law. The time to reach B^* from $B^* - x$ is $\int_0^x dy/S(B^* - y) \propto x^{1/2}$, which is finite. □

This square-root law characterizes the impatient sovereign's approach to a limit it cannot cross at the risk-free price. Beyond B^* the sovereign is in partial default, the price is below q^{rf}, and the drift is negative, so the state returns to B^* from above.

Theorem 2 (the cube-root law at the steady state)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium in which debt reaches B^* from above in finite time, with $d_{\text{first}} < 1$, and suppose that on a right neighborhood of B^* the drift is interior and V is C^1. Write $x = B - B^*$, $\bar{c} = 1 - rB^*$, and*

$$\Pi_0 := (1-\kappa)(\delta+\lambda)\, d_{\text{first}}, \qquad \beta := \left(3\, a(d_{\text{first}})\, B^*\, \Pi_0\, \frac{u'(\bar{c})}{|u''(\bar{c})|}\right)^{1/3}, \qquad \alpha := \frac{3\Pi_0}{2\beta}.$$

Then as $x \downarrow 0$,

$$S(B^*+x) = -\beta\, x^{1/3}\,(1+o(1)), \qquad c(B^*+x) = \bar{c} - \beta\, x^{1/3}\,(1+o(1)),$$
$$q^{\text{rf}} - q(B^*+x) = \alpha\, x^{2/3}\,(1+o(1)),$$

and $V'(B^{*+}) = -u'(\bar{c}) = V'(B^{*-})$. So V is C^1 but not C^2 at the steady state, the drift leaves zero with a vertical tangent, and the price has a vertical tangent from the right.

The cube-root law describes gradual deleveraging: a sovereign over its capacity works its debt down at a speed that falls as it nears capacity, and it arrives in finite time, to leading order $(3/(2\beta))\, x^{2/3}$. In the illustration below this is about 0.6 years from an excess of one percent of output and 2.9 years from ten percent. Even a small excess, however, lowers the price visibly at once. This is the third lesson in its deterministic form: the adjustment above capacity is gradual, where a binary default decision would make it a jump. The constant Π_0 is the flow of debt service lost at the first default, net of the arrears it creates. The price has a vertical tangent to the right of the steady state because a small excursion into default costs lenders this flow at a rate that does not vanish, over a time that is only $O(x^{2/3})$.

The equilibrium with these properties exists whenever the solution above capacity stays regular up to full default, that is, whenever the drift above capacity has no second zero (Proposition B.2 in the Online Appendix). The condition is on an explicit initial-value problem, and the computed solution in the illustration satisfies it. A second zero would be a self-fulfilling partial-default steady state.

Numerical illustration. Figure 3 shows the deterministic equilibrium at the calibrated parameters, with the endowment fixed at its mean. The drift vanishes at the steady state $B^* = B_{\text{pd}}(0, q^{\text{rf}}) = 0.565$, and the default intensity jumps to $d_{\text{first}} = 0.49$ there, as the closed forms say. Consumption falls steeply at the first default, with the vertical tangent of the cube-root law, but it is continuous there, and the value is C^1. The underwriting standard binds from $B = 0.92$, where the price falls through 0.575 (dashed vertical line). Beyond that point issuance is zero, and from $B = 0.93$ default is complete.

Full default. The deterministic model has no recovery, so a sovereign that stays in full default has worthless bonds and cannot issue what would keep its debt from shrinking: debt does not rise in full default (Proposition C.1 in the Online Appendix). Where no standard binds it still

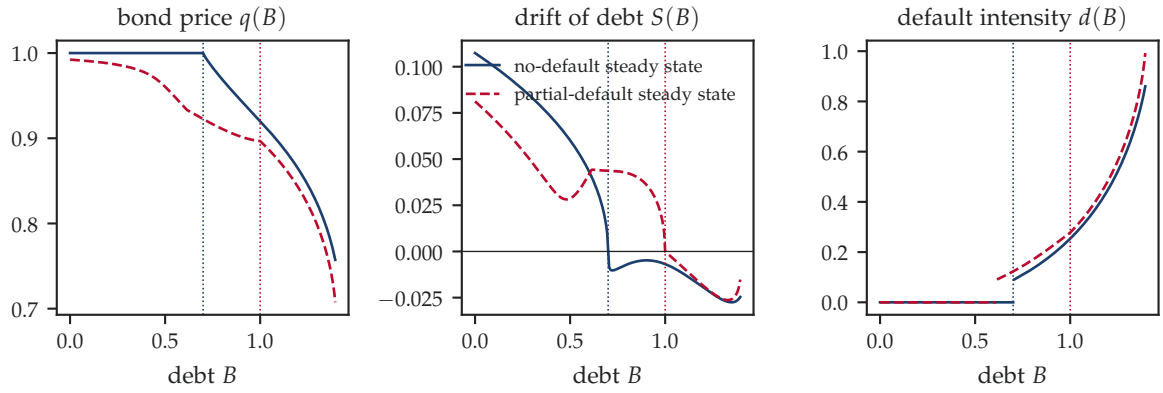


Figure 4: A numerical example of self-fulfilling partial default. Two equilibria of the deterministic model at the same parameters: the no-default steady state and a self-fulfilling partial-default steady state (dotted lines). Parameters: $\gamma_0 = 0.075$, $\gamma_1 = 1.5$, $F(0) = 0.999$, $\rho = 0.06$, $\delta = 0.12$, $\kappa = 0.7$, $r = \lambda = 0.039$, $\nu = 2$, and the standard at 0.7. The impatience wedge is 210 basis points.

issues. In the limit of high debt, and at every debt level when it is risk-neutral, it issues the share $(\rho - r)/(\rho + a)$ of the amount aB that would hold its debt constant (Proposition C.2).

Self-fulfilling partial default. In the equilibrium of Theorem 1, lenders expect full repayment and the price is q^{rf}. Suppose instead that lenders expect the sovereign to settle at a debt B°, typically above its capacity, and to withhold a constant share d° of its debt service there forever. They then pay at B° the price of a bond that defaults at rate d° forever,

$$q_r(d) := \frac{(1-d)(\lambda+\delta)}{\xi(d)},$$

and less than q^{rf} at every lower debt, since every path leads to B°. Two consequences validate the expectation. First, default starts earlier. At a lower price the arrears created by withholding a unit of debt service are worth less, so withholding costs the sovereign less. The threshold $B_{\mathrm{pd}}(0, q)$ falls with the price (Proposition 2), and the sovereign defaults at debt levels at which it would repay in full at the risk-free price. Second, the sovereign can borrow past its old limit. Because the default is priced, the sovereign pays only the risk-free rate on the funds it raises: at the steady state its net payments to lenders, debt service less issuance, are $rq^\circ B^\circ$ per unit of time. Since it discounts at $\rho > r$, it wants to borrow. It stops at B° only because there each further unit of debt lowers the price at which it must roll over the rest, and the impatience wedge balances this dilution exactly (Lemma A.2). At B° the sovereign withholds the share that lenders expected. There is no run and no loss of market access. With curvature the sovereign drifts into a permanent partial default over years, paying an output cost that exists only because lenders expected the default.

Lemma A.2 in Appendix A gives the local conditions at such a point: the price is $q_r(d^\circ)$, its slope to the right is pinned by the impatience wedge, $-q'(B^{\circ+})/q^\circ = (\rho - r)/(a(d^\circ)B^\circ)$, and the

point is stable from above (attracting) only if additional debt dilutes the price strongly enough through additional default. These conditions admit a continuum of candidates, one for every intensity $d^\circ \geq d_{\text{first}}$. Whether a candidate is an equilibrium depends on the whole path from zero debt (a two-point boundary-value problem): the sovereign must be willing to borrow into partial default, at prices below q^{rf}, all the way from zero debt. Two cases can be settled analytically, with the underwriting standard and with a risk-neutral sovereign. Both are governed by the stationary spread at the first default intensity, the spread lenders charge a sovereign that withholds forever the smallest share it ever withholds,

$$s_{\text{first}} := (\lambda + \delta)\left(\frac{1}{q_r(d_{\text{first}})} - 1\right) = \frac{(1-\kappa)(\lambda+\delta)\, d_{\text{first}}}{1 - d_{\text{first}}}.$$

Under the underwriting standard a partial-default steady state must have a price at or above the standard, $q_r(d^\circ) \geq \underline{q}$. Standing still requires issuance: the sovereign must replace the debt that matures and is not replaced by arrears, $\ell = a(d^\circ)B^\circ > 0$, and the standard forbids issuance below that price. Since q_r is decreasing and the static rule never chooses an intensity below d_{first}, a self-fulfilling partial-default steady state can exist only if $q_r(d_{\text{first}}) \geq \underline{q}$, that is, only if the stationary spread at the first default does not exceed the spread at the standard, $s_{\text{first}} \leq (\lambda+\delta)(1/\underline{q} - 1)$. The condition involves neither the discount rate nor the utility function.

A risk-neutral sovereign. With linear utility the sovereign does not smooth consumption, and the existence question has a nearly complete answer. We let issuance be unrestricted in speed, so that the sovereign can make a discrete issuance, moving its debt at once to a higher level at the price of the state it moves to. DeMarzo, He, and Tourre (2023) study a closely related problem in continuous time: an impatient, risk-neutral sovereign that cannot commit to a debt path and borrows from competitive lenders. Define for $d \in [d_{\text{first}}, 1)$ the stationary spread $s(d) := (1-\kappa)(\lambda+\delta)d/(1-d)$ and

$$\Theta(d) := s(d)\,\frac{1 + (\gamma_1 - 1)(d_{\text{first}}/d)^{\gamma_1}}{\gamma_1},$$

the output cost of withholding the share d forever, per unit of the funds the sovereign raises by issuing at once the debt of the steady state with that intensity (Proposition A.6).

Proposition 4 (self-fulfilling partial default with a risk-neutral sovereign)**.** *Let $u(c) = c$ and $\rho > r$ in the deterministic model, with issuance unrestricted in speed.*

(a) In any equilibrium in which debt goes from zero to a partial-default steady state, $\rho - r \geq \Theta^* := \min_{d \geq d_{\text{first}}} \Theta(d)$.

(b) $\Theta(d_{\text{first}}) = s_{\text{first}}$. Θ is increasing, and $\Theta^* = s_{\text{first}}$, if and only if $(\gamma_1 - 1)^2 - 1 \leq \gamma_1 d_{\text{first}}/(1 - d_{\text{first}})$, which holds whenever $\gamma_1 \leq 2$.

The condition $\rho - r \geq \Theta(d)$ is a comparison that a sovereign at zero debt can make directly. Issuing at once the debt of the steady state with intensity d raises its market value. From then on the sovereign pays lenders the risk-free rate on the funds raised, because the default is priced, and it loses $1 - \phi(d, 0)$ of output to the default. Front-loading gains the impatience wedge times the funds raised, per unit of time, and a discrete issuance up to the pessimistic steady state pays when this gain covers the output cost. By part (a), no route from zero debt reaches any partial-default steady state when the wedge is below Θ^*. Part (b) identifies the threshold with the spread implied by the first default: there the output cost equals the payments saved net of the arrears created, and at the stationary price this equals the spread times the value of the debt. With a more convex variable cost, deeper defaults cost less relative to their spread, and the threshold can lie below s_{first}. The condition is also sufficient on a range of wedges above the threshold, up to a bound $\bar{\Omega} := \Omega(d_{\text{first}})$ beyond which the steady state is no longer stable from above, provided the price schedule continues regularly above the steady state (Proposition A.6). When one self-fulfilling steady state exists a continuum does, and lenders' expectations alone select among them. In the numerical example of Figure 4, where $\gamma_1 = 1.5$ and $(\gamma_1 - 1)\xi(d_{\text{first}}) = 0.075 < \delta = 0.12$, the threshold is $s_{\text{first}} = 47$ basis points and $\bar{\Omega} = 80$ basis points.

Risk aversion and uncertainty. Both work against self-fulfilling partial default (Appendix E.4). With CRRA utility the sovereign cannot jump, and it borrows into default gradually. In a numerical example with $\nu = 2$ and a wedge of 210 basis points, a self-fulfilling partial-default steady state at $B = 1$ exists next to the no-default steady state at $B = 0.7$ (Figure 4). In every case we computed, such a steady state is reached from zero debt only above a threshold that exceeds Θ^* and rises with the curvature of utility, about 139 basis points for $\nu = 2$ against 47 for a risk-neutral sovereign. With shocks and a risk-neutral sovereign, for steady debts that do not move with the endowment, volatility narrows the range of supporting wedges from both sides, and at a standard deviation of the log endowment of about 7 percent no equilibrium of this kind remains (Figure E.2 in Appendix E.4). Pessimism alone can start a crisis, the fourth lesson, but only for a sovereign that is impatient enough, and curvature and risk raise the bar. Section 4.3 returns to the question at the calibrated parameters.

3.4 The Chain Model

Uncertainty adds a precautionary motive and a kink in the price schedule, which together decide how defaults begin. Let the log endowment take finitely many values z_j and switch from j to k at rate η_{jk}. The HJB and pricing equations are

$$\rho V_j(B) = \max_{c,d} \left\{ u(c) + S\, V_j'(B) + \sum_{k \neq j} \eta_{jk} \big[V_k(B) - V_j(B) \big] \right\},$$
$$\xi(d_j)\, q_j(B) = (1 - d_j)(\lambda + \delta) + \tilde{S}_j\, q_j'(B) + \sum_{k \neq j} \eta_{jk} \big[q_k(B) - q_j(B) \big].$$

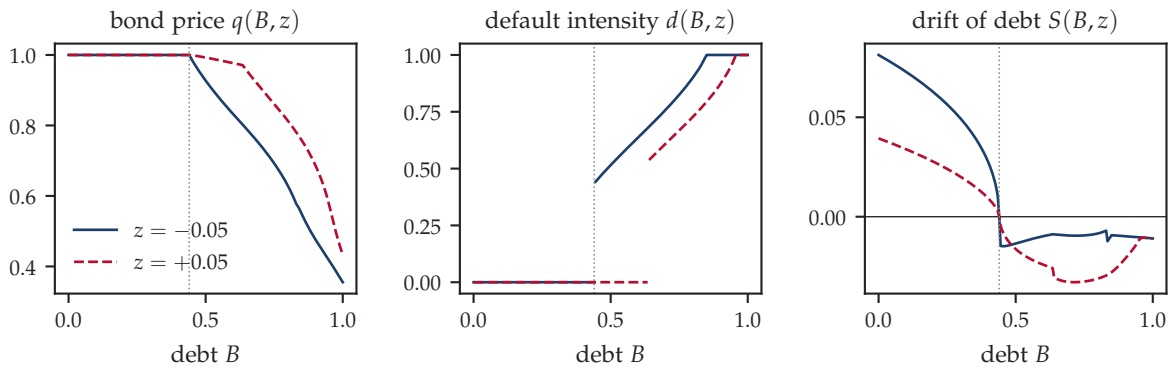


Figure 5: The chain model with two states, $z = \pm 0.05$, that switch at the rate 0.054: bond price, default intensity, and drift of debt by endowment state. Parameters of Table 1. Dotted vertical line: the debt at which both states rest, the node below the low state's default threshold.

Proposition 1 and the thresholds of Proposition 2 hold state by state with $q = q_j(B)$. The precautionary motive can hold a steady state at the risk-free price short of capacity.

Proposition 5 (steady states at the risk-free price in the chain model)**.** *In the chain model, suppose* $[0, \underline{B}]$ *is invariant under the equilibrium dynamics in every state and no state defaults on it. Then* $q_j \equiv q^{\mathrm{rf}}$ *on* $[0, \underline{B}]$ *for every* j *and* $\underline{B} \leq \min_j B_{\mathrm{pd}}(z_j, q^{\mathrm{rf}})$*. If state* j *has a steady state at an interior point* B° *of* $[0, \underline{B}]$*, at which* V_j *is twice differentiable, every* V_k *is differentiable, and the drift cap and the consumption floor are slack, then*

$$(\rho - r)\, u'(c_j) = \sum_{k \neq j} \eta_{jk} \left(u'(c_k) - u'(c_j) \right) \qquad \text{at } B^\circ. \tag{18}$$

In particular, a state whose consumption at B° *is the lowest across states has no steady state in the interior of* $[0, \underline{B}]$*.*

Condition (18) is the Euler equation at a stationary point: the impatience gain from borrowing a little balances the precautionary loss from arriving in a worse state with more debt. In the state with the lowest consumption the precautionary term is nonpositive and nothing offsets the impatience term, so the sovereign cannot rest inside the default-free set and borrows to its edge. In a better state the two terms compete, and a second force works through prices. Beyond the threshold of a state that can be reached in one switch, a switch triggers an immediate first default. So the price schedule of the better state has a kink there, and the marginal revenue of issuance drops. Both forces can hold the better state at or below the threshold of the worse one, and a switch then does not land the sovereign in default.

Figure 5 shows these forces in a chain with two states, $z = \pm 0.05$, that switch at the rate $\eta = 0.054$, half the mean-reversion rate of the calibrated diffusion. No state defaults on the ergodic set. The low state has the lower consumption and borrows up to the node just below its default threshold of 0.445. The high state stops at the same debt, 0.2 short of its own threshold of 0.64. At that debt the impatience term of (18) exceeds the precautionary term, 0.012 against 0.0115, so at a flat price the high state would borrow further. The kink in its price schedule holds

it there, since beyond that debt a switch triggers an immediate default. With more frequent switching the precautionary term takes over, and the high state stops strictly inside the default-free set. With states ± 0.06 that switch at the rate 0.15, the low state rests at $B = 0.42$ and the drift of the high state vanishes at $B = 0.15$, where the two sides of (18) are 0.0115 and 0.0113. This is the second lesson: the sovereign does not borrow into default. Default enters the ergodic set when a shock moves the threshold further than the buffer guards against: a rare jump to a much worse state in a chain, or the continuum of shocks of the diffusion model of Section 4. In a three-state chain with a rare jump to a lower state, default occurs on 0.07 of the ergodic mass, along the paths that follow a jump (Appendix E.1).

4 Quantitative Analysis

This section calibrates the diffusion model to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023) and shows the results of Section 3 at work in this quantitative model. The regions of default and the debt capacity appear in the policies, the stationary debt in the shape of the ergodic set, and the fixed cost in the response to a shock.

4.1 The Diffusion Model

The log endowment follows an Ornstein–Uhlenbeck process, $dz_t = -\mu z_t\, dt + \sigma\, dW_t$, reflected at $\pm\bar{z}$. Its generator is $\mathcal{L}_z = -\mu z\, \partial_z + \frac{1}{2}\sigma^2 \partial_{zz}$, so the HJB and pricing equations are

$$\begin{aligned} \rho V &= \max_{c,\,d} \left\{ u(c) + S\, V_B - \mu z\, V_z + \tfrac{1}{2}\sigma^2 V_{zz} \right\}, \\ \xi(d)\, q &= (1-d)(\lambda+\delta) + \tilde{S}\, q_B - \mu z\, q_z + \tfrac{1}{2}\sigma^2 q_{zz}, \end{aligned} \tag{19}$$

with Neumann conditions $V_z = q_z = 0$ at $\pm\bar{z}$, and the stationary density solves

$$0 = -\partial_B\left[\tilde{S}\, g\right] + \partial_z[\mu z\, g] + \tfrac{1}{2}\sigma^2\, \partial_{zz} g,$$

with zero-flux conditions $\mu z g + \frac{1}{2}\sigma^2 g_z = 0$ at $z = \pm\bar{z}$ and $\tilde{S} g = 0$ at $B = \bar{B}$, and an atom at $B = 0$ that collects the inflow where $\tilde{S}(0^+, z) < 0$. The process is the continuous-time counterpart of the annual AR(1) of Arellano, Mateos-Planas, and Ríos-Rull (2023): an AR(1) with persistence ρ_z and innovation standard deviation σ_ϵ observed at annual frequency corresponds to $\mu = -\ln \rho_z$ and a stationary standard deviation $\sigma_\epsilon / \sqrt{1-\rho_z^2} = \sigma / \sqrt{2\mu}$. Given a continuous price, (19) is a degenerate elliptic equation with a state constraint in B. The value function is its unique constrained viscosity solution and is Lipschitz in B (Soner 1986; Capuzzo-Dolcetta and Lions 1990; Fleming and Soner 2006), and the upwind scheme of Appendix E.3 converges to it (Barles and Souganidis 1991). These results do not cover the underwriting standard, which restricts issuance as a function of the equilibrium price. There we rely on the numerical checks of Appendices E.3 and F.1. We compute the coupled equilibrium as the fixed point of the two equations.

4.2 Calibration and Computation

Utility is CRRA, $u(c) = c^{1-\nu}/(1-\nu)$, and the default cost is (3), the level form of the fixed cost used by Arellano, Mateos-Planas, and Ríos-Rull (2023). We follow their calibration as closely as the continuous-time formulation allows. We fix risk aversion at $\nu = 2$ and the risk-free rate at $r = \ln 1.04$. We also take from them the two parameters of the debt contract, which do not depend on the length of a period. The recovery is $\kappa = 0.7$, the present value at the risk-free rate of the new claims per unit of missed payment, in their model and in ours. Their coupons decay by the factor 0.88 a year, and ours at the rate δ, so $\delta = -\ln 0.88 = 0.128$. Our free parameters are the other seven they estimate: the default cost parameters $\gamma_0, \gamma_1, \gamma_2, y^*$, the discount rate ρ, and the endowment parameters ρ_z, σ_ϵ mapped as above. The targets are the eleven moments of their Table 4, with their weights (three on the mean of debt due to output, 1.25 on the standard deviation of the spread, one on the rest). The loss is the weighted sum of squared proportional deviations.

The level of the underwriting standard is not a free parameter. For every parameter vector it is the lowest level, on a menu with steps of 0.025, at which the sovereign does not issue in full default anywhere on the ergodic set and no mass reaches the borrowing limit. The standard then rules out the accumulation of Section 3.2 and otherwise constrains the equilibrium as little as possible. At the calibrated parameters the rule gives $\underline{q} = 0.575$, a spread of about 1,230 basis points. The highest price of debt in full default on the ergodic set, the counterpart in the diffusion model of the threshold of Proposition 3, is 0.54. In the computations the standard takes effect over a band of prices above $\underline{q}$ instead of at a single price, which makes the policies continuous in the price where it binds (Appendix E.3).

We compute the model's moments with the annual accounting of Arellano, Mateos-Planas, and Ríos-Rull (2023): exactly from the stationary distribution for functions of the state, and from simulated annual data for path functionals (Appendix E.3).

Table 1 reports the calibrated parameters next to their counterparts in Arellano, Mateos-Planas, and Ríos-Rull (2023), and Appendix E.3 describes the search. Relative to their values, the variable default cost is lower in level and more convex. The fixed cost rises less steeply with output and starts at a lower output level. The discount rate is somewhat higher, and the endowment is somewhat more persistent, with smaller innovations.

We compute the equilibrium by the implicit upwind finite difference scheme of Achdou et al. (2022) on a grid of 201 nodes in debt on $[0, 1]$ and 75 nodes in z on ± 2.5 stationary standard deviations. No taste shocks or other smoothing of the decision problem are needed, unlike in discrete-time models with long-term debt (Chatterjee and Eyigungor 2012; Mihalache 2025). Appendix E describes the scheme and its accuracy. The iteration reaches the same equilibrium from very different starting points, among them a price at half its equilibrium level and a flat price below the standard. This does not rule out other equilibria, but the reported one does not depend on where the iteration starts.

Table 1: Calibration

		AMR (2023)	This paper
Fixed			
ν	risk aversion	2	2
r	risk-free rate ($\ln 1.04$)	0.039	0.039
λ	coupon, $= r$	0.039	0.039
κ	recovery of defaulted coupons	0.70	0.70
δ	maturity rate ($-\ln 0.88$)	0.128	0.128
Set by a rule			
$\underline{q}$	underwriting standard	–	0.575
Calibrated			
γ_0	variable default cost, level	0.0476	0.0352
γ_1	variable default cost, curvature	2	2.68
γ_2	fixed default cost, slope	0.12	0.044
y^*	fixed default cost, threshold (level)	0.938	0.803
ρ	discount rate (factor $e^{-\rho}$)	0.047	0.052
ρ_z	annual persistence of the endowment	0.875	0.897
σ_ϵ	annual innovation s.d.	0.052	0.047

Notes: The discrete-time parameters of Arellano, Mateos-Planas, and Ríos-Rull (2023) (Table 3) are mapped to continuous time by $\rho = -\ln(\text{discount factor})$, $\delta = -\ln(\text{annual decay factor of coupons})$, $\mu = -\ln \rho_z$, $\sigma = \sigma_\epsilon \sqrt{2\mu/(1-\rho_z^2)}$. The recovery and the maturity are fixed at their values. The standard is the lowest level at which the sovereign does not issue in full default on the ergodic set. We choose the seven free parameters to minimize the weighted distance to the targets of Table 2, as in their calibration.

4.3 Quantitative Properties

Table 2 reports the targeted moments in the data, in the model of Arellano, Mateos-Planas, and Ríos-Rull (2023), and in ours, and the untargeted moments of their Table 4. The loss at the calibrated point is 0.259, against 0.228 for their model under the same weights. The continuous-time model reproduces the frequency of partial default, the level and the dispersion of debt, the debt service and debt due ratios, and the volatility of output. Partial default is deeper than in the data, and output is somewhat more persistent. The spread is less volatile than the target, and this moment accounts for the largest part of the loss. Among the untargeted moments, the relative volatility of consumption is close to the data. Defaulted coupons in default are lower than in the data, as in their model, and less volatile. The spread is more strongly correlated with output and with debt than in the data, as in their model and more so. With no shocks on the lenders' side, the spread is a function of debt and the endowment alone, so it tracks the state more tightly than a spread that also carries global factors.

Figures 6 and 7 show the value, the policies, the price, and the ergodic distribution. The default policy has the structure of Proposition 2. At the lowest endowment shown, $z = -0.22$, below the fixed-cost threshold $\ln y^* = -0.219$, the sovereign defaults at any positive debt. The seven endowment nodes below the threshold carry about 1.5 percent of the ergodic mass. At higher endowments the sovereign does not default until debt reaches the threshold $B_{\text{pd}}(z, q)$,

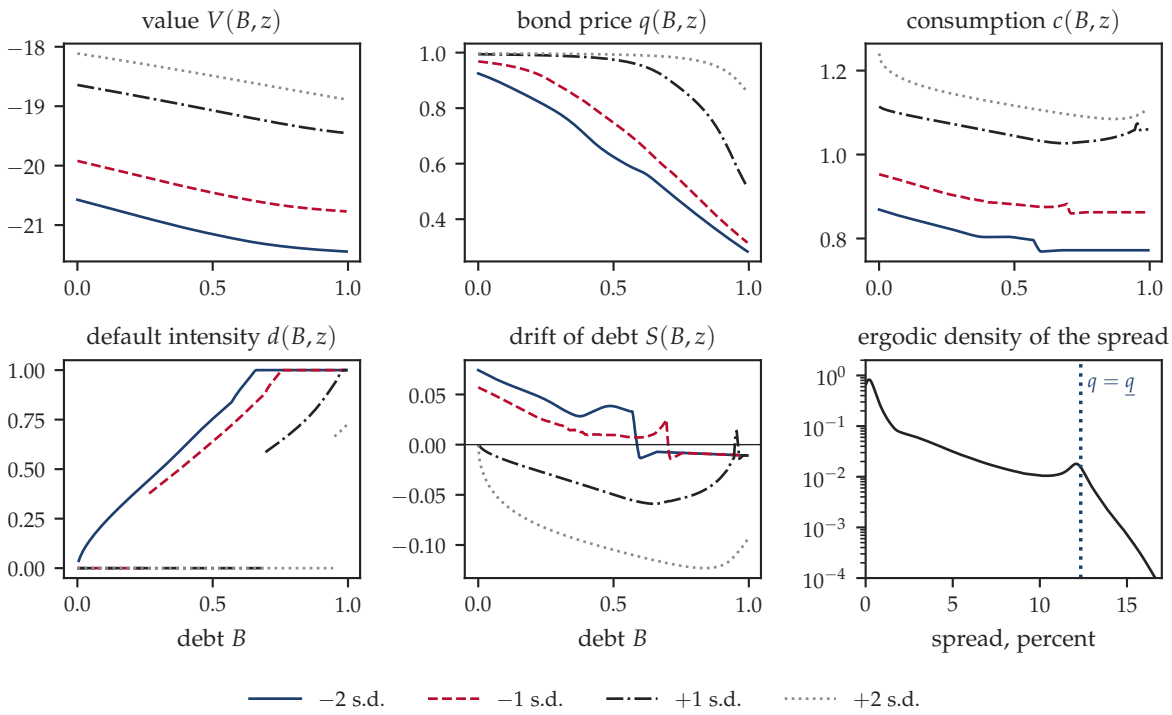


Figure 6: The diffusion model: value, bond price, consumption, default intensity, and drift of debt as functions of debt at four endowment levels, one and two stationary standard deviations below and above the mean, and the ergodic density of the spread, in percent, on a log scale. Dotted: the spread at the underwriting standard.

where the default intensity jumps to the first default intensity, the lumpy entry of the second lesson, and rises thereafter. The threshold rises with the endowment, so the no-default region is a wedge in the (B, z) plane. Full default is reached at high debt, on about 0.1 percent of the ergodic mass. The price falls at the default threshold and declines in the default region, and its contours track the thresholds. Consumption falls steeply at the default threshold and is nearly flat in debt in the default region, where a higher debt is met with a higher default intensity and not with lower consumption.

The drift of debt is positive at low debt and negative at high debt in every endowment state below about one standard deviation above the mean, with a zero-drift locus, the stationary debt. Down to about three quarters of a standard deviation below the mean endowment the locus slopes down, so the stationary debt is higher in bad times than in good times. In good times it lies below the no-default threshold, where precaution holds the sovereign short of the fixed cost, the buffer of Proposition 5. In bad times the threshold has fallen below the current debt, and the sovereign in partial default pays the fixed cost regardless. Its stationary debt is then set by the dilution of its own price, inside the default region. This is the third lesson at work: the episode ends when output recovers and the threshold rises back above the debt, not when the debt is repaid, and the sovereign borrows while it lasts. In worse states the stationary debt is the debt at which the price reaches the band of the standard and issuance is capped. That

Table 2: Moments in Data and Model

Moment	Weight	Data	AMR (2023)	This paper
Targeted				
Partial default: frequency	1	0.36	0.37	0.39
Partial default: mean $\mid d > 0$	1	0.38	0.39	0.48
Partial default: s.d. $\mid d > 0$	1	0.22	0.19	0.26
Debt to output: mean	1	0.32	0.32	0.30
Debt to output: s.d.	1	0.18	0.25	0.20
Debt service to output: mean	1	0.036	0.035	0.033
Debt service to output: s.d.	1	0.021	0.022	0.018
Debt due to output: mean	3	0.049	0.050	0.050
Spread: s.d.	1.25	0.041	0.037	0.029
Output: persistence	1	0.89	0.88	0.93
Output: s.d.	1	0.10	0.12	0.105
Untargeted				
Defaulted coupons to output: mean $\mid d > 0$		0.052	0.040	0.045
Defaulted coupons to output: s.d. $\mid d > 0$		0.064	0.036	0.034
Spread: mean		0.053	0.016	0.020
Spread: correlation with output		−0.17	−0.38	−0.76
Spread: correlation with debt		0.24	0.56	0.84
Consumption s.d. relative to output s.d.		1.00	0.91	0.93

Notes: Data and the discrete-time model from Arellano, Mateos-Planas, and Ríos-Rull (2023), Table 4.

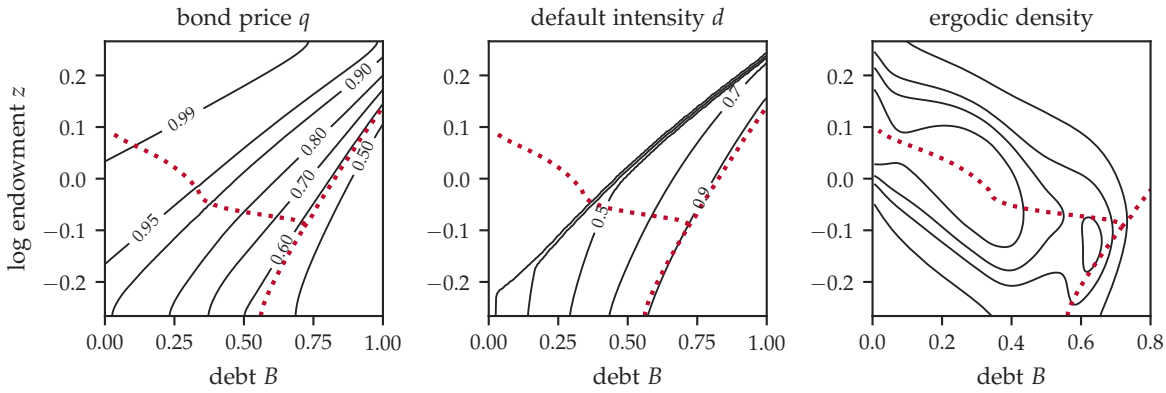


Figure 7: The diffusion model over the state space: contours of the bond price, of the default intensity, and of the ergodic density. The density contours enclose 50, 75, 90, and 99 percent of the mass on positive debt. Dotted: the zero-drift locus, where debt is stationary.

debt falls with the endowment, so the locus bends back, from $B = 0.73$ to 0.56 at the lowest endowment node. Above about one standard deviation above the mean endowment the drift is negative at every debt level the economy visits, and the sovereign repays all the way to zero debt, the atom of the ergodic distribution at $B = 0$. The locus of the standard continues along its price contour to higher endowments and debt levels, which carry no mass. The ergodic set is the region swept by the zero-drift locus as the endowment moves. The density in Figure 7 concentrates along the locus, with most of the mass between zero debt and the deterministic

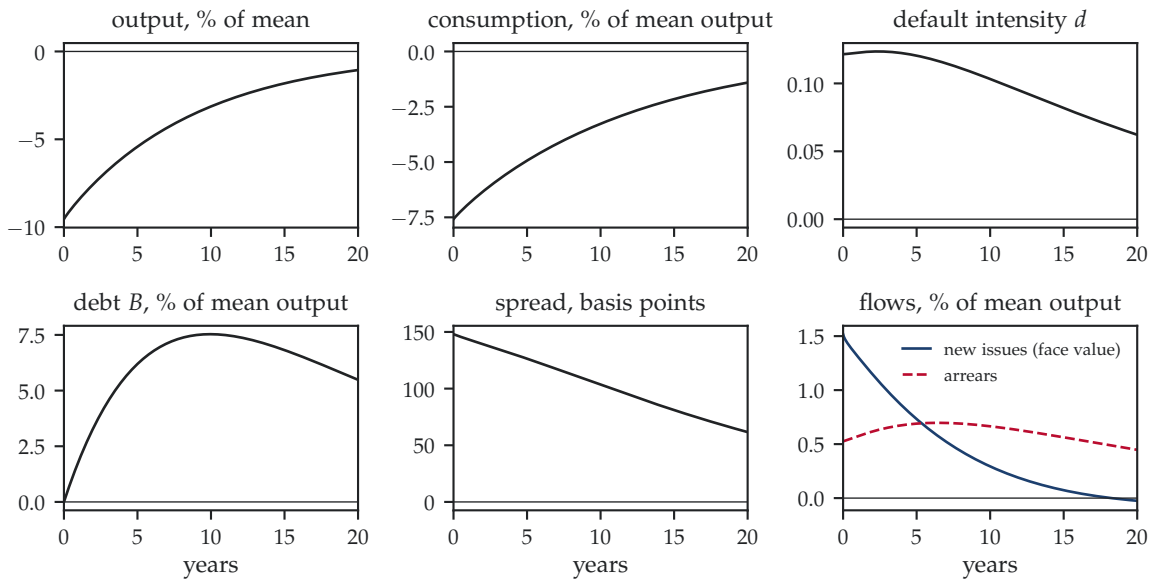


Figure 8: Impulse responses to a negative shock of 0.1 to the log endowment at date zero, from the ergodic distribution: expected path under the forward equation minus the ergodic mean. Flows are in percent of mean annual output, and the spread is in basis points.

debt capacity $B_{\text{pd}}(0, q^{\text{rf}}) = 0.565$ and a tail that extends into the partial-default region at a low endowment. The equilibrium threshold at the average endowment is lower, 0.465, because the price there is 0.91 rather than par (equation (16)). The static comparison of the first lesson thus locates the bulk of the distribution. The tail has a second, smaller mode near $B = 0.6$, where the standard holds the debt in the worst states, and the density thins out above about $B = 0.78$, well before the borrowing limit. There the locus of the standard bounds the ergodic set. Spreads are therefore low most of the time and high in the tail, which is where the model's spread volatility comes from.

Figure 8 shows the response of the economy to a negative shock to the endowment. Starting from the ergodic distribution, we lower z by 0.1 at date zero, about one stationary standard deviation and a ten percent fall in output, and report the expected path of each variable minus its ergodic mean. We compute the expectation with the forward equation (Appendix E.3). Output falls on impact and recovers at the rate μ. Consumption falls by less than output and recovers more slowly, because the sovereign smooths the shock by issuing new debt and withholding debt service. The default intensity jumps on impact and stays elevated for two decades. Debt *rises* for about ten years after the shock, by 7.5 percent of annual output at the peak, before it is worked off. Up to the peak, the shock adds 7.9 percent of annual output to the debt in new issues and 6.5 percent in arrears, the rest of the difference being amortization. The spread rises on impact by about 148 basis points and declines as output recovers. The last panel shows the two flows behind the rise in debt, new issues in face value and arrears. New issues are the larger flow for the first 5.3 years and decline as debt rises and the price falls. Arrears then exceed them, because the default intensity stays elevated and the share κ of every missed coupon returns as

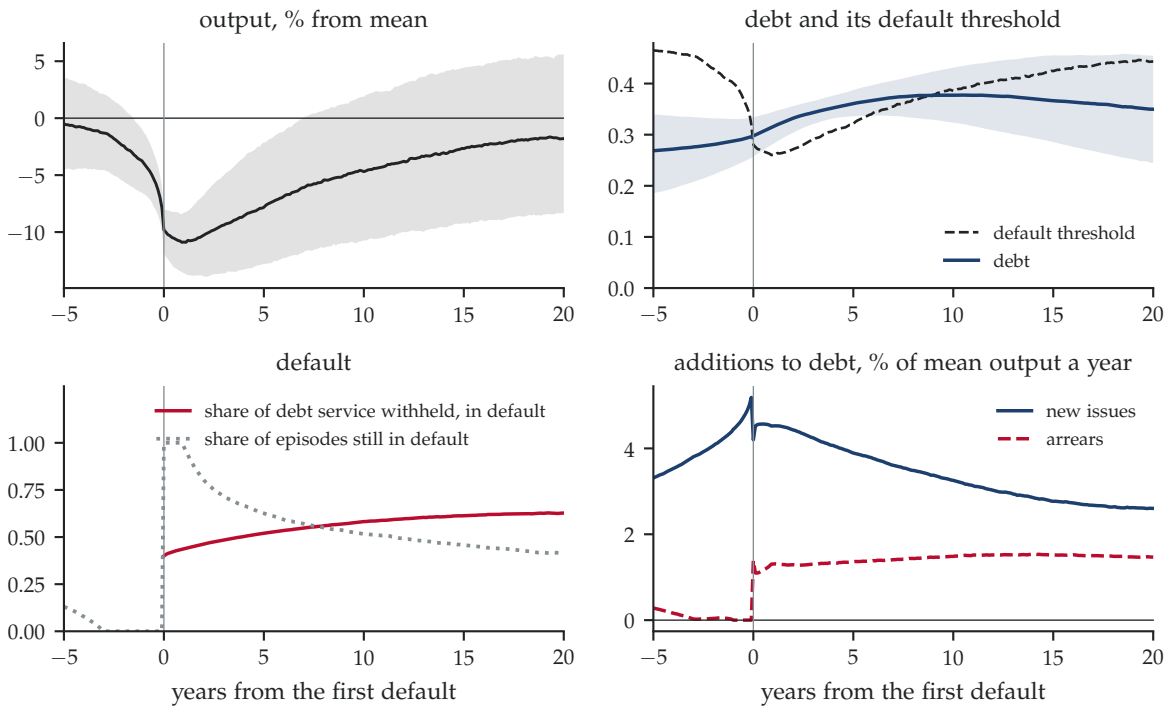


Figure 9: Default episodes in a simulation of the calibrated model, aligned at the first default. Lines: medians across episodes, with the range between the quartiles shaded, for output and debt. The share withheld is the mean across the episodes still in default. The additions to debt are means across episodes.

debt. As sources of funds the two flows are closer than the panel shows. Withholding raises the full defaulted coupons, of which only the fraction κ adds to the debt, and the proceeds of the issues are below their face value. These are the dynamics of partial default episodes documented by Arellano, Mateos-Planas, and Ríos-Rull (2023): default is protracted, debt rises during the episode, and the economy exits the episode with more debt than it entered it. The sovereign in partial default remains a net issuer, and the recovery of output, not repayment, ends the episode.

Figure 9 shows the same dynamics in the default episodes of a long simulation, aligned at the first default. An episode is a spell of default with no interruption of a year or more that lasts at least a year, and we keep those that follow three years without default. Output is about ten percent below its mean at the first default. The threshold falls through the debt there, and the sovereign withholds about forty percent of its debt service at once, the lumpy entry of the second lesson. The median debt then rises for about ten years, by about a quarter. In levels, new issues are the larger addition to the debt throughout. Half of the episodes are over within six years, and they end as the threshold rises back above the debt. The underwriting standard is slack in most episodes. It binds at some point in 11 percent of them, none shorter than ten years, and for 5 percent of the time spent in episodes. Where it binds, it first does so a median of 14 years into the episode.

The response depends on the state. From the mean endowment, debt rises by about the same

amount whether it starts at its ergodic mean or a quarter below or above it, while the default intensity and the spread respond more the higher the debt (Appendix F).

At the calibrated parameters the standard does not rule out the self-fulfilling partial default of Section 3.3 in the deterministic model with the endowment at its mean. The stationary spread at the first default, 483 basis points, is below the spread of 1,235 at the standard, so the standard does not exclude such steady states (Section 3.3). Impatience rules them out: the wedge, 128 basis points, is far below the risk-neutral threshold, 483 basis points with these default costs (the condition of Proposition 4(b) holds), so a risk-neutral sovereign would not reach such a steady state from zero debt (Proposition 4(a)). Curvature raises the threshold further (Section 3.3). The no-default-basin equilibrium of Theorem 1 is not the only equilibrium at these parameters. The cube-root solution of Theorem 2 continues below capacity, where the sovereign defaults at the first default intensity over the last 3.1 percent of output of borrowing, at a spread of up to 43 basis points (Appendix D). Whether the diffusion model admits a self-fulfilling equilibrium is open.

5 Conclusion

In the standard model of sovereign default, the value of market access limits how much a sovereign can borrow, and the limit is the outcome of the whole dynamic problem. In the partial default model of Arellano, Mateos-Planas, and Ríos-Rull (2023) the limit is a static comparison. Withholding part of the debt service raises funds at a price fixed by the recovery on arrears, on terms that do not deteriorate with the sovereign's creditworthiness, against the output cost of default. The debt capacity that follows is a closed form in the default costs, the recovery, the maturity of the debt, and the bond price. Preferences enter only through that price. The same structure fixes the anatomy of a crisis. The fixed cost makes the first default a jump, and a shock that moves the threshold below the sovereign's buffer triggers it. Once the fixed cost is being paid, only the dilution of the sovereign's own price restrains borrowing, so the sovereign keeps issuing, and the episode ends when output recovers. A sovereign above its capacity instead works its debt down along a cube-root law. Pessimism alone can start a crisis, when the gain from front-loading borrowed funds covers the output cost of the cheapest default, and the crisis unfolds over years without a run. Curvature raises that bar, and the calibrated model is far from it.

These results come from a model that is tractable and quantitative at once. Calibrated to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023), with their weights, the continuous-time model fits them about as well as the discrete-time original, and the underwriting standard binds on about 2 percent of the ergodic mass.

Several avenues remain. On the theoretical side, self-fulfilling partial default is nearly settled for a risk-neutral sovereign, up to a stability and a continuation condition. With curvature or shocks we have computed the threshold but not proved it, and a proof would complete the

account of slow-moving debt crises with arrears. With shocks we have also studied only steady debts that do not move with the endowment, and we have studied curvature and uncertainty only one at a time. Existence of the coupled equilibrium in the stochastic models is open. On the quantitative side, the natural extensions are reserves, a maturity structure, and shocks on the lenders' side, which would loosen the tight link between the spread and the state. Each adds a state variable.

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Appendix to "Debt Capacity and the Anatomy of Sovereign Crises"

by Sangdong Kim and Gabriel Mihalache

A Additional Results and Proofs

A.1 Results and Proofs for Section 3.1

We also state and prove here the three properties of the price summarized in Section 3.1, and a lemma used in the Online Appendix.

Proof of Proposition 1. Fix any admissible drift S. The objective $u(\mathcal{R}(d)+qS)+SV_B$ depends on d only through $\mathcal{R}(d)$, u is increasing, and the feasibility constraint $c \geq c_{\min}$ is relaxed by a larger $\mathcal{R}(d)$. Hence for every S the best d maximizes $\mathcal{R}$, and so does the d-component of the joint maximizer. □

Proof of Proposition 2. On $(0,1]$, $\mathcal{R}'(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1-1} + (1-\kappa q)(\delta+\lambda)B$ is strictly decreasing with $\mathcal{R}'(0^+) > 0$, so $\mathcal{R}$ is strictly concave with maximizer d_{int} of (8). The gain from defaulting is $\mathcal{R}(d_{\text{int}}) - \mathcal{R}(0) = \Gamma(B,z,q) - (1-F(z))e^z$, which gives (13). Since d_{int} maximizes $(1-\kappa q)(\delta+\lambda)Bd - \gamma_0 F(z)e^z d^{\gamma_1}$ on $[0,1]$, the envelope theorem gives $\partial_B\Gamma = (1-\kappa q)(\delta+\lambda)d_{\text{int}} > 0$ and $\partial_q\Gamma = -\kappa(\delta+\lambda)Bd_{\text{int}} < 0$ for $B > 0$, whether d_{int} is interior or capped, and $\Gamma = 0$ at $B = 0$; (a), (b), and (d) follow. At $B = B_{\text{fd}}(z,q)$, where $d_{\text{int}} = 1$, $\Gamma = (\gamma_1 - 1)\gamma_0 F(z)e^z$, so $B_{\text{pd}} < B_{\text{fd}}$ if and only if $1 - F(z) < (\gamma_1 - 1)\gamma_0 F(z)$, that is $d_{\text{first}} < 1$. Eliminating B between $\Gamma = (1-F)e^z$ and (8) gives (14), and eliminating d_{int} gives (15); when $d_{\text{first}} \geq 1$ the jump is from 0 to 1, and $\mathcal{R}(1) = \mathcal{R}(0)$ gives the second formula. □

Proposition A.1 (the bond price)**.** *In the general model, fix measurable policies (d,S), let (B_t,z_t) be the induced process, and define q by (9), the bounded solution of (10) wherever that equation holds. Then $0 \leq q \leq q^{\text{rf}}$; $q(B,z) = q^{\text{rf}}$ if and only if $d_t = 0$ for almost every t, almost surely, along the path from (B,z); and $q(B,z) = 0$ if and only if $d_t = 1$ for almost every t, almost surely.*

Proof of Proposition A.1. (9) is the Feynman–Kac representation of (10) with discount rate $\xi(d_t) \geq \xi(1) > 0$. The bound: $(1-d)(\lambda+\delta) \leq q^{\text{rf}}\xi(d)$ for all $d \in [0,1]$ is equivalent to $d\,\xi(1) \geq 0$; integrating, $q \leq q^{\text{rf}}\,\mathbb{E}\int_0^\infty \xi(d_t)e^{-\int_0^t \xi}dt = q^{\text{rf}}$, with equality iff $d_t = 0$ for almost every t, almost surely. The integrand of (9) is nonnegative and vanishes exactly where $d_t = 1$, so $q = 0$ if and only if $d_t = 1$ for almost every t, almost surely. □

Lemma A.1 (the price is nonincreasing in the default path)**.** *In the general model, let d and d' be two adapted default processes with $d'_t \leq d_t$ for all t, almost surely. Then the Feynman–Kac values (9) satisfy $q(d') \geq q(d)$.*

Proof. For $\theta \in [0,1]$ let $d^\theta = d' + \theta(d - d')$ and $Q(\theta)$ the corresponding value. Differentiating under the integral and collecting the effect of d_t^θ on the payment at t and on the discounting of the stream after t (through $\xi'(d) = -\kappa(\delta + \lambda)$), $Q'(\theta) = \mathbb{E}\int_0^\infty e^{-\int_0^t \xi(d^\theta)}(\lambda + \delta)[\kappa q_t^\theta - 1](d_t - d'_t)\,dt \leq 0$, since $\kappa q_t^\theta \leq \kappa q^{\mathrm{rf}} \leq 1$. □

Proposition A.2 (continuity of the price along crossing flows)**.** *In the deterministic, chain, and diffusion models, fix a state (B_0, z_0) and suppose that on a neighborhood of it the equilibrium drift is negative and bounded away from zero, so that from every nearby state debt moves down through B_0 in a time that vanishes with the distance to B_0. Then $q(\cdot, z_0)$ is continuous at B_0: with no further condition in the deterministic and chain models, and in the diffusion model whenever $q(B, \cdot)$ is continuous at z_0 uniformly in B on a neighborhood of B_0. The same holds with positive drift. Hence q can be discontinuous in B only at points where the drift vanishes or changes sign that are not reached in finite time from both sides (and, in the diffusion model, where q is not continuous in z).*

At a stationary point of debt that is approached from below and left from above, the paths on the two sides go to different places, Proposition A.2 does not apply, and the price can jump.

Proof of Proposition A.2. With the drift bounded away from zero, from $(B_0 + x, z_0)$ debt reaches B_0 at a time $\tau(x) \to 0$ on every path along which z stays in the neighborhood. By (9) and the strong Markov property, $q(B_0 + x, z_0) = \mathbb{E}\left[\int_0^\tau e^{-\int_0^t \xi}(1 - d)(\lambda + \delta)\,dt + e^{-\int_0^\tau \xi}\, q(B_\tau, z_\tau)\right]$. The integral is $O(\tau)$ and the discount factor tends to one. In the deterministic model $(B_\tau, z_\tau) = (B_0, z_0)$. For a chain, z switches before τ with probability $O(\tau)$, and on the complement $(B_\tau, z_\tau) = (B_0, z_0)$; since q is bounded, the expectation tends to $q(B_0, z_0)$. For the diffusion, z leaves the neighborhood before τ with vanishing probability and otherwise $B_\tau = B_0$ with $z_\tau - z_0 = O_p(\tau^{1/2})$, so the expectation tends to $q(B_0, z_0)$ when $q(B_0, \cdot)$ is continuous at z_0. This gives continuity from the right. From the left, start at (B_0, z_0) itself: debt reaches $B_0 - x$ at a time $\tau(x) \to 0$, and the same decomposition gives $q(B_0, z_0) = O(\tau) + \mathbb{E}\left[e^{-\int_0^\tau \xi}\, q(B_0 - x, z_\tau)\right]$. In the deterministic and chain models $z_\tau = z_0$ with probability $1 - O(\tau)$, so $q(B_0 - x, z_0) \to q(B_0, z_0)$. In the diffusion model $q(B_0 - x, z_\tau) - q(B_0 - x, z_0) \to 0$ in probability by the uniform continuity in z, and the same conclusion follows. With positive drift the two sides exchange roles. □

Proposition A.3 (autarky and its removal)**.** *In the general model with z in a bounded set: under a zero price, full default is optimal if and only if B is at least $B_{\mathrm{fd}}(z, 0)$ and above $B_{\mathrm{pd}}(z, 0)$, and, without the underwriting standard, $(q \equiv 0, d \equiv 1)$ is an equilibrium of the model restricted to any invariant set contained in $\{B \geq \max\{B_{\mathrm{fd}}(z, 0), B_{\mathrm{pd}}(z, 0)\}\}$, a knife edge. Invariance requires the sovereign to issue at least aB at a zero price. Under the convention that no bonds are traded at a zero price, $q > 0$ at every state in every equilibrium.*

Proof of Proposition A.3. The default rule under $q = 0$ is Proposition 1 with $q = 0$. With $q = 0$ issuance does not enter the budget, so any ℓ is optimal. With $d = 1$ no payment is made and $q = 0$ is the lenders' valuation. Invariance is needed for self-consistency, and with $d = 1$ debt decays at rate a unless $\ell \geq aB$. For the removal: suppose $q(B_0, z_0) = 0$. By Proposition A.1, $d = 1$

a.e. along the path, so $q = 0$ along the path, so $\ell = 0$ along the path and $B_t = B_0 e^{-at}$. Since $B_{\text{fd}}(z,0) = \gamma_0\gamma_1 F(z)e^z/(\delta+\lambda)$ is bounded below by a positive constant on a bounded set of z, so is $\max\{B_{\text{fd}}(z,0), B_{\text{pd}}(z,0)\}$, and in finite time B_t falls below $\inf_z \max\{B_{\text{fd}}(z,0), B_{\text{pd}}(z,0)\}$. There full default is not optimal. With no issuance at a zero price, consumption is $\phi(d,z)e^z - (1-d)(\delta+\lambda)B$ and debt decays at rate $a(d)$. Lowering d for an instant, slightly below 1 if $B < B_{\text{fd}}(z,0)$, where $\mathcal{R}_d(1;B,z,0) < 0$, or to 0 if $B < B_{\text{pd}}(z,0)$, raises consumption now and lowers debt, while returning to $d = 1$ afterwards gives the same consumption $(1-\gamma_0)F(z)e^z$ whatever the debt. The deviation is strictly better, a contradiction. □

A.2 The Model at High Debt

At high debt the model has the knife-edge autarky equilibrium of Proposition A.3, which a convention removes. Where default carries no fixed cost, debt also accumulates into the borrowing limit, and the underwriting standard removes this accumulation.

Accumulation without a fixed cost. In the no-fixed-cost region a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt $-V_B$ close to zero, while the price is positive because output will recover. By (7) it issues at any positive price. Even without the borrowing limit it could not keep doing so forever: debt that stayed above every full-default threshold would never be serviced again and would be worthless, which the convention against trade at a zero price rules out (Proposition A.3).

The underwriting standard. With issuance switched off below $\underline{q}$, the drift beyond the full-default line is at most $-a(d)B < 0$, and the full-default line is crossed downward from both sides. In the deterministic model the price is continuous across it whenever the drift is also negative just below it (Proposition A.2). For any $\bar{B} \geq B^\sharp$ the long-run distribution is confined to $[0, B^\sharp]$ (Proposition A.5(b)). The standard also affects the default decision, because its constraint, $\ell \leq 0$, depends on d through $\ell = S + a(d)B$. We therefore re-examine the separation of Proposition 1 where the standard binds.

Proposition A.4 (the default decision under the standard)**.** *In the general model with the underwriting standard, fix a state with $q = q(B,z) < \underline{q}$, $B > 0$, and $V_B < 0$, and let (d^*, S^*) be the unconstrained maximizer with issuance $\ell^* = S^* + a(d^*)B$.*

(a) If $\ell^* \leq 0$ the standard is slack and Proposition 1 holds verbatim.

(b) If $\ell^* > 0$ and $d^* > 0$ (the unconstrained optimum is in partial default) the constrained maximum has $\ell = 0$, hence $S = -a(d)B$, and the default intensity maximizes

$$J(d) := u(\phi(d,z)e^z - (1-d)(\delta+\lambda)B) - a(d)\,B\,V_B.$$

At a maximizer $d^{**} \in (0,1]$, the corner $d^{**} = 1$ included, with $c^{**} := \phi(d^{**}, z)e^z - (1 - d^{**})(\delta + \lambda)B$,

$$d^{**} = d_{\text{int}}(B, z, \hat{q}), \qquad \hat{q} := \frac{-V_B}{u'(c^{**})} \in (0, q) :$$

the unconstrained rule (8) evaluated at the sovereign's own shadow price of a unit of debt instead of the market price. Since d_{int} is nonincreasing in the price, $d^{**} \geq d_{\text{int}}(B, z, q)$, strictly whenever $d_{\text{int}}(B, z, q) < 1$.

(c) If $\ell^* > 0$ and $d^* = 0$ (the unconstrained optimum repays in full while issuing), the constrained optimum defaults whenever $B > B_{\text{pd}}(z, \hat{q})$, with $\hat{q} := -V_B / u'(c^{**})$ the shadow price at the maximizer d^{**} of J on $(0,1]$, and only if $B > B_{\text{pd}}(z, q_0)$, with $q_0 := -V_B / u'(e^z - (\delta + \lambda)B)$ the shadow price at full repayment. So the standard can induce a default that the market-price rule does not.

Proposition A.5 (the standard removes the accumulation)**.** *In the general model with z in a bounded set Z and the underwriting standard:*

(a) in every equilibrium in which $q < \underline{q}$ at every state where the sovereign is in full default, the drift in full default is $S \leq -aB < 0$, so the full-default region is transient. In the deterministic model, if $V_B \leq 0$, it is left in time at most $a^{-1} \ln\big(B / \max\{B_{\text{fd}}(0,0), B_{\text{pd}}(0,0)\}\big)$;

(b) for any limit $\bar{B} \geq B^\sharp := \max\Big\{0,\ \sup\{B :\ q(B,z) \geq \underline{q} \text{ for some } z\}\Big\}$: whenever $B_t > B^\sharp$ the drift satisfies $S_t \leq -aB_t$, the set $[0, B^\sharp] \times Z$ is invariant, and every stationary distribution is supported in it.

The long-run distribution of debt is confined to the debt levels at which the price can still reach the standard, whatever the borrowing limit above them. The computations use $\bar{B} = 1$, and that the equilibrium below the limit does not depend on its level is a numerical finding (Online Appendix F).

Proofs.

Proof of Proposition A.4. Parametrize the choice by (d, ℓ). The objective is $\mathcal{G}(d, \ell) = u(\tilde{\mathcal{R}}(d) + q\ell) + (\ell - a(d)B)V_B$ with $\tilde{\mathcal{R}}(d) = \phi(d, z)e^z - (1 - d)(\delta + \lambda)B$ the resource function at a zero price, and the constraint is $\ell \leq 0$, which no longer involves d. On $(0,1] \times \mathbb{R}$, $\tilde{\mathcal{R}}$ is concave in d, u is concave and increasing, and the last term is linear, so $\mathcal{G}$ is jointly concave. (a) is immediate. For (b), let (d_c, ℓ_c) be a constrained maximizer on the $d > 0$ branch with $\ell_c < 0$. The segment from it to (d^*, ℓ^*) crosses $\{\ell = 0\}$ at a point at which, by concavity, $\mathcal{G}$ is at least $\min\{\mathcal{G}(d_c, \ell_c), \mathcal{G}(d^*, \ell^*)\} = \mathcal{G}(d_c, \ell_c)$, so that point is a constrained maximizer with $\ell = 0$ on the $d > 0$ branch. The $d = 0$ branch is treated separately, since $\mathcal{G}$ is not concave across the fixed-cost jump at $d = 0$. Its own unconstrained maximizer has issuance $\ell_0^* > \ell^* > 0$, because $d^* > 0$ maximizes $\mathcal{R}$ and $a(d^*) < \delta$. So the same comparison along $\{d = 0\}$ gives $\ell = 0$ there as well, and the constrained optimum is the better of $J(d^{**})$ and $J(0)$. It is $J(d^{**})$: the concavity bound in the proof of (c) below gives $J(d^{**}) - J(0) \geq u'(c^{**})[\mathcal{R}(d^{**}; B, z, \hat{q}) - \mathcal{R}(0; B, z, \hat{q})] > 0$,

since $B > B_{\mathrm{pd}}(z,q) \geq B_{\mathrm{pd}}(z,\hat{q})$. On $\{\ell = 0\}$, $c = \tilde{\mathcal{R}}(d)$ and $S = -a(d)B$, which gives J. At an interior maximizer $J'(d) = u'(c)\tilde{\mathcal{R}}'(d) + \kappa(\delta+\lambda)BV_B = 0$, that is $\tilde{\mathcal{R}}'(d) = \kappa(\delta+\lambda)B\hat{q}$, the first-order condition of $\mathcal{R}(\cdot;B,z,\hat{q})$. At the corner $d^{**} = 1$, $J'(1) \geq 0$, that is $\tilde{\mathcal{R}}'(1) \geq \kappa(\delta+\lambda)B\hat{q}$, which is $d_{\mathrm{int}}(B,z,\hat{q}) = 1$. In both cases $d^{**} = d_{\mathrm{int}}(B,z,\hat{q})$, and the comparison with $d_{\mathrm{int}}(B,z,q)$ follows from $\hat{q} < q$ and the monotonicity of (8), which is strict below the cap. Finally $\partial_\ell \mathcal{G}(d^{**},0) = qu'(c^{**}) + V_B > 0$ because the constraint binds strictly, which is $\hat{q} < q$, and $\hat{q} > 0$ because $V_B < 0$ by hypothesis. For (c), on the $d = 0$ branch the unconstrained issuance is $\ell^* > 0$, so the comparison of (b) along $\{d = 0\}$ gives $\ell = 0$ and the value $J(0)$. On the $d > 0$ branch the unconstrained issuance can have either sign. If it is positive, the argument of (b) gives $\ell = 0$ and the branch value $J(d^{**})$. If it is nonpositive, the constraint is slack on the branch, and its value $\max_{d\in(0,1],\ell\leq 0}\mathcal{G}(d,\ell)$ is at least $\mathcal{G}(d^{**},0) = J(d^{**})$. Either way the best default is worth at least $J(d^{**})$. By the concavity of u, $J(d^{**}) - J(0) \geq u'(c^{**})\big[\tilde{\mathcal{R}}(d^{**}) - \tilde{\mathcal{R}}(0)\big] + \kappa(\delta+\lambda)d^{**}BV_B = u'(c^{**})[\mathcal{R}(d^{**};B,z,\hat{q}) - \mathcal{R}(0;B,z,\hat{q})]$. Since $d^{**} = d_{\mathrm{int}}(B,z,\hat{q})$ maximizes $\mathcal{R}(\cdot;B,z,\hat{q})$ on $(0,1]$, the bracket is positive whenever $B > B_{\mathrm{pd}}(z,\hat{q})$ (Proposition 2), so the constrained optimum defaults there. The condition is sufficient, not necessary: the bound uses the concavity of u, and the default region under the standard can extend slightly below $B_{\mathrm{pd}}(z,\hat{q})$. For the necessary condition, let $c_0 := e^z - (\delta+\lambda)B$ and $q_0 := -V_B/u'(c_0)$. The unconstrained issuance at $d = 0$ is positive, so $q > q_0$. By the concavity of u at c_0, $\mathcal{G}(d,\ell) - \mathcal{G}(0,0) \leq u'(c_0)[\mathcal{R}(d;B,z,q_0) - \mathcal{R}(0;B,z,q_0)] + \ell\, u'(c_0)(q - q_0)$, whose last term is nonpositive for $\ell \leq 0$, so a default can be optimal only if $B > B_{\mathrm{pd}}(z,q_0)$. □

Proof of Proposition A.5. (a) With $d = 1$ and $\ell \leq 0$, $S = \ell - aB \leq -aB$; integrate. Where the standard is slack, full default requires $B \geq B_{\mathrm{fd}}(0,q)$ and $B > B_{\mathrm{pd}}(0,q)$; where it binds, any default requires $B > B_{\mathrm{pd}}(0,q_0)$ and full default requires $B \geq B_{\mathrm{fd}}(0,\hat{q})$ (Proposition A.4), with $q_0, \hat{q} \geq 0$ because $V_B \leq 0$. All these thresholds are nondecreasing in the price, so in the deterministic model debt has left the full-default region by the time it falls below $\max\{B_{\mathrm{fd}}(0,0), B_{\mathrm{pd}}(0,0)\}$. (b) Debt rises only when $\ell > 0$, which the standard permits only at $q \geq \underline{q}$, hence only at $B \leq B^\sharp$. Above $B^\sharp$, $\ell \leq 0$ and $S = \ell - a(d)B \leq -aB$. The rest follows from the continuity of B_t. □

A.3 Results and Proofs for Section 3.3

Proof of Theorem 1. From any $B \leq B^*$ the path rises to B^* and stays, with $d \equiv 0$. Then (9) gives $q = q^{\mathrm{rf}}$, and consumption at B^* is $1 - (\delta+\lambda)B^* + \delta B^* = 1 - rB^*$ forever. At B^* the price is q^{rf} and $d = 0$ is optimal, so $B^* \leq B_{\mathrm{pd}}(0,q^{\mathrm{rf}})$. Suppose $B^* < B_{\mathrm{pd}}(0,q^{\mathrm{rf}})$. Since debt reaches B^* from above in finite time, (9) gives $q(B^*+x) \geq e^{-(r+\delta)\tau(x)}q^{\mathrm{rf}} \to q^{\mathrm{rf}}$, and $B_{\mathrm{pd}}(0,\cdot)$ is continuous and increasing, so for small x, $B^* + x < B_{\mathrm{pd}}(0,q(B^*+x))$ and $d = 0$ at $B^* + x$. Along the path down to B^*, then, $d \equiv 0$ and $q(B^*+x) = q^{\mathrm{rf}}$ exactly. Thus $q \equiv q^{\mathrm{rf}}$ and $d \equiv 0$ on a neighborhood of B^*, where the sovereign faces the constant risk-free price with no default. There, staying at B^* is not optimal because $\rho > r$. The round trip that borrows at rate $m = \iota\tau$ for a time τ and repays at the same rate for the next τ perturbs consumption by $\Delta c_t = m - rmt$ on $[0,\tau]$ and $-m - rm(2\tau - t)$

on $[\tau, 2\tau]$, with first-order gain $u'(\bar{c})\int_0^{2\tau} e^{-\rho t}\Delta c_t\,dt = (\rho - r)u'(\bar{c})\iota\tau^3 + O(\iota\tau^4)$ and second-order loss $O(\iota^2\tau^3)$. For small ι the gain dominates. Since V is the value of the sovereign's problem, staying at B^* is then not optimal, contradicting $S(B^*) = 0$. □

Proof of Theorem 2. Step 1: the drift vanishes. On the right branch the first-order condition is $V' = -qu'(c)$, and with $S = (c - \mathcal{R})/q$, $\mathcal{R}(B) := \max_d \mathcal{R}(d; B, 0, q(B))$, the HJB reads

$$\rho V = u(c) + w\,u'(c), \qquad w := \mathcal{R} - c = -qS \geq 0. \tag{20}$$

Let $x \downarrow 0$: $V \to u(\bar{c})/\rho$, $q \to q^{\mathrm{rf}}$ (because debt reaches B^* in finite time, by (9), as in the proof of Theorem 1) and $\mathcal{R} \to \bar{c}$, because at $B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$ the sovereign is indifferent between $d = 0$ and d_{first}. Along any sequence on which $c \to c^+$, (20) gives $u(\bar{c}) = u(c^+) - (c^+ - \bar{c})u'(c^+)$. The function $f(c) = u(c) - (c - \bar{c})u'(c) - u(\bar{c})$ has $f(\bar{c}) = 0$ and $f'(c) = -u''(c)(c - \bar{c})$, so $f > 0$ for $c \neq \bar{c}$; hence $c \to \bar{c}$, $w \to 0$, $S \to 0$, and $V'(B^{*+}) = -u'(\bar{c})$. The same argument on the left, where $V' = -u'(c)$, gives $V'(B^{*-}) = -u'(\bar{c})$.

Step 2: the exponents. Put $p := q^{\mathrm{rf}} - q$. The pricing equation gives $p' = q\Pi/w$ with $\Pi(B, q) := \xi(d^*)q - (1 - d^*)(\lambda + \delta) \to \Pi_0 > 0$. Differentiating (20), using $V' = -qu'(c)$, $c = \mathcal{R} - w$, and the envelope theorem $\mathcal{R}' = \mathcal{R}_B - \mathcal{R}_q p'$ with $\mathcal{R}_B = -(1 - d^*)(\delta + \lambda) + qa(d^*)$ and $\mathcal{R}_q = a(d^*)B$,

$$w' = \mathcal{R}_B - \mathcal{R}_q p' + \frac{u'(c)}{|u''(c)|}\,\frac{\mathcal{R}_q q\Pi/w - (\mathcal{R}_B + \rho q)}{w} = \frac{M(x)}{w^2} + O(1/w),$$
$$M(x) \to M := \frac{u'(\bar{c})}{|u''(\bar{c})|}\,a(d_{\mathrm{first}})B^*\Pi_0.$$

Since $w \to 0$, $w^2 w' = M + o(1)$ and $w^3 = 3Mx(1 + o(1))$, that is $w = \beta x^{1/3}(1 + o(1))$ with $\beta^3 = 3M$. Then $p' = \Pi_0(1 + o(1))/w$ integrates to $p = \alpha x^{2/3}(1 + o(1))$, and $S = -w/q$, $c = \mathcal{R} - w$ give the stated laws. The second derivative $V'' = -q'u'(c) - qu''(c)c'$ is dominated by $-|u''(\bar{c})|\beta x^{-2/3}/3$. □

Lemma A.2 (local conditions at a partial-default steady state)**.** *In the deterministic model, suppose the equilibrium has a steady state B° in the interior of the partial-default region at which S and q are right-differentiable and V is twice right-differentiable, with $d^\circ = d^*(B^\circ, q^\circ) \in (0, 1)$. Then $q^\circ = (1 - d^\circ)(\lambda + \delta)/\xi(d^\circ)$, the slope of the price is pinned by the impatience wedge,*

$$-\frac{q'(B^{\circ+})}{q^\circ} = \frac{\rho - r}{a(d^\circ)B^\circ},$$

and the slope of the drift follows from the pricing equation,

$$q'(B^{\circ+})\big(S'(B^{\circ+}) - \xi(d^\circ)\big) = (\lambda + \delta)(1 - \kappa q^\circ)\,\frac{d}{dB}d_{\mathrm{int}}(B, q(B))\Big|_{B^{\circ+}},$$

so that the steady state is stable from above, $S'(B^{\circ+}) < 0$, if and only if the dilution term on the right

exceeds $\xi(d^\circ)\,|q'(B^{\circ+})|$.

Proof of Lemma A.2. The first identity is (9) with constant d. Differentiating $\rho V = u(c) + SV'$ with $c = \mathcal{R}^* + qS$ gives $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q' + q'S + qS') + S'V' + SV''$. At $S = 0$ the two terms in S' cancel by the first-order condition $V' = -qu'(c)$, leaving $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q')$. With $V' = -qu'(c)$, $\mathcal{R}_q = a(d)B$, and $\mathcal{R}_B = -(1-d)(\delta+\lambda) + qa(d) = -rq$ by the first identity, this reads $-\rho q = -rq + a(d)Bq'$, the second identity. The third is l'Hôpital's rule applied to $q' = (\xi(d)q - (1-d)(\lambda+\delta))/S$. □

Proposition A.6 (the risk-neutral construction)**.** *Let $u(c) = c$ and $\rho > r$ in the deterministic model, with issuance unrestricted in speed. Write $F = F(0)$ and define for $d \in [d_{\text{first}}, 1)$ the price $q_\rho(d) := (1-d)(\lambda+\delta)/(\rho + a(d))$ of a bond that defaults at rate d forever, discounted at the sovereign's rate, the debt $B(d)$, and the stability bound $\Omega(d)$ below. Here Θ of Section 3.3 is the output cost of the default per unit of the funds raised at the steady state:*

$$B(d) := \frac{\gamma_0\gamma_1 F\, d^{\gamma_1 - 1}\,\xi(d)}{(1-\kappa)(\lambda+\delta)^2},$$
$$\Theta(d) = \frac{1-\phi(d,0)}{q_r(d)B(d)},$$
$$\Omega(d) := \frac{s(d)\,a(d)}{(\gamma_1 - 1)\xi(d) - \kappa(\lambda+\delta)d},$$

with $\Omega(d) = \infty$ when the denominator is not positive.

(a) On any interval of debt through which the sovereign moves at a finite speed it is indifferent, $V' = -q$, and the price solves

$$q'(B) = -\frac{(\rho + a(d))\,\big(q - q_\rho(d)\big)}{a(d)B}, \qquad d = d^*(B, q),$$

with drift $S = a(d)B\,\xi(d)(q_r(d) - q)/\big((\rho + a(d))(q - q_\rho(d))\big)$; debt rises if and only if $q_\rho(d) < q < q_r(d)$.

(b) Every partial-default steady state is a point $(B(d), q_r(d))$, $d \in (d_{\text{first}}, 1)$, with $q_r(d) \geq \underline{q}$ under the standard. The one with intensity d is stable from above if and only if $\rho - r < \Omega(d)$.

(c) If $\Theta(d) \leq \rho - r < \Omega(d)$ and $q_r(d) \geq \underline{q}$, and the solution of (a) through $(B(d), q_r(d))$ can be continued above it up to $\bar{B}$, beyond the full-default threshold or where the standard binds by the full-default branch or the constrained regime of Proposition A.4(b), with negative drift and $-V' \leq q$ throughout, then there is an equilibrium in which the sovereign issues $B(d)$ at once from zero debt and stays there, withholding the share d forever.

(d) $\Omega(d_{\text{first}}) > s_{\text{first}}$ if and only if $(\gamma_1 - 1)\xi(d_{\text{first}}) < \delta$. When this holds and $q_r(d_{\text{first}}) > \underline{q}$, for every wedge $\rho - r$ in $(s_{\text{first}}, \Omega(d_{\text{first}}))$ there are intensities d slightly above d_{first} that satisfy the conditions of (c) other than the continuation condition.

Proof of Proposition A.6. With $u(c) = c$ the maximized Hamiltonian is $\mathcal{R}^*(B,q) + S\,(q + V')$, linear in the drift: where $q + V' > 0$ the sovereign makes a discrete issuance, where $q + V' < 0$ it reduces its debt as fast as it can, and a finite positive speed requires $q + V' = 0$.

(a) On an interval with $q + V' = 0$ the HJB equation reduces to $\rho V = \mathcal{R}^*(B, q(B))$. Differentiating and using the envelope theorem, $\mathcal{R}^*_q = a(d)B$, gives $-\rho q = \mathcal{R}^*_B + a(d)Bq'$ with $\mathcal{R}^*_B = -(1-d)(\lambda+\delta) + qa(d)$, which is the stated equation because $(\rho + a)q - (1-d)(\lambda+\delta) = (\rho + a)(q - q_\rho)$. The pricing equation $\xi(d)q = (1-d)(\lambda+\delta) + Sq'$ gives $S = \xi(d)(q - q_r(d))/q'$, and substituting q' gives the drift. Since $\rho > r$, $q_\rho(d) < q_r(d)$, and the sign follows.

(b) At a steady state the price is $q_r(d^\circ)$ by (9), and d° is interior, so $(1-\kappa q)(\lambda+\delta)B = \gamma_0\gamma_1 F d^{\gamma_1 - 1}$ (Proposition 2). With $(1-\kappa q_r(d))/q_r(d) = (1-\kappa)/(1-d)$, this is $B = B(d)$. At $q = q_r(d)$ the equation in (a) has slope $q' = -q(\rho - r)/(a(d)B)$, the slope of Lemma A.2. Write $P(B,q) := \xi(d^*)q - (1-d^*)(\lambda+\delta)$, so that $S = P/q'$ along the solution and $S' = P_q + P_B/q'$ at the steady state. With $P_d = (1-\kappa q)(\lambda+\delta)$, $\partial_B d_{\text{int}} = d/((\gamma_1 - 1)B)$, and $\partial_q d_{\text{int}} = -\kappa d/((\gamma_1 - 1)(1-\kappa q))$, we have $P_B = (1-\kappa q)(\lambda+\delta)d/((\gamma_1-1)B)$ and $P_q = \xi(d) - \kappa(\lambda+\delta)d/(\gamma_1 - 1)$. If $P_q \le 0$ then $S' < 0$; otherwise $S' < 0$ if and only if $\rho - r < P_B a(d)B/(qP_q)$, which is $\Omega(d)$ after using $(1-\kappa q_r)(\lambda+\delta)d/q_r = s(d)$.

(c) Let $B^\circ = B(d)$ and $q^\circ = q_r(d)$. Set $q = q^\circ$ and $V(B) = V(B^\circ) + q^\circ(B^\circ - B)$ on $[0, B^\circ]$, with $\rho V(B^\circ) = \mathcal{R}^*(B^\circ, q^\circ)$, and above B° the solution of (a) and its continuation. From any $B < B^\circ$ the state moves at once to B° and stays, so the lenders' price there is q°, and the issuance is allowed because $q^\circ \ge \underline{q}$. On $[0, B^\circ]$, $q + V' = 0$, so the sovereign is indifferent among speeds and among discrete issuances that end at or below B°, and the HJB equation requires that standing still not be better: $D(B) := \rho V(B) - \mathcal{R}^*(B, q^\circ) \ge 0$. The function D is concave, because $\mathcal{R}^*(\cdot, q)$ is a maximum of functions affine in B. Moreover, $D(B^\circ) = 0$, and $D'(B^{\circ -}) = (\rho + a(d))\big(q_\rho(d) - q^\circ\big) < 0$. So $D \ge 0$ on $[0, B^\circ]$ if and only if $D(0) \ge 0$. Since $(1-d)(\lambda+\delta) - q^\circ a(d) = rq^\circ$, $\mathcal{R}^*(B^\circ, q^\circ) = \phi(d,0) - rq^\circ B^\circ$ and $D(0) = (\rho - r)q^\circ B^\circ - (1 - \phi(d,0))$, which is nonnegative if and only if $\rho - r \ge \Theta(d)$. A discrete issuance that ends above B°, at B' on the solution of (a), yields $q(B')(B' - B) + V(B')$, which does not exceed $V(B)$ because $V' = -q$ there and q decreases. Above B°, stability makes the drift negative near B°, and the continuation is the hypothesis. The closed form of Θ follows from $1 - \phi(d,0) = 1 - F + \gamma_0 F d^{\gamma_1}$, from $1 - F = (\gamma_1 - 1)\gamma_0 F d_{\text{first}}^{\gamma_1}$ by (14), and from $q_r(d)B(d) = \gamma_0\gamma_1 F d^{\gamma_1 - 1}(1-d)/((1-\kappa)(\lambda+\delta))$.

(d) Since $a(d) + \kappa(\lambda+\delta)d = \delta$, $\Omega(d_{\text{first}}) > s(d_{\text{first}})$ if and only if $\delta > (\gamma_1 - 1)\xi(d_{\text{first}})$, including the case of a nonpositive denominator. With $\Theta(d_{\text{first}}) = s_{\text{first}}$, from the closed form, continuity of Θ, Ω, and q_r gives, for a wedge in $(s_{\text{first}}, \Omega(d_{\text{first}}))$, an intensity d slightly above d_{first} with $\Theta(d) < \rho - r < \Omega(d)$ and $q_r(d) \ge \underline{q}$. □

Proof of Proposition 4. As in the proof of Proposition A.6, the sovereign makes a discrete issuance where $q + V' > 0$ and moves at a finite speed only where $q + V' = 0$.

(a) Along a path on which debt rises, the sovereign makes discrete issuances or moves along the solutions of Proposition A.6(a), and consecutive discrete issuances are one. Let B_e be the first debt level on the path at which $d^* > 0$. If the path reaches B_e along a solution of Proposi-

tion A.6(a), the price q_e there satisfies $q_e > q_\rho(0)$, for the drift to be positive just below, where $d = 0$, and $q_e \le q_r(d_{\text{first}})$, for it to be nonnegative just above, where $d = d_{\text{first}}$. So $q_\rho(0) < q_r(d_{\text{first}})$, which is $\rho - r > s_{\text{first}} = \Theta(d_{\text{first}})$. If the path reaches B_e by a discrete issuance from a point $B_O > 0$ of such a solution, the price is continuous at B_O, since a bond held just below B_O is carried into the issuance. So the landing price q_J equals the price at B_O and exceeds $q_\rho(0)$, while landing with a nonnegative drift requires $q_J \le q_r(d_J) \le q_r(d_{\text{first}})$. Again, $\rho - r > \Theta(d_{\text{first}})$. If the discrete issuance is made at zero debt and lands at (B_J, q_J) with intensity $d_J \in [d_{\text{first}}, 1)$, then $V(0) = q_J B_J + V(B_J)$ with $\rho V(B_J) = \mathcal{R}^*(B_J, q_J)$, and staying at zero debt forever is worth $1/\rho$, so $\mathcal{R}^*(B_J, q_J) + \rho q_J B_J \ge 1$. As $q_J \le q_r(d_J)$, $\mathcal{R}^*(B_J, q_J) \le \phi(d_J, 0) - r q_J B_J$, hence $(\rho - r) q_J B_J \ge 1 - \phi(d_J, 0)$. By the first-order condition for d_J, $(1 - \phi(d_J,0))/(q_J B_J) = (1 - \phi(d_J,0))(1 - \kappa q_J)(\lambda + \delta)/\left(\gamma_0 \gamma_1 F d_J^{\gamma_1 - 1} q_J\right)$, which is decreasing in q_J and equals $\Theta(d_J)$ at $q_J = q_r(d_J)$. So $\rho - r \ge \Theta(d_J) \ge \Theta^*$. If the discrete issuance overshoots, landing above the steady state B° that the path then descends to along a solution of Proposition A.6(a), the price is nonincreasing above B° and $V' = -q$ there. So $V(0) = q_J B_J + V(B_J) \le q^\circ B^\circ + V(B^\circ)$ with $\rho V(B^\circ) = \phi(d^\circ, 0) - r q^\circ B^\circ$, and $V(0) \ge 1/\rho$ again gives $\rho - r \ge \Theta(d^\circ) \ge \Theta^*$.

(b) The definition of Θ gives $\Theta(d_{\text{first}}) = s(d_{\text{first}}) = s_{\text{first}}$. Write

$$\Theta(d) = \frac{(1-\kappa)(\lambda+\delta)\, N(d)}{\gamma_1 (1-d)}, \qquad N(d) = d + (\gamma_1 - 1) d_{\text{first}}^{\gamma_1} d^{1-\gamma_1};$$

The sign of $\Theta'(d)$ is that of $G(d) := (1-d)N'(d) + N(d)$, and $G'(d) = (1-d)N''(d) \ge 0$, so Θ is increasing on $[d_{\text{first}}, 1)$ if and only if $G(d_{\text{first}}) \ge 0$. With $N(d_{\text{first}}) = \gamma_1 d_{\text{first}}$ and $N'(d_{\text{first}}) = 1 - (\gamma_1 - 1)^2$,

$$G(d_{\text{first}}) = \gamma_1 d_{\text{first}} - \left[(\gamma_1 - 1)^2 - 1\right](1 - d_{\text{first}}),$$

which is nonnegative under the condition of the statement, and always when $\gamma_1 \le 2$. If $G(d_{\text{first}}) < 0$, Θ decreases at d_{first} and $\Theta^* < s_{\text{first}}$. □

A.4 Proofs for Sections 3.2 and 3.4

Proof of Proposition 5. On $[0, \underline{B}]$ no state defaults and every path stays in the set, so (9) gives $q_j = q^{\text{rf}}$ there. No default at that price requires $B \le B_{\text{pd}}(z_j, q^{\text{rf}})$ in every state (Proposition 2), so $\underline{B} \le \min_j B_{\text{pd}}(z_j, q^{\text{rf}})$. Near B° the price is $q^{\text{rf}} = 1$ and no state defaults, so consumption in state k is $c_k = e^{z_k} - rB + S_k$, and the HJB equation of state j reads $\rho V_j = \max_S \{u(e^{z_j} - rB + S) + S V_j'\} + \sum_{k \ne j} \eta_{jk}(V_k - V_j)$. The maximand is strictly concave in S and the maximizer is interior, so with V_j twice differentiable at B° the envelope theorem gives, at B°, $\rho V_j' = -r u'(c_j) + S_j V_j'' + \sum_{k \ne j} \eta_{jk}(V_k' - V_j')$, and $S_j(B^\circ) = 0$ removes the second term. The first-order conditions $V_k' = -u'(c_k)$, valid at B° because each V_k is differentiable there and the cap and the floor are slack, turn this into (18). If c_j is the lowest consumption at B°, the right-hand side of (18) is at most zero while the left-hand side is positive, since $\rho > r$. □

Lemma A.3 (the price along a path)**.** *In the general model, fix measurable Markov policies, a state (B,z), and a stopping time σ. The price (9) satisfies*

(a) $q(B,z) \;\geq\; \mathbb{E}_{B,z}\Big[e^{-(r+a)\sigma}\, q(B_\sigma, z_\sigma)\Big]$, *with equality if and only if* $d_t = 1$ *for almost every* $t < \sigma$, *almost surely;*

(b) $q^{\mathrm{rf}} - q(B,z) \;=\; q^{\mathrm{rf}}\,(r+a)\;\mathbb{E}_{B,z}\displaystyle\int_0^\infty \exp\!\left(-\int_0^t \xi(d_s)\,ds\right) d_t\,dt.$

Proposition A.7 (a bond in full default is a claim on the recovery)**.** *In the general model, suppose that from the state (B,z) the sovereign is in full default until a stopping time τ. Then*

$$q(B,z) \;=\; \mathbb{E}_{B,z}\Big[e^{-(r+a)\tau}\, q(B_\tau, z_\tau)\Big] \;\leq\; q^{\mathrm{rf}}\;\mathbb{E}_{B,z}\Big[e^{-(r+a)\tau}\Big].$$

In particular:

(a) in the deterministic model, a sovereign that stays in full default forever has $q(B) = 0$;

(b) in the chain model, if the sovereign is in full default until the endowment leaves state j, which it does at the rate $\eta_j := \sum_{k\neq j}\eta_{jk}$, then

$$q_j(B) \;\leq\; q_j^* := q^{\mathrm{rf}}\,\frac{\eta_j}{r+a+\eta_j},$$

with equality if and only if the debt is priced at q^{rf} at the date of the switch, almost surely;

(c) in the diffusion model, if the sovereign is in full default at least until the endowment first reaches a level $\zeta > z$, then $q(B,z) \leq q^{\mathrm{rf}}\,\varphi(z)/\varphi(\zeta)$, where φ is the positive increasing solution of $\mathcal{L}_z\varphi = (r+a)\varphi$ with $\varphi'(-\bar{z}) = 0$.

Proof of Lemma A.3. (a) Fix a path, write $\Xi_t := \int_0^t \xi(d_s)\,ds$, and let $x \in [0, q^{\mathrm{rf}}]$. For $t \leq \sigma < \infty$ define

$$Y_t := \int_0^t e^{-\Xi_s}(1-d_s)(\lambda+\delta)\,ds + e^{-\Xi_t}\,e^{-(r+a)(\sigma-t)}\,x,$$

the value of a claim that follows the path until t, pays nothing between t and σ, and is worth x at σ. Since $\xi(d) - (r+a) = \kappa(\delta+\lambda)(1-d)$,

$$\frac{dY_t}{dt} = e^{-\Xi_t}\,(1-d_t)\,(\lambda+\delta)\Big[1-\kappa\,x\,e^{-(r+a)(\sigma-t)}\Big] \;\geq\; e^{-\Xi_t}(1-d_t)(\lambda+\delta)\Big(1-\kappa q^{\mathrm{rf}}\Big) \;\geq 0,$$

because $1-\kappa q^{\mathrm{rf}} = (r+a)/(r+\delta) > 0$ by Assumption 1. Hence $Y_\sigma \geq Y_0 = e^{-(r+a)\sigma}x$, with equality if and only if $d_t = 1$ for almost every $t < \sigma$. On paths with $\sigma = \infty$ the inequality $\int_0^\infty e^{-\Xi_s}(1-d_s)(\lambda+\delta)ds \geq 0$ is trivial, with equality under the same condition. By (9) and the strong Markov property, $q(B,z) = \mathbb{E}[Y_\sigma]$ with $x = q(B_\sigma, z_\sigma) \in [0, q^{\mathrm{rf}}]$ (Proposition A.1), and taking expectations gives the claim.

(b) Since $(\lambda+\delta) = q^{\mathrm{rf}}(r+\delta)$ and $(\lambda+\delta)(1-\kappa q^{\mathrm{rf}}) = q^{\mathrm{rf}}(r+a)$, we have $(1-d)(\lambda+\delta) = q^{\mathrm{rf}}[\xi(d) - (r+a)d]$. Substituting in (9) and using $\int_0^\infty e^{-\Xi_t}\xi(d_t)\,dt = 1$, which holds because $\xi \geq r+a > 0$, gives the identity. □

Proof of Proposition A.7. The equality is the case of equality in Lemma A.3(a) with $\sigma = \tau$, and the bound follows from $q \leq q^{\mathrm{rf}}$ (Proposition A.1). (a) With $\tau = \infty$ the expectation is zero. (b) Conditional on the path of debt in state j, the date τ of the switch is exponential with rate η_j, so $\mathbb{E}[e^{-(r+a)\tau}] = \eta_j/(r+a+\eta_j)$. Equality in $q \leq q^{\mathrm{rf}}$ at τ almost surely is the stated condition. (c) Let τ_ζ be the first date at which the endowment reaches ζ. The function φ is positive and increasing, and $e^{-(r+a)t}\varphi(z_t)$ is a bounded martingale up to τ_ζ, so $\mathbb{E}_z[e^{-(r+a)\tau_\zeta}] = \varphi(z)/\varphi(\zeta)$. Since $\tau \geq \tau_\zeta$, $\mathbb{E}[e^{-(r+a)\tau}] \leq \varphi(z)/\varphi(\zeta)$. □

Proof of Proposition 3. (a) Suppose that from (B_0, j) the sovereign stays in full default until the endowment leaves state j, and let B_t be the path of debt in state j. From every (B_t, j) it stays in full default until the switch as well, so $q_j(B_t) \leq q_j^* < \underline{q}$ for every t by Proposition A.7(b). The standard then forbids issuance along the whole path, so $\dot{B}_t = \ell_t - aB_t \leq -aB_t$ and $B_t \leq B_0 e^{-at} \to 0$. But full default requires $B \geq B_{\mathrm{fd}}(z_j, q')$ at a price $q' \geq 0$: the market price where the standard is slack (Proposition 2) and the shadow price where it binds (Proposition A.4), which is nonnegative because V_j is nonincreasing. Since $B_{\mathrm{fd}}(z_j, \cdot)$ is increasing, $B_t \geq B_{\mathrm{fd}}(z_j, 0) = \gamma_0\gamma_1 F(z_j)e^{z_j}/(\delta+\lambda) > 0$ for every t, a contradiction. A stationary point in full default is a debt level from which the sovereign stays in full default until the switch, so there is none.

Now let q_j be nonincreasing, and suppose that the sovereign is in full default at B_0 with $S_j(B_0) \geq 0$. Issuance is $\ell = S_j + aB_0 > 0$, so $q_j(B_0) \geq \underline{q}$, the standard does not restrict the sovereign at B_0, and full default there means $B_0 \geq B_{\mathrm{fd}}(z_j, q_j(B_0))$ and $B_0 > B_{\mathrm{pd}}(z_j, q_j(B_0))$ (Proposition 2). Follow the path in state j. Debt does not fall while the drift is nonnegative, and a path that reaches a debt level above which the drift is negative stays there, at a stationary point. So the path is nondecreasing: it rises forever, or it stops. By the first part the sovereign cannot be in full default at every date, so at some debt level $B_1 > B_0$ on the path it is not. Let q' be the price at which it applies the default rule at B_1: the market price $q_j(B_1)$ if the standard is slack there (Proposition 1) and a shadow price below it if the standard binds (Proposition A.4). Not being in full default at B_1 means $B_1 < B_{\mathrm{fd}}(z_j, q')$ or $B_1 \leq B_{\mathrm{pd}}(z_j, q')$. Both thresholds are increasing in the price and $B_1 > B_0$, so $q' > q_j(B_0)$, hence $q_j(B_1) \geq q' > q_j(B_0)$, which contradicts the monotonicity of q_j. So $S_j < 0$ wherever the sovereign is in full default, and debt falls as long as it is.

(b) Let τ be the date of the switch. By Lemma A.3(a) with $\sigma = \tau$ and the hypothesis, $q_j(B) \geq \mathbb{E}[e^{-(r+a)\tau}q^{\mathrm{rf}}] = q_j^*$, with equality if and only if $d_t = 1$ for almost every $t < \tau$. If $\underline{q} < q_j^*$ then $q_j(B) > \underline{q}$ and (12) does not restrict the sovereign at (B, j). □

Online Appendix to "Debt Capacity and the Anatomy of Sovereign Crises"

by Sangdong Kim and Gabriel Mihalache

B The Branch Above Capacity

This appendix gives the cube-root law of Theorem 2 to second order, and the conditions under which the two branches at capacity form an equilibrium.

Proposition B.1 (the cube-root law to second order)**.** *Under the hypotheses of Theorem 2, let u be real-analytic near $\bar{c}$, as CRRA utility is, and let $\psi(c) := u'(c)/|u''(c)|$. Then*

$$S(B^* + x) = -\beta\, x^{1/3} - \Sigma_1\, x^{2/3} + O(x), \qquad q^{\mathrm{rf}} - q(B^* + x) = \alpha\, x^{2/3} + Q_1\, x + O(x^{4/3}),$$
$$\Sigma_1 = -\frac{1}{4}\left[\beta^2 \frac{u'''(\bar{c})}{|u''(\bar{c})|} + \frac{3\psi(\bar{c})\,(\rho - r + \Pi_0)}{\beta}\right], \qquad Q_1 = -\frac{\Pi_0\,\Sigma_1}{\beta^2},$$

where $u'''(\bar{c})/|u''(\bar{c})| = (\nu+1)/\bar{c}$ for CRRA utility with relative risk aversion ν. Both terms in the bracket are positive when u is prudent and $\rho > r$, so $\Sigma_1 < 0$. The solution above capacity (the right branch of (17)) exists on a right neighborhood of B^*, is unique among solutions that stay bounded in the scaled variables $(q^{\mathrm{rf}} - q)/x^{2/3}$ and $-qS/x^{1/3}$ as $x \downarrow 0$ (hence among those with these asymptotics), and is an analytic function of $(B - B^*)^{1/3}$.

Proof. The proof continues that of Theorem 2, with its notation.

Step 3: the regular system (existence, uniqueness, and analyticity of the branch). In the variable $\chi = x^{1/3}$ the scaled unknowns $Q = (q^{\mathrm{rf}} - q)/\chi^2$ and $\Sigma = -qS/\chi$ solve an analytic system of Briot–Bouquet type. With $p = \chi^2 Q$, $w = \chi\Sigma$, and $d/dx = (3\chi^2)^{-1} d/d\chi$,

$$\chi Q_\chi = \frac{3q\Pi}{\Sigma} - 2Q, \qquad \chi\Sigma_\chi = \frac{3u'}{|u''|}\frac{\mathcal{R}_q q\Pi}{\Sigma^2} - \Sigma - 3\chi\left[\frac{u'}{|u''|}\frac{\mathcal{R}_B + \rho q}{\Sigma} + \frac{\mathcal{R}_q q\Pi}{\Sigma}\right] + 3\chi^2\mathcal{R}_B,$$

with the coefficients evaluated at $c = \mathcal{R}(B^* + \chi^3, q^{\mathrm{rf}} - \chi^2 Q) - \chi\Sigma$ and $(B^* + \chi^3, q^{\mathrm{rf}} - \chi^2 Q)$. For CRRA utility and $d_{\mathrm{first}} \in (0,1)$ all coefficients are analytic near $(0, \alpha, \beta)$. At $\chi = 0$ the right-hand sides vanish exactly at (α, β) and their Jacobian is $\begin{pmatrix} -2 & -3\Pi_0/\beta^2 \\ 0 & -3 \end{pmatrix}$ (the $(2,2)$ entry is $-6(u'/|u''|)a(d_{\mathrm{first}})B^*\Pi_0/\beta^3 - 1 = -3$ by the definition of β). Neither eigenvalue is a positive integer, so the Briot–Bouquet theorem (Hille 1976, Chapter 12) yields a unique solution analytic at $\chi = 0$ with the prescribed value. Since both eigenvalues are negative, every other solution of the linearized system behaves like χ^{-2} or χ^{-3} and is unbounded as $\chi \downarrow 0$, which is the sense in which the branch is unique. The nonlinear system leads there too: in $s = \ln\chi$ the Σ equation gives $d(\Sigma^3 - 3M)/ds = -3(\Sigma^3 - 3M) + O(\chi)$, so every branch with $\Sigma > 0$ that stays bounded as $\chi \downarrow 0$ converges to (α, β), where the linearization applies.

Step 4: the second-order coefficients. The analytic solution has $Q = \alpha + Q_1\chi + O(\chi^2)$ and $\Sigma = \beta + \Sigma_1\chi + O(\chi^2)$. In the coefficients of the system only $\psi(c)$, evaluated at $c = \bar{c} - \beta\chi + O(\chi^2)$, varies at order χ; the others differ from their values at (B^*, q^{rf}) by $O(\chi^2)$. Collecting the terms of order χ in the two equations, with $3\psi(\bar{c})a(d_{\mathrm{first}})B^*\Pi_0 = \beta^3$, $1/\psi + \psi'/\psi = u'''/|u''|$, and $R_B + \rho = \rho - r + \Pi_0$ for $R_B := -(1 - d_{\mathrm{first}})(\delta + \lambda) + q^{\mathrm{rf}}a(d_{\mathrm{first}})$, gives Σ_1 as stated and $Q_1 = -\Pi_0\Sigma_1/\beta^2$. Finally $-S = w/q = \chi\Sigma\,(1 + O(\chi^2))$ and $q^{\mathrm{rf}} - q = \chi^2 Q$. □

Proposition B.2 (existence of the no-default-basin equilibrium)**.** *In the deterministic model with the underwriting standard, let $F(0) < 1$ and $d_{\mathrm{first}} < 1$, let u be real-analytic near $\bar{c}$, as CRRA utility is, let $\underline{q} < q^{\mathrm{rf}}$ exceed the full-default prices, and let $B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$ and $\bar{c} = 1 - rB^*$.*

(a) The left branch. *There is a unique drift $S > 0$ on $[0, B^*)$ with $S(B^*) = 0$ that solves $SS' = rS - (\rho - r)\psi(1 - rB + S)$, $\psi := u'/|u''|$. With $q \equiv q^{\mathrm{rf}}$, $d \equiv 0$, $c = 1 - rB + S$, and V given by $V(B^*) = u(\bar{c})/\rho$ and $V' = -u'(c)$, it satisfies the HJB and pricing equations on $[0, B^*]$, as long as $S < \bar{S}$.*

(b) The right branch. *The solution of (17) that leaves B^* along the cube-root expansion of Theorem 2 exists and is unique near B^*. Suppose its maximal continuation keeps $S^* < 0$ and $c^* > c_{\min}$ up to the full-default threshold, and that beyond it buybacks are not optimal, $-V_B \le qu'(c)$. Then it extends to all of $[B^*, \bar{B}]$. Where the standard binds before full default, on $\{q < \underline{q},\ \ell^* > 0\}$, the system switches to the constrained regime of Proposition A.4(b), $S = -a(d^{**})B$ with d^{**} at the shadow price, and consumption steps down at the switch; beyond the full-default threshold the sovereign neither issues nor buys back, debt decays as $B_t = B_0 e^{-at}$, the price is the power $q(B) \propto B^{-\xi(1)/a}$, and V is given by quadrature.*

(c) The equilibrium. *Under the conditions of (a) and (b), the two branches together form a no-default-basin equilibrium on $[0, \bar{B}]$ in which debt reaches B^* from above in finite time, the only one whose right branch has the cube-root asymptotics of Theorem 2. In it q is continuous on $[0, \bar{B}]$, and V is C^1 below the point where the standard starts to bind.*

Proof of Proposition B.2. (a) In the parametrization $dB/d\vartheta = S$, $dS/d\vartheta = rS - (\rho - r)\psi(1 - rB + S)$ the basin equation is a regular system, and at $(B^*, 0)$ its vector field is $(0, -(\rho - r)\psi(\bar{c})) \neq 0$, since $\rho > r$. So exactly one trajectory passes through $(B^*, 0)$, and it crosses $S = 0$ downward: for $\vartheta < 0$ it has $S > 0$ and $B < B^*$. Followed backward in ϑ, B falls while $S > 0$, and S cannot return to zero, because on $S = 0$ the field points downward, so the trajectory could only reach $S = 0$ moving forward in ϑ. The system has no rest point with $S \ge 0$, so B reaches 0 at a finite ϑ, and the trajectory is the graph of a drift $S(B) > 0$ on $[0, B^*)$, the unique one with $S(B^*) = 0$. Define c and V as in the statement. Then $\frac{d}{dB}(\rho V - u(c) - SV') = -(\rho - r)u'(c) + Su''(c)(S' - r)$, which vanishes by the basin equation, since $S(S' - r) = -(\rho - r)u'/|u''|$. By construction, $\rho V(B^*) - u(\bar{c}) = 0$. So the HJB equation holds on $[0, B^*]$ with the drift S, which is the maximizer given $V' = -u'(c)$ and $q = q^{\mathrm{rf}}$ by concavity, as long as it is below the cap. Moreover, $d = 0$ is optimal there by Proposition 2, since $B \le B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$, and $S(0) > 0$ satisfies

the state constraint. Finally, every path reaches B^* in finite time (Lemma 1) and stays, without default, so (9) gives $q \equiv q^{\mathrm{rf}}$. (b) Local existence and uniqueness near B^* are Step 3 of the proof of Proposition B.1, which concerns the system (17) alone. Away from B^* the system (17) is a regular ODE as long as $S^* < 0$, $c^* > c_{\min}$, and $d^* \in (0,1)$. The HJB equation, which is implicit in V' through c^*, can be solved for V' by the implicit function theorem because its derivative in V' is $S^* \neq 0$, and $(B,q) \mapsto d^*, \mathcal{R}^*$ are smooth there. So Picard–Lindelöf continues the branch uniquely from the cube-root start up to the first point where one of these fails or the standard starts to bind. On $\{q < \underline{q}\}$ with $\ell^* > 0$ the policy is the constrained one of Proposition A.4(b), $S = -a(d^{**})B < 0$, and the HJB equation $\rho V = u(c^{**}) - a(d^{**})BV'$ together with the pricing equation is again a regular system in (V,q), continued in the same way up to the full-default threshold. At the switch, consumption drops by about $q\ell^*$ and V' may have a kink. Beyond the full-default threshold, with $\underline{q}$ above the full-default prices the sovereign cannot issue, and by hypothesis it does not buy back, so $S = -aB$. Then the pricing equation $\xi(1)q = -aBq'$ gives the power law, and $V(B) = \int_0^\tau e^{-\rho t}u((1-\gamma_0)F(0))dt + e^{-\rho\tau}V(B_f)$ with $\tau = a^{-1}\ln(B/B_f)$, where B_f is the debt at which the default rule, at the shadow price where the standard binds, reaches full default. Continuity of q on $(B^*, \bar{B}]$ follows from Proposition A.2, as the drift is negative throughout. (c) The value is C^1 at B^*: on the left $V'(B^{*-}) = -u'(\bar{c})$, and on the right $V' = -qu'(c) \to -u'(\bar{c})$, by Step 1 of the proof of Theorem 2, which uses only the branch. So the HJB equation holds at B^* with $S = 0$, and the pieced pair solves the HJB and pricing equations on $[0, \bar{B}]$, with the policies of the two branches attaining the maximum. It is of no-default-basin type by construction. Debt reaches B^* from above in finite time, since by the cube-root law the time to reach B^* from $B^* + x$ is of order $x^{2/3}$. The price is continuous at B^*, which is reached in finite time from both sides, and on each branch. Uniqueness: by Theorem 1 every no-default-basin equilibrium in which debt reaches B^* from above in finite time has its steady state at B^*. Its left branch is the unique trajectory of (a), and a right branch with the cube-root asymptotics is the unique branch of Step 3 of the proof of Proposition B.1, continued uniquely as in (b). □

C The Standard in Full Default

This appendix completes the results on issuance in full default of Section 3.3 and on the recovery-claim threshold of Section 3.2, and proves them.

The deterministic model. The endowment is constant at z. Write $F = F(z)$, $\hat{c} := (1-\gamma_0)Fe^z$ for consumption in full default without issuance. The full-default threshold at a zero price is $B_{\mathrm{fd}}(z,0) = \gamma_0\gamma_1 Fe^z/(\delta+\lambda)$.

Proposition C.1 (debt does not rise in full default)**.** *In the deterministic model, let (V,q) be an equilibrium in which V is nonincreasing in debt, under the convention against trade at a zero price or under a standard $\underline{q} > 0$.*

(a) There is no debt level from which the sovereign stays in full default forever. In particular, no stationary point of debt is in full default.

(b) If q is nonincreasing in debt, the drift of debt is negative at every debt level at which the sovereign is in full default, and debt falls until the sovereign leaves full default.

The sovereign can still issue in full default while its debt falls. How much it issues has a closed form.

Proposition C.2 (issuance in full default)**.** *In the deterministic model, let (V, q) be an equilibrium and I an interval of debt levels at which the sovereign is in full default, the drift is negative, and consumption and the drift are interior, so that (7) holds and the standard does not restrict issuance. Let $\mathcal{R}_1(B) := \mathcal{R}(1; B, z, q(B)) = \hat{c} + a\,q(B)\,B$ be consumption at zero drift, $v(B) := -S(B)/(aB)$ the speed at which debt falls relative to a, $\psi := u'/|u''|$, and $K(B) := \psi(c(B))/(a^2 q(B) B)$. On I:*

(a) The value. *$\rho V = u(\mathcal{R}_1) + g$ with $0 < g \leq (\mathcal{R}_1 - c)(u'(c) - u'(\mathcal{R}_1))$ and $\mathcal{R}_1 - c = a\,q B\,v$.*

(b) The price. *The elasticity of the price in debt is*

$$-\frac{Bq'}{q} = \frac{r+a}{a\,v} = \frac{\rho+a}{a} + \frac{c'(B)\,S(B)}{a\,\psi(c(B))}.$$

(c) The speed. $B\,v' = \dfrac{K}{v}\left(\dfrac{r+a}{v} - (\rho+a)\right) - (1-v)\left(\dfrac{r+a}{a\,v} - 1\right)$.

(d) Once the sovereign issues, it issues at every higher debt. *If $\ell(B_2) \geq 0$ at some $B_2 \in I$, then $0 < \ell < aB$ at every higher debt level in I.*

(e) The limit. *Suppose that $\ell(B_2) \geq 0$ at some $B_2 \in I$, that $c_{\min} \leq \hat{c}$, that the price is nonincreasing in debt above B_2, and that neither the standard nor the cap on the drift restricts the sovereign above B_2. Then the solution of (17) through $(B_2, V(B_2), q(B_2))$ continues to every debt level above B_2, in full default and with a negative drift. The equilibrium coincides with it up to the debt limit $\bar{B}$, and as $B \to \infty$*

$$\frac{\ell(B)}{aB} \to \frac{\rho - r}{\rho + a}, \qquad -\frac{Bq'(B)}{q(B)} \to \frac{\rho+a}{a}, \qquad \frac{S(B)}{B} \to -\frac{a\,(r+a)}{\rho+a}, \qquad c(B) \to \hat{c}.$$

For a risk-neutral sovereign the first three hold with equality at every debt level of I (Proposition A.6(a) at $d = 1$).

In full default the sovereign pays nothing, and a unit of debt costs it only at the date at which it leaves full default. By part (a) its value is the value of holding its debt constant forever at the current price, up to a term of second order in qS, the market value of the reduction of its debt. So the sovereign is close to indifferent about how much to issue, and the pricing equation determines the amount, as it does for the risk-neutral sovereign of Proposition A.6. Lenders

discount at r and the sovereign at ρ. Indifference requires a price that falls with debt with the elasticity $(\rho+a)/a$, and lenders pay such a price only if debt falls at the rate $a(r+a)/(\rho+a)$. The sovereign therefore issues the share $(\rho-r)/(\rho+a)$ of the amount aB that would hold its debt constant. The share is below one, so debt falls, as Proposition C.1 requires, and Proposition C.2 gives the rate. The share is positive: without a standard the sovereign issues at every debt level in full default from the first one at which it does not buy back. The curvature of u enters through $1/K = a\,(a\,qB)/\psi(c)$, the amount at stake relative to consumption, which vanishes as debt rises.

Proposition C.3 (rest points in the chain model)**.** *In the chain model, let $B^\circ > 0$ be a stationary point of debt in state j, with default intensity d°. Define, for $\eta \geq 0$ and $d \in [0,1]$,*

$$q_\eta(d) := \frac{(1-d)(\lambda+\delta) + \eta\, q^{\mathrm{rf}}}{\xi(d)+\eta},$$

which is decreasing in d with $q_\eta(0) = q^{\mathrm{rf}}$ and $q_\eta(1) = q^{\mathrm{rf}}\,\eta/(r+a+\eta)$. Then:

(a) $q_j(B^\circ) = \dfrac{(1-d^\circ)(\lambda+\delta) + \sum_{k\neq j}\eta_{jk}\,q_k(B^\circ)}{\xi(d^\circ)+\eta_j} \;\leq\; q_{\eta_j}(d^\circ)$*, with equality if and only if $q_k(B^\circ) = q^{\mathrm{rf}}$ for every k with $\eta_{jk} > 0$;*

(b) under the standard, $q_j(B^\circ) \geq \underline{q}$, hence

$$d^\circ \;\leq\; d_j^\sharp := \frac{(q^{\mathrm{rf}} - \underline{q})\,(r+\delta+\eta_j)}{(\lambda+\delta)\,(1-\kappa\underline{q})},$$

and $d_j^\sharp < 1$ if and only if $\underline{q} > q_j^*$;

(c) if $d^\circ \in (0,1)$, then $B^\circ = B_{\mathrm{fd}}\big(z_j, q_j(B^\circ)\big)(d^\circ)^{\gamma_1-1} \leq B_j(d^\circ)$, where

$$B_j(d) := B_{\mathrm{fd}}\Big(z_j, q_{\eta_j}(d)\Big)\,d^{\gamma_1-1} = \frac{\gamma_0\gamma_1 F(z_j)e^{z_j}\; d^{\gamma_1-1}\big(\xi(d)+\eta_j\big)}{(\lambda+\delta)\big(r+a+\eta_j(1-\kappa q^{\mathrm{rf}})\big)};$$

(d) if $\underline{q} > q_j^*$ and B_j is increasing on $(0, d_j^\sharp]$, which holds when $(\gamma_1-1)(\xi(d)+\eta_j) > \kappa(\lambda+\delta)d$ there, every stationary point of state j in default has

$$B^\circ \;\leq\; B_j^\sharp := B_j(d_j^\sharp) = B_{\mathrm{fd}}(z_j,\underline{q})\,(d_j^\sharp)^{\gamma_1-1} \;<\; B_{\mathrm{fd}}(z_j,\underline{q}).$$

With $\eta_j = 0$ part (b) is the condition of Section 3.3. We call $B_j^\sharp$ the sovereign's *debt capacity in the crisis* (not the $B^\sharp$ of Proposition A.5): the largest debt at which it can rest in default while the endowment is in state j. It is a closed form in the standard, the recovery rate of the endowment, the default costs, and the contract, and it is reached when the recovery is priced at par and the sovereign borrows up to the floor. The threshold q_j^* is the level of the standard at which the debt capacity in the crisis meets the full-default threshold: $d_j^\sharp = 1$ and $B_j^\sharp = B_{\mathrm{fd}}(z_j,\underline{q})$ exactly at $\underline{q} = q_j^*$.

Corollary C.1 (recurring crises)**.** *In the chain model with two states, let the crisis state b carry no fixed cost of default, let the endowment leave it at the rate η, and let it return at the rate η'. Define*

$$Q_b := \frac{\eta\,(\lambda+\delta)}{(r+a+\eta)(r+\delta+\eta')-\eta\eta'},$$

the price in the crisis of a bond on which the sovereign pays nothing in every crisis and pays in full otherwise. Then $Q_b \le q_b^*$, with equality if and only if $\eta' = 0$. If, on the paths from (B,b), the sovereign does not default while the endowment is in the normal state, then $q_b(B) \ge Q_b$, with equality if and only if it is in full default in every crisis, and the spread at (B,b) is at most $(\lambda+\delta)/Q_b - (\delta+r) = (r+a)(r+\delta+\eta')/\eta$. So if $\underline{q} < Q_b$ the standard is slack at (B,b), and by Proposition 3(a) the sovereign does not stay in full default in a crisis if $\underline{q} > q_b^*$.

Proofs for the deterministic model.

Proof of Proposition C.1. (a) Suppose that from B_1 the sovereign stays in full default forever, and let B_t be the path of debt. No payment is ever made along it, so $q(B_t) = 0$ for every t by Proposition A.1. Under the convention against trade at a zero price this contradicts Proposition A.3, by which the price is positive at every state. Under a standard $\underline{q} > 0$ the price is below the floor along the whole path, so the sovereign cannot issue, $\dot{B}_t = \ell_t - aB_t \le -aB_t$, and $B_t \le B_1 e^{-at} \to 0$. But full default requires $B \ge B_{\mathrm{fd}}(z,q')$ at a price $q' \ge 0$: the market price where the standard is slack (Proposition 2) and the shadow price where it binds (Proposition A.4), which is nonnegative because V is nonincreasing. Since $B_{\mathrm{fd}}(z,\cdot)$ is increasing, $B_t \ge B_{\mathrm{fd}}(z,0) > 0$ for every t, a contradiction. A stationary point in full default is a debt level from which the sovereign stays in full default forever.

(b) Let q be nonincreasing, and suppose that the sovereign is in full default at B_1 with $S(B_1) \ge 0$. Its issuance is $\ell = S + aB_1 > 0$, so $q(B_1) \ge \underline{q}$, the standard does not restrict it at B_1, and full default there means $B_1 \ge B_{\mathrm{fd}}(z, q(B_1))$ and $B_1 \ge B_{\mathrm{pd}}(z, q(B_1))$ (Proposition 2). Follow the path from B_1. Debt does not fall while the drift is nonnegative, and a path that reaches a debt level above which the drift is negative stays there, at a stationary point. So the path is nondecreasing. By (a) the sovereign is not in full default at every date, so at some debt level $B_2 > B_1$ on the path it is not. Let q' be the price at which it applies the default rule at B_2: the market price $q(B_2)$ if the standard is slack there (Proposition 1) and a shadow price below it if the standard binds (Proposition A.4). Not being in full default at B_2 means $B_2 < B_{\mathrm{fd}}(z,q')$ or $B_2 \le B_{\mathrm{pd}}(z,q')$. The first threshold is increasing and the second nondecreasing in the price, and $B_2 > B_1$, so $q' > q(B_1)$, hence $q(B_2) \ge q' > q(B_1)$, which contradicts the monotonicity of q. So the drift is negative wherever the sovereign is in full default, and debt falls as long as it is. □

Proof of Proposition C.2. (a) By (4) with $d = 1$, $c = \mathcal{R}_1 + qS$. With (7) the HJB equation (6) reads

$$\rho V = u(c) + SV' = u(c) - u'(c)\,qS = u(c) - u'(c)(c - \mathcal{R}_1). \tag{21}$$

So $g = u(c) - u(\mathcal{R}_1) - u'(c)(c - \mathcal{R}_1) = \int_c^{\mathcal{R}_1} (u'(c) - u'(x))dx$. Since $S < 0$, $c < \mathcal{R}_1$, and u' is decreasing, which gives the two bounds. Finally $\mathcal{R}_1 - c = -qS = a\,qB\,v$.

(b) The pricing equation in (17) with $d^* = 1$ is $q'S = (r+a)q$, so $-Bq'/q = -(r+a)B/S = (r+a)/(av)$. On I the system (17) is regular, so V is twice and c once continuously differentiable. Differentiating (21), with $\mathcal{R}_1' = a(q + Bq')$,

$$\rho V' = -u''(c)\,c'(c - \mathcal{R}_1) + u'(c)\,a\big(q + Bq'\big).$$

Substituting $V' = -u'(c)q$ and $c - \mathcal{R}_1 = qS$ and dividing by $u'(c)q$ gives $-\rho = c'S/\psi(c) + a(1 + Bq'/q)$, which is the second equality.

(c) Since $c = \mathcal{R}_1 + qS = \hat{c} + a\,qB\,(1 - v)$ and $(qB)' = q(1 + Bq'/q)$,

$$c' = a\,q\left[(1-v)\left(1 + \frac{Bq'}{q}\right) - Bv'\right].$$

Substitute this, $S = -aBv$, and $-Bq'/q = (r+a)/(av)$ into the second equality of (b), and multiply by aK/v.

(d) $\ell = S + aB = aB(1 - v)$, so $\ell \geq 0$ if and only if $v \leq 1$, and $\ell < aB$ because $v > 0$. At a debt level with $v = 1$, (c) gives $Bv' = -K(\rho - r) < 0$. So v cannot return to 1 from below, and $v < 1$ at every debt level of I above B_2.

(e) Let $\bar{v} := (r+a)/(\rho+a) < 1$. *Continuation.* While the solution continues with $0 < v \leq 1$, (b) gives $-Bq'/q \geq (r+a)/a > 1$, so q and qB are decreasing, and $c = \hat{c} + a\,qB\,(1 - v)$ lies in $[\hat{c}, \hat{c} + a\,q(B_2)B_2]$. On this interval $\psi \geq \psi_0 > 0$, so $K \geq \underline{K}(B) := \psi_0/(a^2 q(B)B)$, which is increasing in B. For $v < \bar{v}$ the first bracket of (c) is $(\rho+a)(\bar{v} - v)/v$, and the last term of (c) is larger than $-(r+a)/(av) = -(\rho+a)\bar{v}/(av)$, so

$$B\,v' > \frac{\rho + a}{v}\left(K\,\frac{\bar{v} - v}{v} - \frac{\bar{v}}{a}\right) \qquad \text{for } v < \bar{v}. \tag{22}$$

The right side is positive for $v < v_0 := \bar{v}/(1 + \bar{v}/(a\underline{K}(B_2)))$. Hence $v \geq \min\{v(B_2), v_0\} > 0$, and by (d) $v \leq 1$. So the drift stays negative, consumption stays at or above $\hat{c} \geq c_{\min}$, and full default stays optimal, because q is decreasing and the thresholds of Proposition 2 are increasing in the price. The system (17) therefore remains regular, as in the proof of Proposition B.2(b), and its solution continues to every debt level above B_2. The equilibrium coincides with it. Its price is nonincreasing above B_2, so the sovereign is in full default there, by the thresholds of Proposition 2, with a negative drift, by Proposition C.1(b). The right side of (21) is strictly decreasing in c on $c < \mathcal{R}_1$, so consumption is determined by V and q, and the equilibrium solves the same regular system from the same data at B_2.

The speed falls below $\bar{v}$. Where $\bar{v} \leq v < 1$ the first term of (c) is nonpositive and $(r+a)/(av) - 1 \geq r/a$, so $Bv' \leq -(1 - v)\,r/a$: the distance $1 - v$ grows at least like $B^{r/a}$. By (d) it is positive just above B_2. So v reaches $\bar{v}$ at a finite debt level. At $v = \bar{v}$, (c) gives $Bv' = -(1 - \bar{v})\,\rho/a < 0$, so $v < \bar{v}$ from then on.

The limit. Since $-Bq'/q \geq (r+a)/a$, $qB \leq q(B_2)B_2\,(B/B_2)^{-r/a} \to 0$. So $c \to \hat{c}$ and $\underline{K} \to \infty$. Fix $\epsilon \in (0, \bar{v})$. For B large enough $\underline{K}(B)\,\epsilon/\bar{v} \geq 2\bar{v}/a$, and then (22) gives $Bv' > (\rho+a)/a$ wherever $v \leq \bar{v} - \epsilon$. So v rises above $\bar{v} - \epsilon$ at a finite debt level and stays above it. Hence $v \to \bar{v}$, and the four limits follow from $\ell = aB(1-v)$, (b), and $S = -aBv$.

For $u(c) = c$, Proposition A.6(a) at $d = 1$, where $q_\rho(1) = q_r(1) = 0$, gives $q' = -(\rho+a)q/(aB)$ and $S = -a(r+a)B/(\rho+a)$. □

Proofs for the chain model.

Proof of Proposition C.3. (a) The state stays at B° with the intensity d° until the endowment leaves state j, at an exponential date τ with rate η_j, for state k with probability η_{jk}/η_j. By (9) and the strong Markov property, $q_j(B^\circ) = \mathbb{E}[\int_0^\tau e^{-\xi(d^\circ)t}(1-d^\circ)(\lambda+\delta)\,dt + e^{-\xi(d^\circ)\tau} q_{z_\tau}(B^\circ)]$, which is the expression, and $q_k \leq q^{\mathrm{rf}}$ gives the bound and its case of equality. The derivative of q_η is $q'_\eta(d) = -(\lambda+\delta)(1-\kappa q_\eta(d))/(\xi(d)+\eta) < 0$, and the two values follow from $(\lambda+\delta) = q^{\mathrm{rf}}(r+\delta)$. (b) At a stationary point issuance is $\ell = a(d^\circ)B^\circ \geq aB^\circ > 0$, which the standard permits only at $q_j(B^\circ) \geq \underline{q}$. With (a), $q_{\eta_j}(d^\circ) \geq \underline{q}$, and since q_{η_j} is decreasing, $d^\circ \leq d_j^\sharp$, the solution of $q_{\eta_j}(d) = \underline{q}$:

$$d\,(\lambda+\delta)(1-\kappa\underline{q}) = (\lambda+\delta) + \eta_j q^{\mathrm{rf}} - \underline{q}(r+\delta+\eta_j) = (q^{\mathrm{rf}} - \underline{q})(r+\delta+\eta_j).$$

Moreover $d_j^\sharp < 1$ if and only if $q_{\eta_j}(1) < \underline{q}$, that is $q_j^* < \underline{q}$. (c) At a stationary point the drift is zero, so consumption is $\mathcal{R}(d; B^\circ, z_j, q_j(B^\circ))$ by (4), and among the policies that keep the state at B° the sovereign chooses the intensity that maximizes $\mathcal{R}$ at the market price. With $d^\circ \in (0,1)$ this is $d^\circ = d_{\mathrm{int}}(B^\circ, z_j, q_j(B^\circ))$, which reads $B^\circ = B_{\mathrm{fd}}(z_j, q_j(B^\circ))(d^\circ)^{\gamma_1 - 1}$ by (8). $B_{\mathrm{fd}}(z_j, \cdot)$ is increasing and $q_j(B^\circ) \leq q_{\eta_j}(d^\circ)$, which gives the bound. The closed form of B_j follows from

$$1 - \kappa q_\eta(d) = \frac{r + a + \eta(1 - \kappa q^{\mathrm{rf}})}{\xi(d) + \eta}.$$

(d) By (b), $d^\circ \leq d_j^\sharp < 1$, and by (c) and the monotonicity of B_j, $B^\circ \leq B_j(d^\circ) \leq B_j(d_j^\sharp)$. Since $q_{\eta_j}(d_j^\sharp) = \underline{q}$, $B_j(d_j^\sharp) = B_{\mathrm{fd}}(z_j, \underline{q})(d_j^\sharp)^{\gamma_1 - 1}$. Finally, B_j is increasing where the derivative of $d^{\gamma_1-1}(\xi(d) + \eta)$ is positive, and

$$\frac{d}{dd}\Big[d^{\gamma_1-1}(\xi(d)+\eta)\Big] = d^{\gamma_1-2}[(\gamma_1 - 1)(\xi(d)+\eta) - \kappa(\lambda+\delta)d].$$

□

Proof of Corollary C.1. Let Q_b and Q_g be the prices, in the two states, of the bond with default intensity $d'_t = \mathbb{1}\{z_t = z_b\}$. By (10), $(r+a+\eta)Q_b = \eta Q_g$ and $(r+\delta+\eta')Q_g = (\lambda+\delta) + \eta' Q_b$, which gives the expression. The denominator equals $(r+a+\eta)(r+\delta) + \eta'(r+a)$, so $Q_b = q^{\mathrm{rf}}\eta/(r+a+\eta+\eta'(r+a)/(r+\delta)) \leq q_b^*$, with equality if and only if $\eta' = 0$. Under the hypothesis the equilibrium default path from (B, b) satisfies $d_t \leq d'_t$ at every date, so $q_b(B) \geq Q_b$

by Lemma A.1, whose proof gives a strict inequality unless $d_t = d'_t$ for almost every t, almost surely. For the spread, $(\lambda+\delta)/Q_b = [(r+a)(r+\delta+\eta') + \eta(r+\delta)]/\eta$. □

D A Second Equilibrium Along the Path from Zero Debt

The no-default-basin equilibrium of Proposition B.2 is not the only equilibrium of the deterministic model. The cube-root branch of Theorem 2 continues below capacity, into a *pessimistic approach*: the sovereign defaults on a last stretch of debt before capacity, and the price is below q^{rf} on the whole path from zero debt. Write $b_0 := B_{\mathrm{pd}}(0,0) > 0$ for the no-default threshold at a zero price.

Proposition D.1 (the pessimistic approach)**.** *In the deterministic model with the underwriting standard:*

(a) Under the hypotheses of Proposition B.2, the branch of Proposition B.1, analytic in $(B-B^*)^{1/3}$, continues to $B < B^*$, where, with $x = B^* - B$,

$$S(B^*-x) = \beta x^{1/3}(1+o(1)), \qquad q^{\mathrm{rf}} - q(B^*-x) = \alpha x^{2/3}(1+o(1)),$$
$$c(B^*-x) = \bar{c} + \beta x^{1/3}(1+o(1)), \qquad \tau(B^*-x) = \frac{3}{2\beta}\, x^{2/3}(1+o(1)),$$

where τ is the time to reach B^*, and the sovereign is in partial default, $d = d_{\mathrm{int}}(B, q(B)) \geq d_{\mathrm{first}}$.

(b) Continued downward, the branch of (a) leaves the default region at a tie $B_2 = B_{\mathrm{pd}}(0, q(B_2)) > b_0$ and continues without default. If it keeps $S > 0$ and $c > c_{\min}$ down to $B = 0$, and $q \geq \underline{q}$ on $[B_2, B^*]$, then together with the right branch of Proposition B.2(b) it is an equilibrium, a pessimistic approach, and the only one whose branches stay bounded in the scaled variables of Proposition B.1.

Proof. (a) Step 3 of the proof of Proposition B.1 constructs the right branch as an analytic function of $\chi = (B-B^*)^{1/3}$ on a neighborhood of $\chi = 0$, as a solution of (17) with $d^* = d_{\mathrm{int}}$. The function is defined for $\chi < 0$, where $x = -\chi^3 > 0$, and the expansions $S = -\beta\chi - \Sigma_1\chi^2 + O(\chi^3)$ and $q^{\mathrm{rf}} - q = \alpha\chi^2 + Q_1\chi^3 + O(\chi^4)$ give the leading terms stated, with $c = \mathcal{R}^* + qS$. The system (17) encodes the first-order condition $V' = -qu'(c)$, which is sufficient for the drift because the objective is concave in S, and the static rule with $d^* = d_{\mathrm{int}}$. The only point to check on the left is that d_{int} is optimal, that is $B > B_{\mathrm{pd}}(0, q(B))$ (Proposition 2). The threshold is C^1 and increasing in the price near q^{rf}, with slope $C := \partial_q B_{\mathrm{pd}}(0, q^{\mathrm{rf}}) > 0$, so $B_{\mathrm{pd}}(0, q(B)) = B^* - C\alpha x^{2/3} + o(x^{2/3}) < B^* - x = B$ for small x. The standard, the drift cap, and the consumption floor are slack since $q \to q^{\mathrm{rf}} > \underline{q}$, $S \to 0$, and $c \to \bar{c}$, and V is obtained from $V' = -qu'(c)$ and $V(B^*) = u(\bar{c})/\rho$ as in Proposition B.2. The arrival time is $\int_0^x dy/S(B^*-y)$.

(b) Away from B^* the system (17) is a regular ODE while $S > 0$, $c > c_{\min}$, and the regime is fixed, as in Proposition B.2(b), so Picard–Lindelöf continues it. The default region is left where the static rule switches, at the first $B_2 < B^*$ with $\mathcal{R}(d_{\mathrm{int}}; B_2, q) = \mathcal{R}(0; B_2, q)$, which lies

above b_0 because below b_0 defaulting is never optimal. At the switch $\mathcal{R}^*$, hence c, S, and V', are continuous, and on $[0, B_2]$ the system continues with the no-default resource function and $q' = (r+\delta)(q - q^{\mathrm{rf}})/S < 0$. Under the stated conditions every constraint is slack along the branch, the pieced (V, q) solves the HJB and pricing equations on $[0, \bar{B}]$ with the right branch of Proposition B.2(b), the price is continuous at B^*, which both sides reach in finite time, and the steady state is B^*. Uniqueness among branches bounded in the scaled variables is the uniqueness statement of Step 3 of the proof of Proposition B.1, which does not depend on the sign of χ. □

The proposition takes a positive drift down to zero debt as given. Whether other equilibria exist along the path from zero debt is open at the calibrated parameters, where the standard does not exclude partial-default steady states (Section 4.3). Figure D.1 shows the two equilibria at the calibrated parameters near capacity.

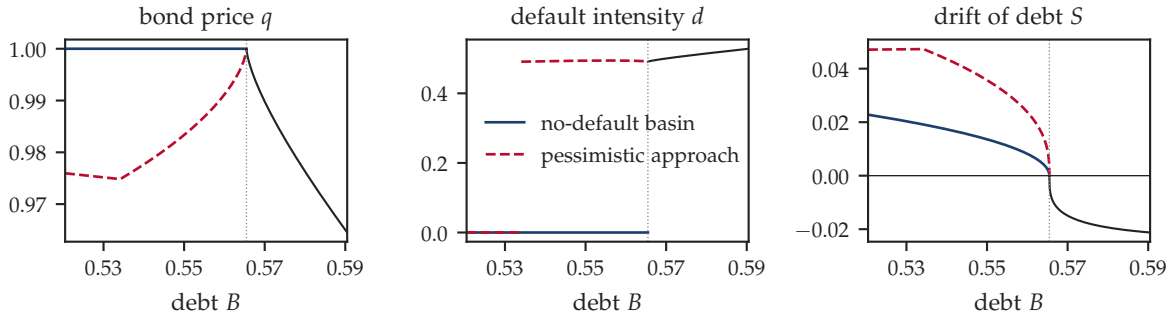


Figure D.1: The two equilibria of Proposition D.1 near capacity at the calibrated parameters: the no-default basin (solid) and the pessimistic approach (dashed) below B^* (dotted vertical line), with the common cube-root branch above it.

The pessimistic approach of Proposition D.1 is computed by integrating (17) downward from the two-term expansion of Proposition B.1 at $B^* - 10^{-6}$, in the form obtained by differentiating the HJB equation along the branch,

$$-\rho q = \mathcal{R}^*_B + \mathcal{R}^*_q\, q' + S q\, c'/\psi(c), \qquad q' = \frac{\xi(d) q - (1-d)(\lambda+\delta)}{S}, \qquad S' = \frac{c' - \mathcal{R}^*_B - \mathcal{R}^*_q q' - q' S}{q},$$

with $\psi = u'/|u''|$; on the right of B^* the same system reproduces the time-stepped solution to grid accuracy. At the calibrated parameters the default stretch is $(0.534, 0.565)$, 3.1 percent of output, and takes 1.1 years; the price is lowest at its start, 0.975, a spread of 43 basis points; the drift stays positive down to zero debt, and the sovereign reaches capacity in 10.9 years instead of 14.5. On a grid of 4,251 nodes on $[0, 0.85]$, away from B^*, the residuals of the two candidates under the scheme of Appendix E are 10^{-5} and 3×10^{-6} in (23) for the value, and 2×10^{-2} and 2×10^{-4} for the price, the larger one at the edge of the stretch, where the default intensity jumps. The policy kernel chooses partial default on the 157 nodes of the stretch. Started from the pessimistic candidate, the time stepping moves toward the basin equilibrium: after 4,000 steps default remains on five nodes below capacity. The value at zero debt is lower in the pessimistic

approach by 0.06 percent of consumption.

E Numerical Implementation

Discrete equations. Debt takes values on a grid B_i, $i = 0, \ldots, n_B - 1$, with step ΔB, and the log endowment on z_j, $j = 0, \ldots, n_z - 1$. One-sided differences are $D^+V_{ij} = (V_{i+1,j} - V_{ij})/\Delta B$ and $D^-V_{ij} = (V_{ij} - V_{i-1,j})/\Delta B$. At a node, for a given one-sided derivative V_B and price q, consumption is linear in the drift at fixed d, $c(S, d) = \mathcal{R}(d) + qS$. So the Hamiltonian $u(c(S, d)) + SV_B$ is concave in S with maximizer $S = (c^* - \mathcal{R}(d))/q$, $c^* = (u')^{-1}(-V_B/q)$, clamped to the admissible interval. The interval comes from $c \geq c_{\min}$, the cap $S \leq \bar{S}$ (applied also to $-S$), the sign restriction of the candidate, and the underwriting standard, $S \leq -a(d)B$ where $q < \underline{q}$ or, with the band of (24), $S \leq L(q) - a(d)B$. Over d the maximized Hamiltonian is concave on $(0, 1]$ and is maximized by golden-section search (28 iterations, a tolerance of about 10^{-6} in d), with the candidates $d = 0$, $d = 1$, and the closed form (8) also evaluated. Where the closed form is below one and neither the standard nor the consumption floor binds at it, it is the maximizer over $d > 0$ and the search is skipped. The consumption floor is $c_{\min} = 0.05\, e^z$ and the drift cap $\bar{S} = 1$, and neither binds at any reported solution. We form three candidates at every node: forward (D^+V, $S \geq 0$), backward (D^-V, $S \leq 0$), and zero drift. The one with the largest Hamiltonian is the policy. At B_0 the backward candidate is dropped and at B_{n_B-1} the forward one (state constraints). For a concave Hamiltonian this rule reduces to the standard upwind rule. With the fixed cost of default it is the monotone extension of that rule. We discretize the z-operator with upwinding of the exogenous drift and central second differences, so that every off-diagonal coefficient is nonnegative. With A the generator built from the policy, the discrete equilibrium is the root of

$$R_V \equiv \rho V - u(c) - AV = 0, \qquad R_q \equiv \xi(d)\, q - (1 - d)(\lambda + \delta) - Aq = 0, \tag{23}$$

and all convergence statements refer to $\|R_V\|_\infty$ and $\|R_q\|_\infty$ at the current (V, q) with the policy recomputed from them, never to step-to-step changes. Where two candidates attain the same Hamiltonian to within 10^{-6}, the policy is the one the iteration holds (Appendix E.3).

E.1 Time Stepping for the Finite-State Models

We solve the finite-state models by joint implicit upwind time stepping with the policy explicit from (V^k, q^k),

$$\Big[(\rho + 1/\Delta t)I - A^k\Big] V^{k+1} = u^k + V^k/\Delta t,$$
$$\Big[\operatorname{diag}\Big(\xi(d^k)\Big) + I/\Delta t - A^k\Big] q^{k+1} = (1 - d^k)(\lambda + \delta) + q^k/\Delta t,$$

with both solves sparse direct. The time step grows from 0.015 to 0.5 by two percent a step. We update the value and the price jointly at each implicit step, because the alternating iteration between them is unstable. The iteration starts from the value of servicing the debt at the risk-free rate forever, $u(e^z - rB)/\rho$, and from the stationary price implied by the interior default policy at the risk-free price. It stops when $\|R_V\|_\infty < 10^{-7}$ and $\|R_q\|_\infty < 10^{-8}$. We place the upper end of the grid where the drift is negative in every state, so that the truncation is exact, and we check this placement on the solution. The pricing equation uses the same generator as the HJB equation, so its transport term is upwinded with the equilibrium drift: forward at B_0, where the drift is nonnegative, and backward at B_{n_B-1}, where the state constraint makes it nonpositive. The stationary distribution solves $A^\top g = 0$ with the converged generator.

The chain illustrations. Figure 5 uses two states, $z = \pm 0.05$, that switch at the rate $\eta = 0.054$, on 201 debt nodes on $[0, 1]$. The iteration converges to residuals of 10^{-7} and 4×10^{-12} on the whole grid. So does the chain with states ± 0.06 and the rate 0.15. Above the low state's default threshold its computed solution has further stationary points in default, at debt levels the economy does not reach. The three-state chain mentioned in Section 3.4 has states $z = -0.05$, 0, and 0.05. The low state rises to the middle one, and the two upper states switch with each other, at half the mean-reversion rate $\mu = -\ln \rho_z$ of the diffusion model. Both upper states jump down to the low state at the rate 0.03, once every three decades. Every stationary point sits at a default threshold, the ergodic distribution has atoms at two of them, and default occurs on 0.07 of the ergodic mass, all of it in the low state.

E.2 Deterministic Model

We compute the deterministic solutions at the calibrated parameters (Figures 3 and E.1) by the time stepping of Appendix E.1 with a single endowment state, on a grid with 500 nodes per unit of debt on $[0, 1.5]$, to residuals of 10^{-7} and 10^{-11}. The computed steady state and first default agree with the closed forms (15) and (14) to within a grid step. The discrete system also admits an exact solution by reverse shooting outward from the steady state, node by node, which we use as a check and to compute Figure 4. The replication package documents it. At the calibrated parameters the partial-default candidates have prices up to $q_r(d_{\text{first}}) = 0.776$, above the standard, so the standard does not exclude them (Section 3.3). Reverse shooting cannot settle whether a candidate partial-default steady state is reached from zero debt (Section 3.3), because a whole family of solutions enters each candidate from below. We solve that problem, and the continuum of Section 3.3, by integrating forward from zero debt.

The cube-root law at the calibration. Figure E.1 checks Theorem 2. On log-log axes it compares the computed drift and price gap to the right of the steady state, and the drift to its left, with the two-term expansions of Proposition B.1 and with the square-root law, the constants computed from the primitives, $\beta = 0.113$, $\alpha = 0.33$, $\Sigma_1 = -0.131$, and $Q_1 = 0.254$. The leading terms alone

describe the solution only well inside the first grid step, because the second term is large: it is half the first already at $x = 0.1$, mostly through Π_0, since the first default is large here. With two terms the expansion matches the computed drift to within 8 percent and the price gap to within 12 percent up to five percent of output above the steady state, except within a few grid steps of the steady state, where the comparison is limited by the grid's placement of the steady state, which the law itself says is off by $O(h)$. What the grid shows as a drop in consumption at the threshold is, to leading order, $\beta h^{1/3}$ for a grid step h.

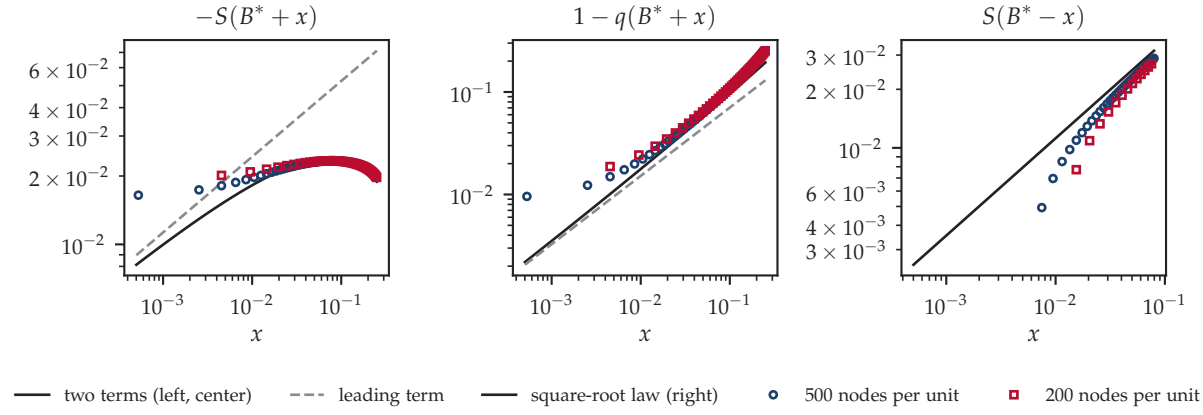


Figure E.1: The cube-root law at the steady state. Markers: the computed drift and price gap at a distance x above the steady state (left and center) and the drift at a distance x below it (right), on two grids. Lines: the two-term expansions of Proposition B.1 (solid) and their leading terms (dashed), with constants computed from the primitives. In the right panel, the line is the square-root law.

E.3 Diffusion Model

The diffusion model uses the same time stepping on the two-dimensional grid, with the z-operator discretized as above and Neumann conditions at $\pm\bar{z}$. Two features make the iteration converge at every node. The standard takes effect over a band of prices: issuance is capped by a function L of the price, (24) below, which is zero at the floor, as in (12), and grows without bound toward the top of the band. The floor (12) itself switches issuance on and off on a locus that no grid node hits. And a node at which two upwind candidates attain the same Hamiltonian to within 10^{-6} keeps the candidate of the previous step. A node whose candidate has changed three times within thirty steps keeps it for the rest of the iteration. At the reported solution 66 of the 15,075 nodes are held in this way, with ergodic mass of about 10^{-2}. The two linear systems of a step are solved by line Gauss–Seidel sweeps, an exact tridiagonal solve along the endowment at each debt level, upward and then downward in debt, to a relative residual of 10^{-13}.

$$\ell_t \le L(q(B_t, z_t)), \qquad L(q) = \begin{cases} 0, & q < \underline{q}, \\ \bar{\ell}\,\dfrac{w}{1-w}, \quad w = \dfrac{q-\underline{q}}{\omega}, & \underline{q} \le q < \underline{q} + \omega, \\ \infty, & q \ge \underline{q} + \omega, \end{cases} \tag{24}$$

with a band of $\omega = 0.03$ and $\bar{\ell} = 0.1$.

Other starting points. The reported solution starts from the solution at the starting point of the calibration search. We solve again at the calibrated parameters from four other starting points: half the equilibrium price with the equilibrium value, and flat prices of 0.62, of 0.30, and of one, each with the value of consuming the endowment net of interest on the debt forever. All four converge to the tolerances of the main solution, in about 600 steps. The mean of debt, the mass in default, and the mean and the standard deviation of the spread agree with the reported solution to three decimals. The solutions differ in the candidates held at tied nodes, and prices differ by at most 0.005 on the nodes with ergodic mass.

We minimize the calibration loss with two independent population searches (Hansen and Ostermeier 2001), of 24 points per generation, from the same starting point with different seeds and initial step sizes, on the grid of the main solution. Each evaluation solves the model from the best solution so far to residuals of 10^{-6}, at the level of the standard that the rule of Section 4.2 selects: it moves down the menu while a level is admissible and up otherwise. The searches made 5,016 evaluations and ended at nearly the same parameters. We report the best point, re-solved to the tolerances of Appendix E.1. The HJB residual is 2×10^{-9} and the pricing residual 9×10^{-9} at every node, after 672 steps. At the looser tolerances of the search the rule had selected the next lower level, 0.55, with a loss of 0.241; at the tolerances of the main solution the iteration does not converge at that level. The impulse responses of Section 4.3 are expectations under the forward equation $\dot{g} = A^\top g$, started from the ergodic distribution pushed down in z by the shock (split linearly between neighboring nodes) and advanced with a Crank–Nicolson step of 0.01 years.

Moments. The model's moments follow the annual accounting of Arellano, Mateos-Planas, and Ríos-Rull (2023). Debt due is $(\delta + \lambda)B$ per year, debt service is $(1 - d)$ times debt due, defaulted coupons are d times debt due, and the annual partial default is defaulted coupons over debt due within the year, as in their equation (4). The debt level is B, its face value, which equals its risk-free value since $q^{\mathrm{rf}} = 1$. Output and consumption are annual integrals, logged and linearly detrended. A year is in partial default if any coupon was defaulted on during it. At the calibrated point the instantaneous probability of being in default under the stationary distribution is 0.3, against an annual frequency of 0.39, a difference due to time aggregation. We compute moments that are functions of the state (debt, debt service, and debt due to output, the spread and its correlations) exactly from the stationary distribution of the discretized process, by solving the adjoint system $A^\top g = 0$. We compute moments that are path functionals under annual accounting (the

partial default frequency, mean and standard deviation, the persistence and volatility of output, and consumption) from sixteen simulated paths of 25,000 years each, by Euler–Maruyama steps with bilinearly interpolated policies, 100 steps per year, annual aggregation of flows, and reflecting boundaries for z. The same paths give the untargeted moments conditional on default.

A resolution artifact. The scheme obtains V_B by differencing, with an error of about $\frac{1}{2}|V_{BB}|\Delta B$, and the drift responds to that error with a gain of $1/(q^2|u''(c)|)$: consumption responds to the ratio V_B/q and the drift is the consumption gap divided by q again. Where the *resolution ratio*

$$\frac{\frac{1}{2}|V_{BB}|\,\Delta B}{q^2\,|u''(c)|\,|S|} \tag{25}$$

exceeds about one half, the scheme cannot determine the sign of the drift, and when the scheme loses the sign it tends to produce a positive drift, which carries the state to the top of the grid. The remedy is a grid whose spacing is proportional to q^2. Table F.1 reports the ergodic mass on unresolved nodes, those where the ratio exceeds one half, excluding stationary nodes ($|S| \leq 10^{-3}$).

E.4 Self-Fulfilling Partial Default with Curvature and Shocks

Risk aversion. From below, the state arrives at a candidate steady state along a square-root law with a nearly flat price, and away from the candidates risk aversion slows the borrowing. We find the candidates reached from zero debt by integrating (17) forward from zero debt, over a grid of starting prices and consumption levels. In the example of Figure 4, computed by reverse shooting (Appendix E.2), the sovereign withholds 28 percent of its debt service at the partial-default steady state, at a price of 0.9, a spread of 183 basis points. Default starts at $B = 0.62$, lower than under the optimistic price, because the threshold rises with the price. This steady state is one of a continuum: at this wedge the steady states reached from zero debt fill the interval from 0.93 to 1.02. The left panel of Figure E.2 shows the smallest wedge at which any is reached, as a function of the curvature. It approaches Θ^* as the curvature vanishes, but slowly, so even a little curvature raises it markedly.

Uncertainty. Shocks also work against self-fulfilling partial default. We study them with a risk-neutral sovereign and extend the construction of Proposition A.6 to the diffusion model, as described below. Lenders expect the sovereign to settle at a debt B° that does not depend on the endowment, and to stay there. The default intensity and the price at B° then vary with the endowment. A sovereign at zero debt makes a discrete issuance of B° when that beats staying debt-free, and above B° it is indifferent, as in Proposition A.6(a). The right panel of Figure E.2 shows where such equilibria exist, at the parameters of Figure 4. Volatility narrows the range of supporting wedges from both sides. From below, the discrete issuance must beat staying debt-free in every endowment state, and in bad states the pessimistic steady state carries a deeper and

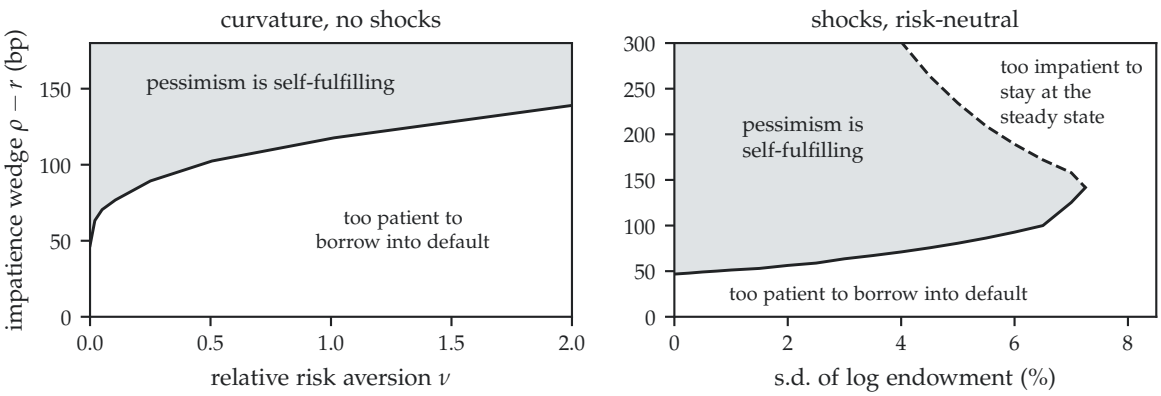


Figure E.2: Where pessimism is self-fulfilling. Shaded: impatience wedges $\rho - r$ at which a self-fulfilling partial-default equilibrium exists, at the parameters of Figure 4. Both solid lines start at the risk-neutral threshold $s_{\text{first}} = 47$ basis points. The panels have different vertical scales. Left: the deterministic model with CRRA utility, against the relative risk aversion ν. The upper boundary is not computed. Right: the risk-neutral diffusion model with mean reversion $\mu = 0.225$, against the standard deviation of the log endowment. Above the dashed line the steady state is no longer stable.

costlier default. From above, the steady state must be stable in every state, and the dilution that holds the sovereign there is weaker in some states. At a standard deviation of the log endowment of about 7 percent the two boundaries meet, and no equilibrium of this kind remains.

The construction. We compute the right panel of Figure E.2 with the following construction. The sovereign is risk-neutral, $u(c) = c$, and issuance is unrestricted in speed, as in Proposition A.6. The log endowment follows the diffusion of Section 4.1. The construction gives sufficient conditions for a self-fulfilling partial default, and we find the region numerically.

The steady debt. At B° the drift is zero and the default intensity is the static choice $d^\circ(z) = d^*(B^\circ, z, q^\circ(z))$. The price solves an ordinary differential equation in z,

$$\xi(d^\circ)\, q^\circ = (1 - d^\circ)(\lambda + \delta) + \mathcal{L}_z q^\circ,$$

and the value solves $\rho V^\circ = \mathcal{R}^*(B^\circ, z, q^\circ) + \mathcal{L}_z V^\circ$. With $\sigma = 0$ these reduce to $q^\circ = q_r(d^\circ)$ and $\rho V^\circ = \mathcal{R}^*$. We index the steady debts by the intensity d of the deterministic candidate, $B^\circ = B(d)$ of Proposition A.6, and follow the root of the price equation from $q_r(d)$.

Above the steady debt. On an interval through which the sovereign moves at a finite speed it is indifferent, $V_B = -q$. Differentiating $\rho V = \mathcal{R}^*(B, z, q) + \mathcal{L}_z V$ in B, with the envelope theorem, gives the price equation of Proposition A.6(a) with a diffusion term,

$$a(d) B\, q_B = (1 - d)(\lambda + \delta) - (\rho + a(d))\, q + \mathcal{L}_z q, \qquad d = d^*(B, z, q).$$

The equation is parabolic in B, so we integrate it upward from B°. The drift follows from the pricing equation, $S\,q_B = \xi(d)q - (1-d)(\lambda+\delta) - \mathcal{L}_z q$. The steady debt is stable from above if $S_B < 0$ at $B^{\circ+}$ in every state. As in Lemma A.2, this slope follows from the two equations at B°. The continuation condition is that $S < 0$ and $q_B < 0$ up to the full-default region.

Below the steady debt. A sovereign below B° makes a discrete issuance that takes it to B° at the price $q^\circ(z)$. So $q(B,z) = q^\circ(z)$ and $V(B,z) = V^\circ(z) + q^\circ(z)(B^\circ - B)$. Standing still must not be better. As in the proof of Proposition A.6(c), the gap $\rho V - \mathcal{R}^* - \mathcal{L}_z V$ is concave in B and vanishes at B°. The condition is therefore that the gap be nonnegative at zero debt in every state,

$$\mathcal{R}^*(B^\circ, z, q^\circ) + \rho\, q^\circ B^\circ - e^z - B^\circ\, \mathcal{L}_z q^\circ \geq 0.$$

With $\sigma = 0$ this is $\rho - r \geq \Theta(d)$. Finally, issuance at B° requires $q^\circ \geq \underline{q}$ in every state.

Waiting and computation. A sovereign at zero debt could also wait for a better endowment before it makes the discrete issuance. We allow this by solving the corresponding optimal-stopping problem at zero debt and checking that the sovereign at B° does not prefer to buy back its debt and wait; waiting does not change the boundaries of the region. The endowment grid has 201 nodes on ± 3 stationary standard deviations with reflecting boundaries. We solve the price equation at B° by Newton's method, or by the monotone map $q \mapsto [\xi(d^*(q)) - \mathcal{L}_z]^{-1}(1 - d^*(q))(\lambda+\delta)$ where Newton's method stalls at the jump of d^*, and we integrate above B° in debt, implicitly in $\mathcal{L}_z$. The region does not change when the grid is refined from 101 to 801 nodes, and with $\sigma = 0$ the construction reproduces the conditions of Proposition A.6. Its lower boundary depends on how far the endowment grid extends, because the condition on the discrete issuance binds in the worst states.

F Robustness

F.1 The Standard and the Grid

Without the standard. We solve the model at the calibrated parameters of Table 1 without the standard, on debt grids extending to $\bar{B} = 1$ and to $\bar{B} = 2$. The iteration does not settle without the standard, so these are the solutions after 8,000 steps of the plain scheme, with pricing residuals of 0.3 and 0.7. On both grids the sovereign issues in full default in the low endowment states, at prices up to 0.49 and 0.53, and about 1.4 percent of the ergodic mass is in full default. Debt runs up: on $[0,1]$ the top tenth of the grid holds 0.4 percent of the mass, and on $[0,2]$ mass reaches $B = 1.28$. The moments depend on the limit. The standard deviation of the spread is 0.053 on $[0,1]$ and 0.056 on $[0,2]$, and the loss 0.278 and 0.376. With the standard the spread volatility is 0.029 and the loss 0.259, and both change little across the grids of Table F.1. The standard binds

in partial default on about 2 percent of the ergodic mass, where the default intensity exceeds the market-price rule by 3 percent of that rule on average and by up to 7 percent.

Robustness of the calibrated point under the standard. Table F.1 re-solves the calibrated point from a warm start on a finer debt grid, on a non-uniform grid of the kind of Appendix E.3, with the borrowing limit at $\bar{B} = 1.5$, and with fewer and more endowment nodes. Across these grids the moments move by amounts comparable to the sampling noise of the simulated moments, the top of the grid carries no mass, and the loss is unchanged to the second digit. With the limit at $\bar{B} = 1.5$ the iteration does not reach the tolerance at debt above 1.12, where full default begins in the best states. No mass lies there, and on the nodes with mass the residuals are those of the main solution.

The last two rows widen the endowment range from ± 2.5 to ± 3 and ± 3.5 stationary standard deviations, at the same node spacing. The main grid has seven nodes below the calibrated threshold $\ln y^*$, with 1.5 percent of the ergodic mass. The wider grids have 14 and 22 such nodes, with 1.9 and 2.2 percent of the mass. The moments of debt, of default, and of the spread change little, and the largest debt that carries mass is the same. The loss is higher because the endowment is truncated less. The standard deviation of output rises from 0.105 to 0.109 and 0.11, away from its target, and this term accounts for most of the increase.

Table F.1: The calibrated point under the standard on other grids

Grid	Debt/y mean	Debt/y s.d.	Spread s.d.	PD freq.	PD mean	Loss	Unresolved
main grid	0.300	0.204	0.029	0.389	0.477	0.259	0.029
debt step halved	0.300	0.203	0.029	0.388	0.477	0.260	0.011
non-uniform debt grid	0.300	0.203	0.029	0.389	0.478	0.261	0.008
limit at $\bar{B} = 1.5$	0.300	0.204	0.029	0.389	0.477	0.259	0.028
$n_z = 51$	0.301	0.205	0.030	0.395	0.474	0.261	0.034
$n_z = 101$	0.300	0.203	0.029	0.387	0.478	0.259	0.025
endowment range ± 3 s.d.	0.300	0.206	0.030	0.390	0.484	0.267	0.026
endowment range ± 3.5 s.d.	0.300	0.205	0.030	0.391	0.483	0.265	0.028

Notes: Each row re-solves the calibrated point from a warm start. Main grid: 201 debt nodes (step 0.005) on $[0, 1]$ and 75 endowment nodes; debt step halved: 401 debt nodes; non-uniform: 760 debt nodes with steps from 0.0003 to 0.004, proportional to the square of the lowest price at each debt level; these rows use 75 endowment nodes on ± 2.5 stationary standard deviations. The rows $n_z = 51$ and $n_z = 101$ keep the debt grid and the endowment range of the main solution. The last two rows keep the debt grid of the main solution and the spacing of its endowment nodes, with 89 and 105 nodes. Their loss is higher mostly through the standard deviation of output, since the range of the endowment grid is part of the calibrated output process. Unresolved: ergodic mass on nodes where the resolution ratio (25) exceeds one half, excluding stationary nodes ($|S| \leq 10^{-3}$).

The response to a shock by initial state. Figure F.1 applies the same shock at the mean endowment from three debt levels, the ergodic mean and 25 percent below and above it, and reports each response against the path from the same state without the shock. The response of debt is nearly the same from the three states: it rises by some eight percent of output and peaks after ten years. The responses of the default intensity and of the spread are not. From the ergodic

mean and from above it the shock moves the threshold below the debt at once, and the intensity jumps to the first default and declines from there. From below the mean the economy enters default over the first years, as the endowment keeps moving and the threshold with it, and less deeply. The spread rises by more than twice as much from above the mean as from below. This state dependence echoes Fernández-Villaverde, Hurtado, and Nuño (2023), in whose economy the same shock has larger effects when leverage is high. Along all three shocked paths arrears are the smaller part of the gross flow of new debt, at most about a third in the first decade. The last panel shows this share along the shocked path, not the difference from the path without the shock as in Figure 8.

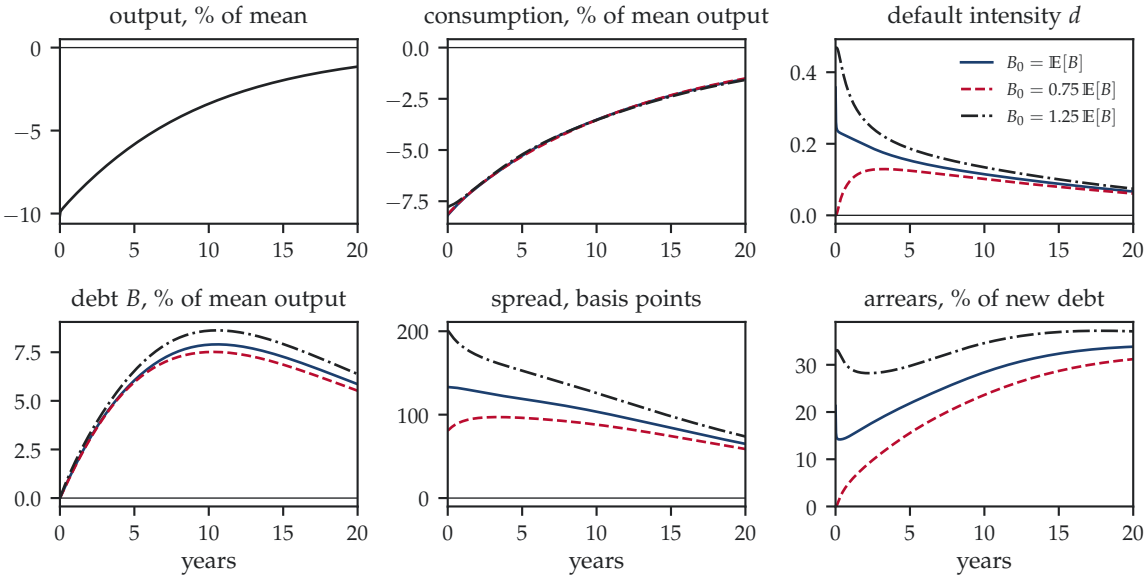


Figure F.1: The same shock from three states. The economy is at the mean endowment with debt at its ergodic mean (solid), 25 percent below it (dashed), or 25 percent above it (dash-dotted): expected path under the forward equation from that state minus the path from the same state without the shock. The last panel shows arrears as a share of the gross additions to debt, new issues plus arrears, along the shocked path.

F.2 Saving

As in Arellano, Mateos-Planas, and Ríos-Rull (2023), the main text restricts the sovereign to borrowing, $B \geq 0$. Here we relax the restriction at the calibrated parameters of Table 1, without recalibrating, as a check on the model's behavior. Below zero debt the sovereign holds a risk-free asset at the price q^{rf} and has nothing to default on. When its debt crosses zero, it retires its outstanding bonds, or issues new ones, at the market price, so bonds trade only at $B \geq 0$ and the bond price has the same form as without saving. We solve on 401 debt nodes on $[-1, 1]$, with the step of the main grid and 75 endowment nodes, warm-started from the main solution, to residuals of 9×10^{-8} and 7×10^{-7}. The bottom of the grid carries about 0.15 percent of the ergodic mass. The moments use gross debt, $\max\{B, 0\}$, and in states with assets the spread is

the one at which the sovereign would issue at zero debt, as at the atom at zero debt in the main solution.

Figure F.2 compares the ergodic density with and without saving. The difference is confined to the left of the distribution. The atom at zero debt, which holds 6 percent of the ergodic mass without saving, becomes a ridge of assets in good times: above about 0.7 stationary standard deviations of the endowment the stationary debt is negative, and 21 percent of the mass holds assets, 0.27 of output on average. On positive debt the density keeps its stationary debt, the bend where the price reaches the standard, its modes, and its tail into default. Table F.2 reports the moments: saving does not change the depth and dispersion of partial default, the defaulted coupons, the volatility of the spread, or the output process. Partial default is somewhat less frequent and debt somewhat lower on average, because time that the sovereign spends at zero debt without saving is spent holding assets with it, and consumption is a little smoother relative to output. The mechanisms of Sections 3 and 4 operate on positive debt and are not affected.

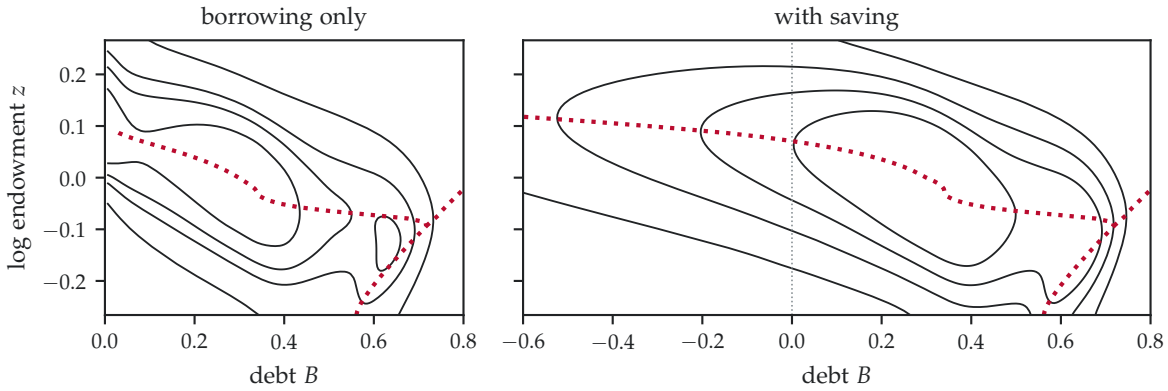


Figure F.2: The ergodic density of debt and the endowment at the calibrated parameters, when the sovereign only borrows (left, as in Figure 7) and when it can also save (right). The contours enclose 50, 75, 90, and 99 percent of the mass shown, which leaves out the atom at zero debt in the left panel. Dotted: the zero-drift locus. The panel widths are proportional to the range of debt they show.

Figure F.3 is the phase diagram of the model with saving. The arrows give the expected direction of motion, from the drift of debt and the mean reversion of the endowment. Debt is stationary on the solid line. In good times the stationary debt is negative, and the sovereign holds assets. It rises as the endowment falls, and in bad times it is the debt at which the price reaches the standard. The thin line that continues along the standard is where the standard stops issuance in better times, at debt levels the economy does not visit.

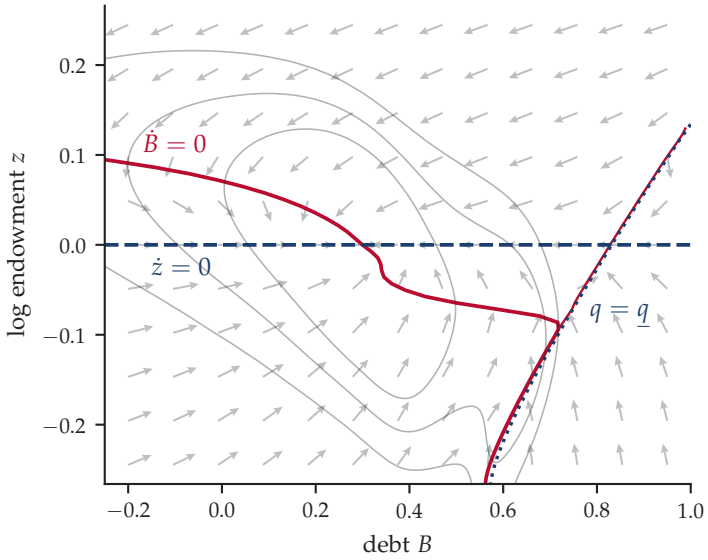


Figure F.3: Phase diagram of the model with saving at the calibrated parameters. Arrows: the expected direction of motion, $(S, -\mu z)$, drawn at a common length. Solid: the zero-drift locus of debt. Dashed: the zero-drift line of the endowment. Dotted: the price contour of the underwriting standard. Gray: contours that enclose 50, 75, and 90 percent of the ergodic mass, without the mass at the asset limit. The figure shows debt from -0.25.

Table F.2: Moments when the sovereign only borrows and when it can also save

Moment	Borrowing only	With saving
Partial default: frequency	0.39	0.36
Partial default: mean $\mid d > 0$	0.48	0.48
Partial default: s.d. $\mid d > 0$	0.26	0.26
Debt to output: mean	0.30	0.27
Debt to output: s.d.	0.20	0.22
Debt service to output: mean	0.033	0.028
Debt service to output: s.d.	0.018	0.021
Debt due to output: mean	0.050	0.044
Spread: s.d.	0.029	0.029
Output: persistence	0.93	0.93
Output: s.d.	0.105	0.105
Defaulted coupons to output: mean $\mid d > 0$	0.045	0.045
Defaulted coupons to output: s.d. $\mid d > 0$	0.034	0.034
Spread: mean	0.020	0.018
Spread: correlation with output	−0.76	−0.73
Spread: correlation with debt	0.84	0.83
Consumption s.d. relative to output s.d.	0.93	0.91

Notes: The calibrated parameters of Table 1, not recalibrated. The moments above the line are the targets of Table 2, and those below it are its untargeted moments, computed in the same way.